INVESTIGATION OF INTERACTIONS BETWEEN β-LACTAMASES AND β-LACTAMASE INHIBITORY PROTEINS (BLIPS) USING MASS SPECTROMETRIC APPROACHES

HUANG LIWEN

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Department of Applied Biology and Chemical Technology

Investigation of Interactions between β-Lactamases and β-Lactamase Inhibitory Proteins (BLIPs) using Mass Spectrometric Approaches

Huang Liwen

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Certificate of Originality

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Abstract

β-Lactamase inhibitory protein (BLIP) can effectively inactivate a variety of class A β-lactamases, but with very different degrees of potency. Understanding the roles of BLIP in class A β-lactamases inhibition can provide insights for novel inhibitor design. However, this problem was poorly solved based on the static structures obtained by X-ray crystallography.

In this study, ion mobility mass spectrometry, hydrogen deuterium exchange mass spectrometry and molecular dynamics simulation revealed the conformational dynamics of three class A β-lactamases with varying inhibition efficiency by BLIP. A more extended conformation of PC1 were shown compared to TEM1 and SHV1. Localized dynamics showed differences in several important loop regions, i.e., the protruding loops, H10 loops, Ω loops and SDN loops of the three β-lactamases. Upon binding with BLIP, these loops cooperatively rearranged to enhance the bindings and inactivate the catalytic sites. Unfavorable changes in conformational dynamics were found in the protruding loops of SHV1 and PC1. Intriguingly, single mutations on BLIP could compensate the unfavored changes in this region, and thus exhibited enhanced inhibition towards SHV1 and PC1. Additionally, the H10 region was revealed as an important

allosteric site that could modulate the inhibition of class A β-lactamases. It was suggested that the rigid protruding loop and flexible H10 region might be determinants for the effective inhibition of TEM1. Our findings provided unique and explicit insights into the conformational dynamics of β-lactamases and their bindings with BLIP.

The wide range of ability to bind the β-lactamases is also still a puzzle. The structural determinants to the significant enhancement of BLIP with single mutations are poorly understood yet. To further the understanding of these inhibitory interactions, we investigated the conformational dynamics of BLIP and three enhanced mutants (Y50A, E73M and K74G) upon their bindings to three clinically prevalent class A β-lactamases (TEM1, SHV1 and PC1), with varying potency between subnanomolar and micromolar. Hydrogen deuterium exchange mass spectrometry revealed that the flexibility of interdomain region was significantly suppressed upon the strong binding to TEM1 β-lactamases. By contrast, such change was not significant upon the weak bindings to SHV1 or PC1. Single mutations E73M and K74G on the interdomain region with improved potencies towards SHV1 and PC1, respectively, showed significantly improved flexibility of the interdomain region compared to the wild type. In contrast, more rigid structure of the β-hairpin loop 135-145 was observed upon individual mutation on BLIP. These results indicated that higher flexibility of the interdomain linker while more rigid interfacial β-hairpin loop 135-145 could be desirable for enhanced inhibitory bindings. Molecular dynamics simulation of these mutations of BLIP exhibited significant changes in the flexibility of the β-hairpin loop 46-51 and the distant β1'-β2' loop 115-125 compared to the wild type.

This study can be extended to other β-lactamases of interest, especially the inhibitor resistant β-lactamases. Together, these findings provided a general guideline for studying the interactions between β-lactamases and inhibitory proteins. Unique insights and deep understandings were obtained for the design of BLIP-derived inhibitors.

Publications Arising from This Thesis

Huang, L., So, P.K., Chen, Y.W., Leung, Y.C. and Yao, Z.P. The Interdomain Flexibility of β-Lactamase Inhibitory Protein (BLIP) Modulates Its Binding to Class A β-Lactamases, to be submitted.

Huang, L., So, P.K., Chen, Y.W., Leung, Y.C. and Yao, Z.P. Conformational Dynamics of Helix 10 Region as an Allosteric Site in Class a β-Lactamase Inhibitory Binding. *J Am Chem Soc*, 2020, 142(32), 13756–13767.

Huang, L., So, P.K. and Yao, Z.P. Protein Dynamics Revealed by Hydrogen Deuterium Exchange Mass Spectrometry: Correlation between Experiments and Simulation. *Rapid Commun Mass Spectrom,* 2019, 33(S3), 83-89.

Huang, L., So, P.K., Chen, Y.W., Leung, Y.C. and Yao, Z.P., Investigating the Conformational and Dynamic Aspects of the β-Lactamases Inhibitory Resistance by Integrated Mass Spectrometric Methods. China Mass Spectrometry Conference, Guangzhou, China, 23-26 November 2018 (**Excellent paper award**)

Huang, L., So, P.K., Chen, Y.W., Leung, Y.C. and Yao, Z.P., Understanding Antibiotic Resistance from Conformational and Dynamical Structures of β-Lactamases upon Inhibitory Interactions. HKSMS

Symposium 2018, City University of Hong Kong, 30 June 2018 (**Best poster award**)

Huang, L., So, P.K., Leung, Y.C. and Yao, Z.P., Investigation of β-Lactamases and β-Lactamase Inhibitory Protein by Hydrogen/Deuterium Exchange Mass Spectrometry. Structural Mass Spectrometry Workshop, Weizmann Institute of Science, Israel, 3-8 May 2017 (**IMSF fellowship award**)

Huang, L., So, P.K., Chen, Y.W., Leung, Y.C. and Yao, Z.P., Conformational Dynamics of Inhibitory Interactions between Class A β-Lactamases and BLIP: Insights from HDX-MS and MD Simulation. The 8th Asia-Oceania Mass Spectrometry Conference, Macau, 3-8 January 2020 (**Oral**)

Huang, L., So, P.K., Chen, Y.W., Leung, Y.C. and Yao, Z.P., Conformational Dynamics of Inhibitory Interactions between Class A β-Lactamases and BLIP: Insights from HDX-MS and MD Simulation. The 2nd International Conference on Hydrogen Deuterium Exchange Mass Spectrometry, Banff, Canada, 21-24 May 2019 (Poster)

Huang, L., So, P.K., Leung, Y.C. and Yao, Z.P., Interactions of β-Lactamase Inhibitory Protein (BLIP) and B-lactamases: Studies of Conformational Dynamics Using Mass Spectrometric Approaches. The 25th

Symposium on Chemistry Postgraduate Research in Hong Kong, The Hong Kong University of Science and Technology, 28 April 2018 (**Oral**)

Huang, L., So, P.K., Leung, Y.C. and Yao, Z.P., Interactions between β-Lactamases and β-Lactamase Inhibitory Protein: Insights from Ion Mobility and Hydrogen/Deuterium Exchange Mass Spectrometry. The Sunney & Irene Chan Lecture in Chemical Biology, Chinese University of Hong Kong, 13 November 2017 (Poster)

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Huang, L., So, P.K., Leung, Y.C. and Yao, Z.P., Investigation of Interactions between β-Lactamases and β-Lactamase Inhibitory Protein Using Hydrogen/Deuterium Exchange Mass Spectrometry. The 65th ASMS Conference on Mass Spectrometry and Allied Topics, Indianapolis, IN, USA, 4 – 8 June 2017 (Poster)

Huang, L., So, P.K., Leung, Y.C. and Yao, Z.P., Interactions of β-Lactamases and β-Lactamase Inhibitory Protein Interactions: Insight from Hydrogen/Deuterium Exchange Mass Spectrometry. The 1st International Conference on Hydrogen Deuterium Exchange Mass Spectrometry, Gothenburg, Sweden, 16-17 May 2017 (Poster)

Huang, L., So, P.K., Leung, Y.C. and Yao, Z.P., Investigation of β-Lactamases and β-Lactamase Inhibitory Protein by Hydrogen/Deuterium Exchange Mass Spectrometry. The Sunney & Irene Chan Lecture in Chemical Biology, The Hong Kong Polytechnic University, 3 Oct 2016 (Poster)

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Abbreviations

BLIP	β-Lactamase Inhibitory Protein
CID	Collision Induced Dissociation
Da	Dalton
ECD	Electron Capture Dissociation
ESI	Electrospray Ionization
ETD	Electron Transfer Dissociation
HDX	Hydrogen Deuterium Exchange
IM	Ion Mobility
IR	Infrared
m/z	Mass to Charge Ratio
MALDI	Matrix-assisted Laser Desorption Ionization
MD	Molecular Dynamics
MS	Mass Spectrometry
MT	Mutant
NMR	Nuclear Magnetic Resonance
SDS-PAGE	Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis
TOF	Time-of-Flight
UPLC	Ultra Performance Liquid Chromatography
UV	Ultraviolet
WT	Wild Type

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1. Introduction

1.1 Antibiotic resistance

Antibiotics are a type of substances that inhibit the reproduction of microbes. Alexander Fleming found the first antibiotic penicillin G in his antibacterial mold broth filtrate in 1928.[1] Not until 1940s, this antibiotic has been clinically used to treat the affections by different types of bacteria. Meanwhile, penicillin resistance by several types of Gram-negative bacteria has been observed.[2] Interestingly, antibiotic resistance could occur naturally without any use of clinical antibiotics. Several metagenomic studies showed that a variety of ancient genes encoding antibiotic resistance to vancomycin were identified by structure-function analysis.[3-4]. More importantly, these naturally occurring antibiotic-resistant genes could be transferred through plasmids to other bacterial pathogens. Thus, antibiotic resistance can immigrant from nature to the clinics[5]. Moreover, abuse of antibiotics in developing regions can accelerate the evolution of antibiotic-resistant bacteria and eventually cause global health threats.[6-7] Amongst various antibiotics, β-lactams antibiotics are the predominantly used broad-spectrum antibiotics, e.g., penicillins and cephalosporins, showing strength in safety and efficacy. Nevertheless, multidrug resistance (MDR) towards β-lactams has been increasingly identified in Gram-negative bacterial infections and this problem has drawn worldwide attention.[7]

In 1940, an enzyme from Bacillus coli that inactivates penicillin was identified as the first β-lactamase.[2] It disrupts the amide bond of β-lactam ring as a major mechanism of β-lactam resistance amongst Gram-negative bacteria. In the next 70 years, over 890 unique β-lactamases were characterized in natural bacterial species.[8] These β-lactamases are classified into four different classes (from A through D) [9-10] based on their primary sequences. Considering the emergence of a large diversity of class A β-lactamases, a structure-based classification was proposed for class A β-lactamases based on phylogeny analysis (Figure 1.1).[11] Two different subclasses, i.e., A1 and A2, were designated corresponding to the conserved residues (Figure 1.2 & Table 1.2).[12] Alternatively, β-lactamases are classified into four primary groups in terms of their substrates and catalytic functions (Table 1.1).[8, 13-15] Although most of the β-lactamases in the same group have similar primary sequences and catalytic sites, substitution of a single residue could contribute to significant changes in their activities towards antibiotics, suggesting the complexity of the mechanism of β-lactam resistance.

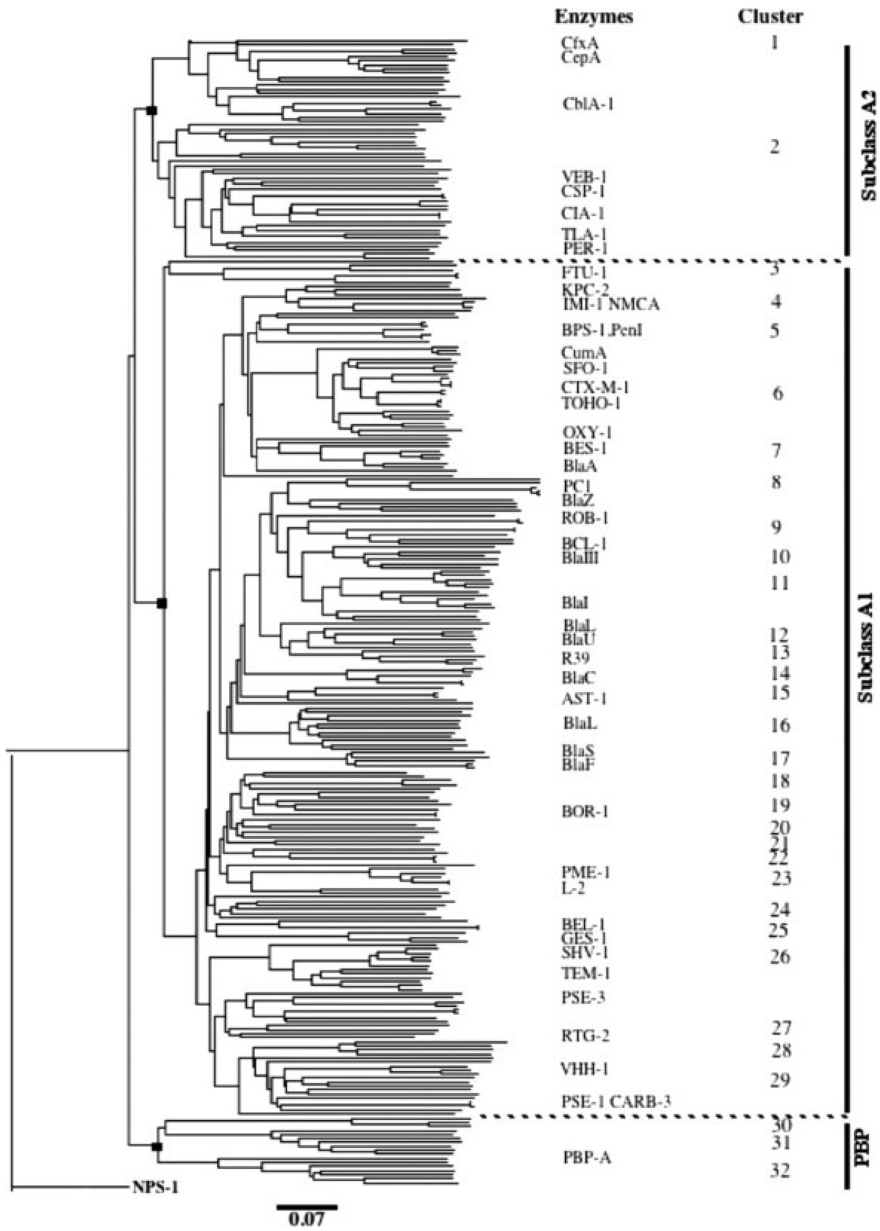

Figure 1.1 Rooted phylogram for 285 representative and putative class A "β-lactamases." (Reprint from the literature[11])

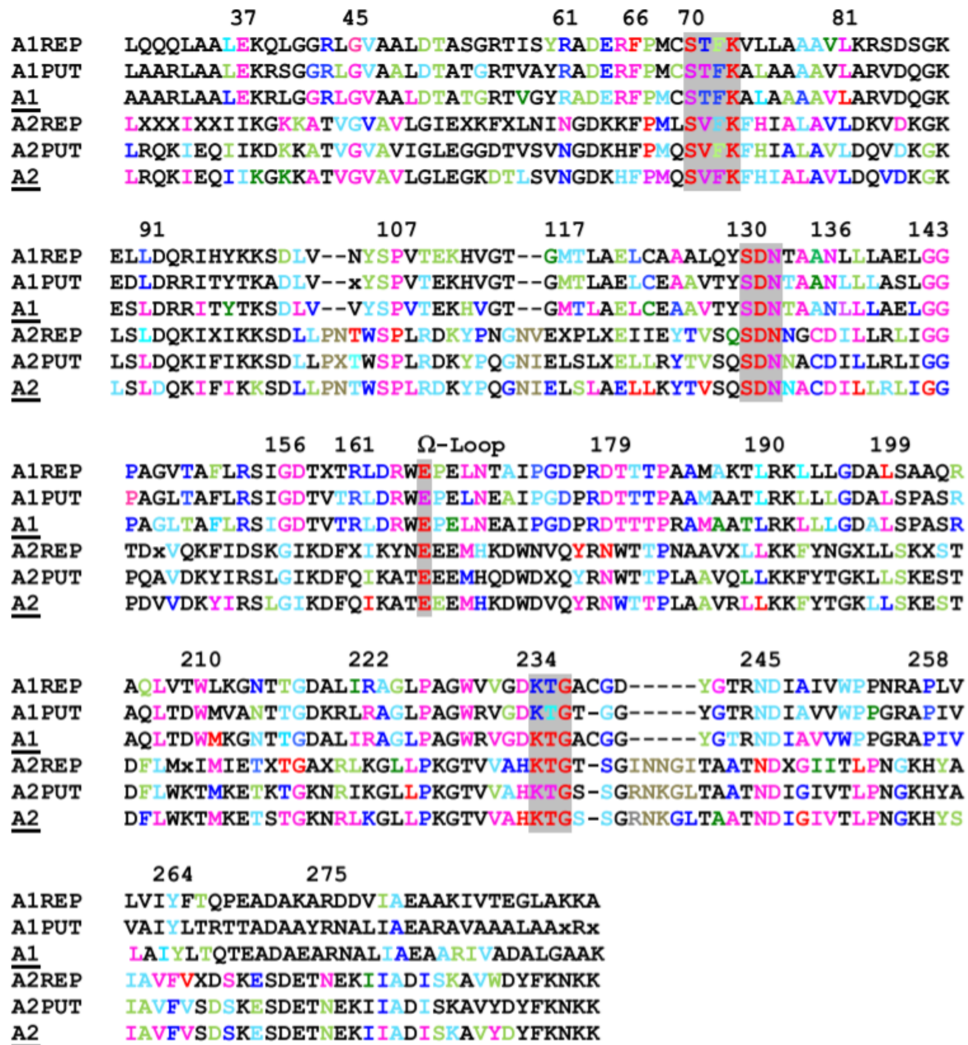

Figure 1.2 ABL consensus sequences after multiple alignments of the amino-acid sequences of the subclass A1 (132 REP for representative, 447 PUT for putative) and subclass A2 (14 REP, 107 PUT) β-lactamases. (Reprint from the literature[12])

Table 1.1 Classification of different types of β-lactamases

Type	Molecular class	Bush-Jacoby-Medeiros group	Representative enzymes
Penicillinase	A	2a	PC1
		2b	TEM-1, SHV-1
Extended β-lactamase	A	2be	TEM-3, SHV-2, CTX-M-15, PER-1, VEB-1
		2e	FEC-1
	D	2d	OXA-1
Cephalosporinase	C	1	AmpC, CMY-2
		1e	GC-1, CMY-10
	D	2de	OXA-11
Carbapenemase	A	2f	KPC-2, SME-1, IMI-1
	B	3a	IMP-1, VIM-1, IND-1
		3b	CphA
	D	2df	OXA-23

Table 1.2 Molecular characteristics of class A β-lactamases according to subclass

Subclass	Cluster	Typical enzyme(s)	Residue(s) at amino acid position(s):										
			70	73	77	130–132	136	166	179	233	234–236	237–238	245–246
A1a	Gram positive	PC1, BlaU, BlaL	S	K		SDN[a]	N	E	D	D	KT[b]G	AG	ND
	MYC1	BlaC	S	K		SDG	N	E	D	D	KTG	T[b]G	ND
A1b	LSBL1/4	TEM-1, SHV-1	S	K	C[c]	SDN	N	E	D	D	KT[b]G	AG	G[d]I[e]
	LSBL2	PSE-1, CARB-3	S	K	C[c]	SDN	N	E	D	D	RSG	AG	G[d]I[e]
	LSBL3	BlaP, RTG-2	S	K	C[c]	SDN	N	E	D	D	RTG	AG	G[d]I[e]
A1c	ESBL1/2	CTX-M, BES-1	S	K		SDN	N	E	D	D	KT[b]G	T[f]G	ND
	CARBA	NMCA, KPC-2	S	K		SDN	N	E	D	D	KTG	TC[g]G	ND
	ESBL3	GES-1, BEL-1	S	K		SDN	N	E	D	E[h]	KTG	T[b]C	ND
	BURK	BPS-1, PenI	S	K		SDN	N	E	D	D	KTG	TG	ND
A2a		PER-1, VEB-1	S	K		SDN	D	E	N	H[i]	KTG	T[b]S[i]	ND

[a] Ser for some Bacillus species.
[b] Or an analogue (Ser).
[c] Presence of a disulfide bridge between Cys77 and Cys123.
[d] Or an analogue (Ala) or Ser.
[e] Or an analogue (Val).
[f] Ala or Gly for some Enterobacteriaceae (K. oxytoca andC. amalonaticus).
[g] Presence of a disulfide bond between Cys69 and Cys238.
[h] Or an analogue (Asp).
[i] Or an analogue (Arg).
[j] Or Glu for CblA and CfxA types and some CepA types.

β-lactamase inhibitor was developed for clinical use to address the problem of β-lactam resistance. The first β-lactamase inhibitor named clavulanic acid was isolated from *Streptomyces clavuligerus* in 1977.[16] It can irreversibly inhibit β-lactamase through blocking the active sites. Other commercially available β-lactamase inhibitors include sulbactam,[17] tazobactam,[18] and avibactam,[19] However, due to the abuse of β-lactamase inhibitors, they have become less effective towards inhibitor-resistant β-lactamases that have been emerging since 1987.[20-22] Among the various inhibitor-resistant β-lactamases, production of hundreds of derivatives from TEM-1 β-lactamases reflects the most prevalent cases.[23] Such resistance towards inhibitors could be investigated by several methods, such as molecular modeling and site-directed mutagenesis.[24] To date, it has become a challenging task to deal with inhibitor-resistant bacterial infections, which remain a global health concern.[25]

1.1.1 β-Lactamases

β-Lactamases can hydrolyze β-lactam antibiotics to render Gram-negative bacteria the ability to survive antibiotic treatment. Several clinically prevalent β-lactamases have become a major threat to infectious treatment (Figure 1.3).

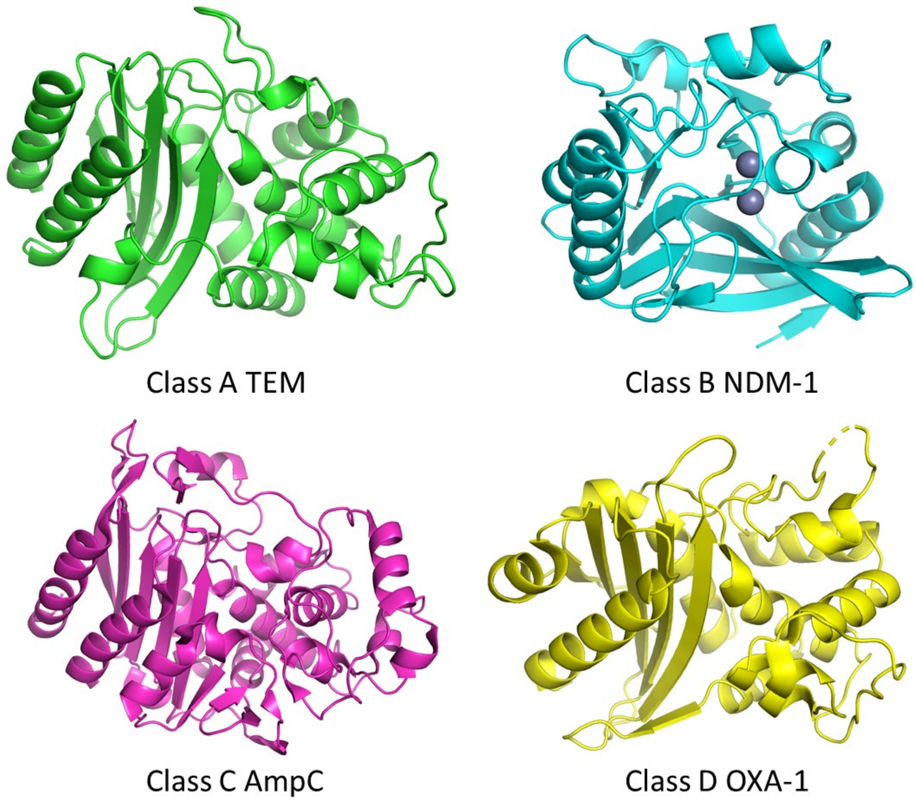

Figure 1.3 Structural representation of class A, B, C and D β-lactamases.

Penicillinases normally belong to class A or group 2a/2b β-lactamase[13] and inactivate penicillins and early generation of cephalosporins. They include over 16 TEM variants and 37 SHV variants β-lactamases and are of high prevalence in hospital and community settings. This group of β-lactamases can be effectively inhibited by small-molecule inhibitors. As a result, infections can be treated with a combination of β-lactam and β-lactamase inhibitor. Extended-spectrum β-lactamases have emerged as a major problem after the widespread use of broad-spectrum β-lactams.[26-28] They mainly comprise class A and class D β-lactamases or group 2be/2e and group 2d.[13-14]

Class A extended-spectrum β-lactamases include more than 89 TEM variants, 47 SHV variants and 90 CTX-M variants as well as a few PER and VEB types, among which the sequences are quite identical. These enzymes confer resistance to penicillins, cephalosporins and monobactams and show low susceptibility to inhibitors. Carbapenems and cephamycins are tolerant of those β-lactamases and thus play an important role in treating extended-spectrum β-lactamase-producing bacterial infections. Compared with TEM and SHV variants, CTX-M-type extended-spectrum β-lactamases have been prevalently identified since 1990,[29] especially in community settings. Class D extended-spectrum β-lactamases are known as the OXA-type that show increased activity towards cloxacillin or oxacillin and variable sensitivity to clavulanic acid, whereas most OXA-type extended-spectrum β-lactamases are resistant to inhibitors.[30-33]

Cephalosporinases, also known as AmpC enzymes, contribute significantly to multidrug resistance in Gram-negative bacteria.[34-38] This type of enzymes, which belongs to class C and class D or group 1/1e and group 2de β-lactamase,[14] hydrolyzes cephalosporins and cephamycins. Moreover, it can resist the inhibition by clavulanic acid. Though expressed in low levels in many organisms, AmpC enzymes could be induced by β-lactams and β-lactamase inhibitors. The AmpC structures are similar to those of class A β-lactamases while the binding pocket is more open to accommodate a broader spectrum of β-lactams.[37] Functional residues for AmpC include Lys67, Tyr150, Asn152, Lys315, and Ala318, in addition to Serine at Ambler number 64.[39] Class D cephalosporinases are OXA-type enzymes for extended-spectrum cephalosporins, such as ceftazidime and cefpodoxinme.

Carbapenemases hydrolyze preferentially carbapenems, which are known as the last-resort β-lactams in clinical treatment of infections.[40-44] They represent the most versatile family of β-lactamases, including class A or group 2f, class B or group 3a/3b and class D or group 2df.[14] Notably, most of the carbapenemases cannot be effectively inhibited by clavulanic acid or tazobactam.

Class A carbapenemases mainly refer to chromosomally encoded SME-(*Serratia marcescens* enzyme), NMC-(not metalloenzyme carbapenemase) and IMI-(imipenem hydrolyzing β-lactamase)type and plasmid-encoded GES-(Guiana extended spectrum) and KPC-(*Klebsiella pneumoniae* carbapenemase)type β-lactamases,[45-46] among which KPC β-lactamases have been rapidly spreading all around the world.[47] Class A carbapenemases demonstrate efficient hydrolysis of almost all β-lactam antibiotics and they are so far sensitive to clavulanic acid.

Class B or group 3 metallo-β-lactamases are structurally different from the other three classes (A, C and D) due to their requirement for zinc ions at the active position.[48-49] They are commonly isolated from bacteria that also produce other serine β-lactamases. Functionally, metallo-β-lactamases hydrolyze carbapenems with high efficiency while these enzymes show weak activities towards other β-lactams, monobactams in particularly. Their apparent resistance to β-lactamase inhibitors has been a great concern. The prevalence of metallo-β-lactamases is encoded in plasmids that can be transferred through different bacterial species. Since they are highly transferable these enzymes are named after their countries of origin, e.g. NDM-(New Deli, India), VIM-(Verona, Italy), SPM-(Sao Paulo, Brazil), GIM-(Germany) and SIM-(Seoul, Korea) metallo-β-lactamases.

Class D OXA-type carbapenemases, group 2df β-lactamase, have weak ability to hydrolyze carbapenems but show the highest activity towards imipenem.[50-51] These enzymes are typically unsusceptible to inhibition by clavulanic acid. As a new subgroup of prevalent β-lactamases, OXA-type carbapenemases can be further divided into nine clusters.

Inhibitor resistance may occur naturally, or it may emerge with the treatment of β-lactam and β-lactamase inhibitor combinations. Inhibitor resistance was identified after the introduction of clavulanic acid into clinical use.[20] Several mechanisms of inhibitor resistance have been proposed, including overproduction of plasmid-encoded TEM β-lactamases,[52-53] limited penetration of β-lactams mediated by outer membrane,[54] production of mutants of class A TEM β-lactamases (SHV-type, KPC-type and CTX-M-type)[55-57] and other enzymes like class D OXA-type β-lactamases,[58] class B metallo-β-lactamases,[59] and class C AmpC-type β-lactamases.[41] More importantly, inhibitor resistance can be developed as a consequence of co-production of several β-lactamases.[60] For instance, bacteria co-producing CTX-M-15, OXA-1, and TEM-1 can decrease the susceptibility to amoxicillin/clavulanic acid and lead to severe dissemination of bacterial infections.

1.1.2 β-Lactamase inhibitors and inhibitory proteins

β-Lactamase inhibitors were developed to cope with the growth of antibiotic resistance. Initially, screening methods were used to investigate the inhibitory activity of a variety of microorganisms. Observation of such activity could be found in culture filtrates of several streptomyces species, e.g., *S. gedanensis*, *S.*

olivaceus and the well-known *S. clavuligerus*. As a successful story in drug discovery, clavulanic acid was found in *S. clavuligerus* in the middle of 1970s. It can inhibit a broad spectrum of class A β-lactamases, such as TEM and SHV β-lactamases. Based on the structure of clavulanic acid, several penicillanic acid sulfones, e.g., sulbactam and tazobactam, were also successful in providing a better inhibitory profile for class C β-lactamases. In the past decade, several novel inhibitors, including the diazabicyclooctane (DBO) and boronic acid, have been recruited in clinical studies. A recently commercialized DBO named avibactam has a broader spectrum target β-lactamases, spanning class A, C and D.[61] Together, these inhibitors normally work in combination with β-lactam antibiotics to cope with the antibiotic resistant bacteria. Due to the occurrence of stimulating inhibitor resistance, novel strategies are urgently needed to win the battle with the death-threatening bacteria.

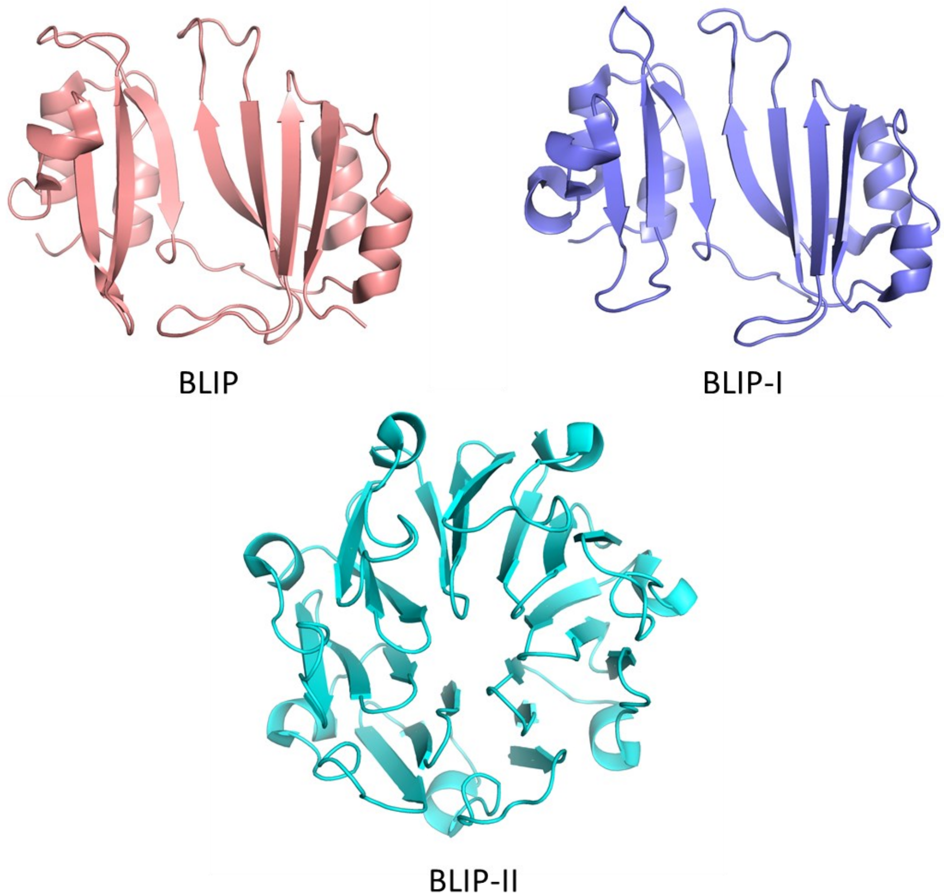

Figure 1.4 Structural representation of β-lactamase inhibitory proteins

β-Lactamase inhibitory protein (BLIP) was discovered in the culture filtrates of *Streptomyces clavuligerus*, a species of Gram-positive bacterium that also produces several β-lactams and the β-lactamase inhibitor clavulanic acid.[62] It constitutes around 10% of the exocellular protein. The BLIP gene encodes a 201-amino-acid precursor protein including a 36-amino-acid N-terminal signal sequence for translocation from the cytosol. The mature form of BLIP consists of 165 amino acids, four of which at position 30, 42, 109 and 131 are cysteines forming two disulfide bonds. BLIP can inhibit a variety of β-lactamases while the inherent biological role of BLIP remains uncertain. Two hypotheses have been proposed to explain the production of BLIP by *S. clavuligerus*. One speculated that it functions as a response to β-lactamase-producing organisms which are resistant to the antibiotics produced by *S. clavuligerus*. The other highlighted that BLIP could regulate the cell wall synthesis by interacting with penicillin-binding proteins. Importantly, the understanding of the original roles of BLIP in biological organisms can improve the design of novel inhibitors and further the development of new strategies towards antibiotic resistance.

BLIP-I from *Streptomyces exfoliatus* overlaps with 37% of the amino acid sequence of BLIP and is also capable to inhibit a similar profile of β-lactamases with BLIP.[63] In contrast, the sequence of precursor BLIP-I is shorter than BLIP and consists of a 157 amino-acid mature polypeptide chain following a 29 amino-acid signal sequence. Notably, the corresponding loop region β3'-β4' in BLIP-I is less extended than that in BLIP and might have different effects on the inhibition towards β-lactamases (Figure 1.4). The biological function of BLIP-I is speculated to be related to the morphological differentiation of the bacterium through regulation of cell wall synthesis. Another inhibitory protein called BLIP-II is also isolated from *S. exfoliatus*.[64] The gene encoding BLIP-II produces a 332 amino-acid mature protein preceded by a 40 amino-acid signal sequence. In contrast to BLIP and BLIP-I, this inhibitory protein is a more potent one with inhibition constants at femtomolar to picomolar level. Importantly, BLIP-II shows no sequence or structural homology to neither BLIP nor BLIP-I but a unique scaffold with a seven-bladed β-propeller where β-turns bind the interface from β-lactamases (Figure 1.4).

In summary, further investigation of the interactions between β-lactamases and β-lactamase inhibitory proteins can modify our understandings of the biological roles of these naturally occurring enzymes and protein inhibitors in bacteria, the basics of protein-protein binding and recognition, the evolutionary pathways of a diversity of enzymes and even the strategies of inhibitor design to overcome antibiotic resistance.

1.2 Approaches for studying the interactions between β-lactamase and BLIP

TEM β-lactamases account for one of the most prevalent subtypes of class A β-lactamase. The parental TEM β-lactamase (TEM1) can be inhibited by BLIP with a constant value at subnanomolar level. Numerous studies on the interactions between TEM1 and BLIP have been conducted to reveal the underlying mechanisms of inhibition. Besides, this pair of proteins is exemplified as a well-recognized model for studying high-affinity protein-protein interactions.

1.2.1 X-ray crystallography

The crystallographic structure of the TEM1-BLIP complex was determined at 1.7 Å resolution for the first time in 1996 (Figure 1.5).[65] No dramatic changes were observed in TEM1 or BLIP upon binding with each other. The value of RMSD on main-chain atoms for TEM1 between the native and bound form is as low as 0.35 Å. In addition to the maintained conformation of protein molecules, the positions of two conserved water molecules also remained unchanged in the complexed form. Notably, the interfacial area was one of the largest among protein complexes, contributed by a convex surface from TEM1 and a concave surface from BLIP. This could be the major cause for the high binding affinity between the β-lactamase and the inhibitory protein.

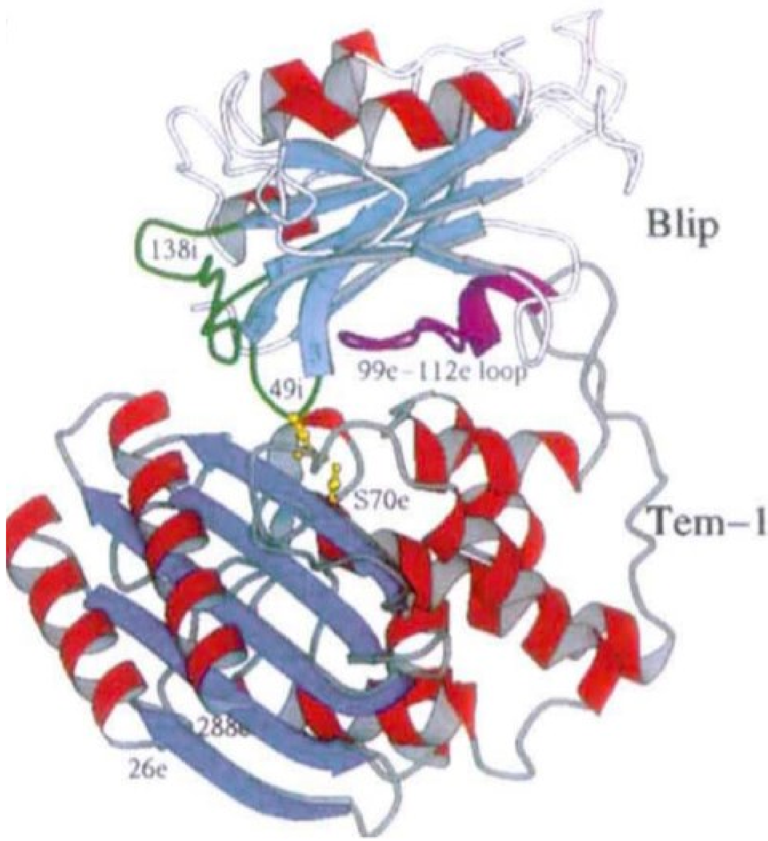

Figure 1.5 Structural representation of the TEM1-BLIP complex.
The structure is shown with red helices and blue β-strands. Ser70 from TEM1 and Asp49 from BLIP are shown in yellow sticks. (Reprint from the literature[65])

In detail, there are two major groups of interactions identified in this complex. One involves the β-hairpin 46-51 in BLIP and the active pocket in TEM1, and the other is mediated by highly ordered waters between the loop-helix motif (99-112) in TEM1 and the polar residues lining on the concave surface of BLIP. The active sites in the enzyme include several conserved residues, including Ser70, Lys73, Ser130, Asn132, Glu166, Asn170, Lys234, Ser235 and Arg244. Four of them form essential hydrogen bonds with Asp49 in the inserting β-hairpin of the inhibitor. The other group of interactions is dominant by the contacts between Tyr105 in the loop-helix motif and two residues from each domain of BLIP, i.e., Lys74 from the N-terminal and Phe142 from the C-terminal. It is also obvious that the loop-helix motif from TEM1 and polar residues from BLIP contribute two thirds of the hydrogen bonds on the binding interface. These interactions could extensively increase the binding interface and prevent the complex from dissociation. Meanwhile, four salt-bridges can reinforce the interaction and ascertain the high affinity between TEM1 and BLIP.

To summarize, the complex of TEM1 and BLIP is visualized by a crystallographic image. Although little conformational changes can be observed, the interaction is governed by two remarkable clusters of interaction. The protruding loop from BLIP inserts into the active cavity of TEM1, suggesting the mechanism of inactivation. In addition, the loop-helix motif from TEM1 shows proximity to the concave of BLIP, forming a massive amount of hydrogen bonds lining the interface. Together, these interactions provide extensively large interface in the tightly bound complex. To our knowledge, this complex provides a well-built model to elucidate the interactions between other class A β-lactamases, e.g. SHV-1 and PC1, and β-lactamase inhibitory proteins. Following this initial study, many investigations have been conducted and further understanding of these interactions has been greatly modified.

1.2.2 Molecular docking

Molecular docking programs had been developed prior to the determination of the crystalized complex and these algorithms was later scored based on the crystallographic results.[66] Six different groups using various docking algorithms successfully predicted the general binding mode of TEM1 and BLIP. The docking was optimized by several aspects, including shape complementarity of the surface, minimal local energy, electrostatic potential and buried surface area. The resulted coordinates were evaluated by calculating the r.m.s. deviation of the main-chain atoms relative to the experimental ones. In spite of the correct prediction of the binding mode, the algorithms failed to predict any conformational changes in neither TEM1 nor BLIP. The side-chain rearrangement on binding negatively changed the precision of these algorithms, indicating more efforts might be put into the optimization of local dynamics during docking. In addition, global hinge-bending motion was observed in BLIP and large conformational change in the loop-helix motif in TEM1. Neither of these changes were observed by the six docking algorithms. The presence of ordered water molecules failed to be predicted by those computational programs. Although it is challenging to identify the roles of these water molecules, more attention should be given to have a further understanding of their positions. Due to the limited knowledge on the interaction of these proteins, only a few restraints could be used to optimize the docking. The criteria to evaluate the resulted coordinates appears to be subjective. Due to the shortage of computing power, more information could be input for more objective prediction.

Docking results could serve as a convenient tool for validation of X-ray crystal structure and might accelerate the interpretation of the crystallographic structures. In addition, simulation provides essential conformational changes from global motion as well as local flexibility. This advantage is significantly useful for visualization of protein dynamics. Again, it is notable to feed enough structural restraints to facilitate a more authentic structure. More robust docking algorithms should be capable of self-correction from tested coordinates. Recent advances in deep learning algorithm have gained increasing attention in the community of protein-protein docking.[67] The accuracy of prediction has been greatly improved regardless the sequence homology of target protein and solved ones. This improvement benefits from the development of neutral network in computing science. And we could expect very promising and practical application of these algorithms in predicting protein complexes. However, protein complexes in silico could be fundamentally different compared to those in solution. Validation by other methods is essentially necessary to get meaningful information from molecular simulations.

Molecular docking can provide another dimension to elucidate the protein-protein interaction. In this study, it is satisfactory to obtain the expected interacting mode and possibly unexpected dynamic orientation. Special attention should be paid to the global and local conformational changes as well as conserved water molecules since the optimization of global score and local ranking is often a balance. Such optimization might require more restraints to gain convincing results. Recent progress in artificial intelligence have revolutionarily broaden our understanding of computational methods, which could bring novel insights into molecular simulations.

1.2.3 Phage display

Phage display has been employed to determine critical residues in TEM1 for binding BLIP.[68] The residues spanning the loop-helix motif (99-114) and β3 (235-240) on TEM1 were randomly mutated. The resulted mutants were displayed on the surface of bacteriophages and then subjected to binding BLIP. A great advantage of this technique is high-throughput screening for significant residues that are either highly conservative or potentially favored amino acids during the binding.

The results provided further understanding of the binding mechanism regarding TEM-type β-lactamases and BLIP. Most residues from both regions of interest showed biased binding towards BLIP. 4 out of 16 residues on loop-helix region were identified to be conserved substitutions, including Gln99, Glu104, Tyr105 and Val108. This suggests that residues in these positions are critical for tight binding with BLIP. The preferred substitutions in TEM1, K111T, L113A and T114R, showed higher affinities with BLIP compared with wild-type TEM1. After another round of screening on phage libraries 106-108 and 112-114, the sequences converged as S-P-V and H-A-R, respectively. In addition to the loop-helix region, β3 strand includes conserved residues, Ser235, Gly236, Ala237 and Gly238. Interestingly, Glu240 exhibits stronger affinity to BLIP upon the substitution to lysine.

To further validate the findings from phage display, several mutated β-lactamases were expressed and purified for inhibition assays. Mutant L113A/T114R was found to bind BLIP six-fold tighter than the wild-type. However, the catalytic efficiencies towards cephalosporin and cephaloridine decreased due to lower k_{cat} and K_m. This finding indicated that mutation at these positions might have been ruled out during the evolution of β-lactamases under the selective pressure of antibiotics. The increased binding affinity could show us some deep insights into the mechanism of this binding mode. The mutation E240K on β3 also exhibited similar improvement in binding with BLIP. This binding was tightened by 11-folds compared to that between wild-type TEM1 and BLIP. More interestingly, the β-lactamase mutant, L113A/T114R/E240K, improved the affinity with BLIP more than that expected from additivity principles. The determined dissociation constant was 0.42 pM, indicating the binding is 550-fold tighter than binding by wild-type TEM1, whereas it is expected that this binding is 75-fold tighter after the triple mutation. The underestimated binding affinity reveals a possible mechanism that L113A/T114R and E240K refer to different binding interfaces. According to these non-additive interactions, two interfaces could be deduced. One refers to the loop-helix region with concave in BLIP and the other refers to β3 strand with an inserting loop in BLIP. Mutations in these regions could facilitate cooperative conformational changes and thus generate a synergic effect on the binding. The critical residues identified on the loop-helix region also indicates comparable observation from molecular modeling,[66] which revealed the imperfect interface mediated by conserved water molecules. The L113A/T114R mutant might narrow this gap between the loop-helix motif and the concave of BLIP. Notably, these critical residues at 113, 114 and 240 are at the periphery of the interface. Such observation indicated an optimized interaction spanning the edge of the interface could provide significant enhancement of the binding.

The screening of high-affinity TEM mutants indicates that BLIP has the potential to be an efficient inhibitor to a variety of TEM-type β-lactamases. This will inspire the development of novel inhibitors based on such insights into the binding mechanism of the complex.

1.2.4 Kinetics and thermodynamics

To examine the binding kinetics between TEM1 and BLIP, several techniques can be employed.[69] Enzymatic inhibition assay determines the activity of enzyme, thus evaluating the inhibition constant of the inhibitor. It should be noted that the reliability of this technique relates to the enzymatic activities of β-lactamase. Fluorescence and isothermal titrations measure the equilibrium constants and are irrelevant to the enzymatic properties of β-lactamase. In addition, association rates and dissociation rates reveal the kinetics of binding by SPR detection. Although the association rate is moderate for TEM1 and BLIP, the dissociation rate is very low, indicating that this interaction is a slow and tight mechanism-based binding. Consideration on these parameters might uncover some insights beneath the interaction of interest.

The kinetic and equilibrium constants were determined at different pH. The results showed that a tightest binding was observed at pH 7.5, with a K_d value at 0.4 nM. Notably, the value of association rate decreased as the pH dropped while the dissociation rate had a minimal value at pH 7.5. Besides, the effect of salt on the binding kinetics was also examined to evaluate the contribution of electrostatic forces. The binding was not obviously affected by addition of 0.5 M NaCl, indicating little contribution from strong electrostatic forces.

Four critical residues on TEM1, Ser130, Lys234, Ser235 and Arg244, which form hydrogen bonds with Asp49 on BLIP, were substituted by alanine. The removal of two positively charged residues resulted in significant decrease of the association constant. By contrast, the interaction of BLIP D49A with TEM K234A showed faster association rate compared to the wild types when a negative residue and a positive residue from each side were removed. These results provided strong evidence of the effect of net charge to the association rate. This can be explained that long-range electrostatic forces are predominant in the association. In this case, BLIP D49A almost neutralized the protein, thus there was less repulsive effects during binding. And even TEM mutant was more negatively charged, the overall repulsion was eliminated.

To summarize, this interaction is not mediated by strong electrostatic forces as indicated by the identical binding affinity under varied ionic strength. However, mutations on the charged residues showed altered binding because the electrostatic forces impact the interaction from a long distance. Interestingly, the mutations of TEM1 alone decreased the association rate while both mutations of TEM1 and BLIP increased the association rate. The explanation is satisfactory but not perfect. It is suggested that the experiment should include mutations Asp49Asn besides Asp49Ala, because the length of the side chain should be maintained in order to make comparisons. This would be more reasonable to conclude that the electrostatic forces instead of hydrogen bonds or other forces make a difference here. After all, it is mentioned that the truncation of Asp49 showed increased dissociation rate because the loss of four hydrogen bonds among Asp49 and residues from TEM1.

Overall, this study performed a well-established method for investigating the electrostatic interactions between two proteins by measuring the binding kinetics. Mutations on charged residues provide a clear view on the role of these positions, thus facilitating a further understanding of these interactions.

1.2.5 Binding hot spots and the modular architecture

Hot spots are defined to describe those residues that cause substantial changes in binding energy after mutations on those positions.[70] Alanine scanning is a conventional method to map the hot spots on binding interface. In addition to alanine mutation, replacement with other natural amino acids provides further insights into the binding mechanisms. Multiple mutant cycle analysis can examine the cooperative contribution of critical residues upon binding, thus revealing the basics of interactions between two proteins. In the case of complexation of TEM1 and BLIP, the binding interface is known as of the largest ones. The binding ability of BLIP is extensively wide towards TEM1 and its numerous mutants, such as SHV1 and SME1. As a model of these interactions, interactions of TEM1 and BLIP were thoroughly investigated by multiple mutations. Modular architecture of these interactions was established and further investigated by multiple mutant cycle.[71-72]

A well-resolved interface contact map was derived from REDUCE and PROBE programs. This map demonstrated clearly six clusters of interactions, including C1 (D49 on BLIP and S130, S234, K235 and R243 on TEM1), C2 (E104, Y105 on TEM1 and K74, F142, Y143 on BLIP) and C3–C6 with diminishing significance. Inter and intra-cluster relations was revealed by comparison of changes in free energy after mutations. It showed that interactions within a cluster are nonlinearly additive while those of different clusters are linearly additive.

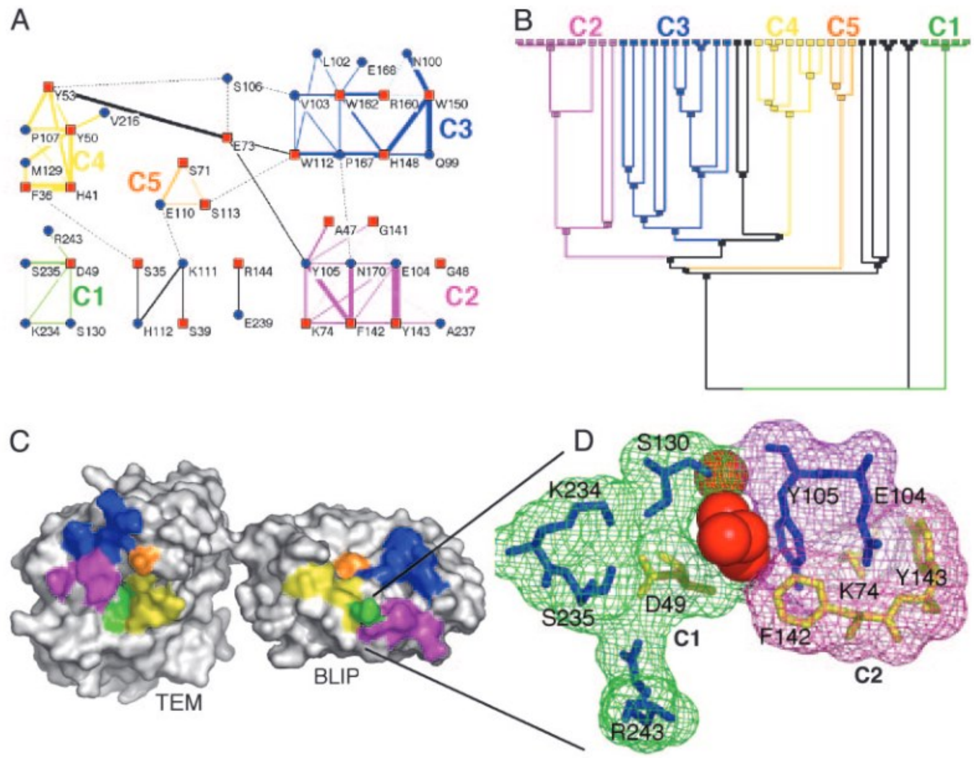

Figure 1.6 Cluster analysis of the TEM1–BLIP interface. (Reprint from the literature[71])

Detailed studies into analysis of hot spots within C1 and C2 showed two distinguishable mechanisms in these interactions. In cluster C1, D49 on BLIP and four residues on TEM1, S130, K234, S235 and R243, did not show significant conformational change upon binding, indicating a *Lock and Key* based binding interaction. By contrast, obviously cooperative interaction was observed within cluster C2 and the conformational rearrangement of sidechains of C2 indicated a *Fit-Induced* mechanism for these interactions. The mutations of TEM1 E104Y-Y105N and BLIP D49A further confirmed different structural arrangement upon binding based on the crystallized structure. C3 and C5 also underwent conformational changes to various extent.

To summarize, interaction map was produced by computational programs and three interaction modes were analyzed, including backbone-backbone, backbone-sidechain, and sidechain-sidechain interactions. The relationship of individual clusters is additive while the influence is nonadditive within clusters. Two binding mechanisms were proposed for cluster C1 and C2, *Lock and Key* and *Fit-Induced*, respectively. The architecture of C1 is relatively rigid. The position of Asp49 on BLIP is conserved and surrounded by four residues on TEM1 in the complex. By contrast, there is obvious conformational change upon binding in the case of C2. This observation indicates that interactions within C2 might undergo induced fit to TEM1 upon the complexation. The flexibility of binding interface contributes essentially to the tight interactions and this results from the rearrangement of the backbone and side chains.

Due to the complexity of the binding between TEM1 and BLIP, it showed a hybrid binding mechanism. It is not surprising that this binding demonstrates slow association and tight complex, because the binding could be initiated by lock and key mechanism. Subsequent to this interaction, other interfaces could undergo conformational rearrangement to fill the gaps between interfacial residues, thus facilitating a tight binding. Probably due to the flexibility of TEM1 and BLIP, the interaction between them is quite tolerant to some mutations on TEM1. This could explain the wide interactions between BLIP and various β-lactamases.

It is an efficient way to study the binding hot spots in light of modular architecture, especially for the complicated interactions. Combined with multiple mutant cycle analysis, it is possible to locate the critical residues on the interface and to distinguish two binding mechanisms based on the cooperative conformational rearrangement. Mutation to residues other than alanine could provide more detailed information to the interacting modes. To further confirm the results, crystallization of mutant proteins is a direct way to investigate the structural rearrangement.

1.2.6 In vitro evolution using yeast surface display

In this study, yeast surface display in combination with deep sequencing was applied for studying TEM1 and BLIP interactions.[73] Randomized TEM1 sequences were created by single-nucleotide mutagenesis to mimic natural evolution and BLIP was used for selection. The kinetic binding of TEM1 mutants were selected by BLIP with low concentration and short incubation time. The sorted TEM1 mutants were sequenced under low stringency and were selectively expressed for validation. The binding assays were performed to determine the kinetic rate constants for further comparison.

Almost all selected mutations are located on the surface where most (11/17) of the mutations are located on the interface, especially charged residues. Since the electrostatic interactions occur in long distance, these mutations can effectively change the association rate constants. Besides, most of the negative selections indicate that buried residues are less tolerant to mutagenesis due to possible loss of protein stability. Notably, three interfacial residues, G236, G238 and R243, are quite conserved in TEM1 and BLIP interactions. Taken together, interfacial residues are less conserved than core residues, indicating the plasticity of the binding interface. The selection for fast binder shows a common feature of the mutations, an increase in positive charge by 2. This result is consistent to that electrostatic forces are crucial for fast binding. By contrast, the selected mutations show little effect on the dissociation rate constants.

To summarize, this study revealed that the interface between TEM1 and BLIP is of high tolerance to mutations and fast association variants could be selected under pre-equilibrium conditions.

1.2.7 Summary

X-ray crystallography solved the 3-D structure of the complex between TEM1 and BLIP, providing a detailed view of the binding pattern. The high affinity of this interaction is derived from two major groups of interactions, forming an extensively large binding interface. Prior to the determination of the crystal structure, molecular docking had been applied to simulate the binding and later proven to be generally correct. However, this technique failed to predict the orientation of several critical residues as well as the positions of conserved water molecules upon binding. The critical residues were then determined by phage display. The results revealed several conserved residues on the loop-helix region and β3 in TEM1. Furthermore, a triple mutation on TEM1, demonstrated an ultra-high binding affinity with BLIP, 550-fold increment compared to the wild-type TEM1. In addition to the experiments under equilibrium conditions, characterization of binding kinetics was also conducted based on SPR (Surface plasmon resonance) and stop-flow fluorescence. It was reported that the slow and tight interaction might not be dominated by electrostatic forces, however, the varied charges on protein surface apparently has an impact on the association rate. Interestingly, a modular architecture of the interactions was established in the complex of TEM1 and BLIP. Notably, there is limited influence across individual clusters while changes within the clusters are cooperative, indicating a new and effective method to elucidate the complex interactions between proteins. What surprised us is that two mechanisms are referred to this interaction. One refers to the *lock and key* based interaction between the active cavity in TEM1 and protruding loop 46-51 in BLIP. The other refers to the *induced fit* mediated binding between loop-helix and polar residues on the concave from TEM1 and BLIP, respectively. The interface of the complex was deeply investigated by an artificial evolution. The binding affinity, association rate, protein stability and enzymatic activity were examined after several rounds of low-stringency selection. The findings showed that most interfacial residues are tolerant to one or two mutations expect for G326, G238 and R243, indicating plasticity of this interface. Fast binders selected from randomized mutations exhibited increased positive charge. Taken together, we summarized the current understanding of TEM1 and BLIP interactions to advance our knowledge prior to this project.

1.3 Hydrogen deuterium exchange and mass spectrometry

1.3.1 Hydrogen deuterium exchange

Hydrogen exchange is an isotopic labeling method introduced into protein study by Kaj Ulrik Linderstrøm-Lang in 1950s,[74] following Pauling's discovery of α-helix and β-sheet.[75-76] In 1972, S. W. Englander presented a comprehensive review of the principles and knowledge of hydrogen exchange for the previous twenty years.[77] With limited chemical techniques and understanding of proteins, hydrogen exchange, in combination with infrared spectroscopy or proton magnetic resonance, was the most fashionable method to study protein structures. The "breathing" concept proposed at that time has been evolved into the most accepted theory nowadays.[78] Several factors should be taken into consideration, e.g. temperature, solvent, isotope effects and polarity of the polymer. These conditions could affect the structure of macromolecule thus lead to variation of hydrogen exchange rate.

The two-step model shown below as proposed by Linderstrøm-Lang is widely accepted for the reaction of HDX.

$$NH_{close} \xrightarrow{K_{op}} NH_{open} \overset{k_{int}}{\longleftrightarrow} ND_{open} \xleftarrow{K_{op}} ND_{close}$$

This is based on the hypothesis that amide hydrogen atoms undergo exchange when the hydrogens are accessible to solvent and are not protected by hydrogen bonding. Since the equilibrium state is dynamic, namely there is fluctuation between the closed form and the open form, the exchange rate is determined by this equilibrium state and the intrinsic exchange rate (k_{int}, i.e., the exchange rate of unprotected amide hydrogens). Thus, the exchange rate k_{ex} can be defined as shown in Equation 1.

$$k_{ex} = \frac{k_{op} \times k_{int}}{k_{clo} + k_{int}} \qquad \text{(Equation 1)}$$

If $k_{clo} \gg k_{int}$, the exchange rate could be defined as Equation 2.

$$k_{ex} = \frac{k_{op} \times k_{int}}{k_{clo}} \qquad \textbf{(EX2 kinetics}\text{, Equation 2)}$$

If $k_{clo} \ll k_{int}$, the exchange rate could be defined as Equation 3.

$$k_{ex} = k_{op} \qquad \textbf{(EX1 kinetics,} \text{ Equation 3)}$$

Since the exchange is catalyzed by acid, base and water, the exchange rate is dependent of pD in solution. The equation given by S. W. Englander in 1993 is shown in Equation 4.[79]

$$k_{int} = k_A \times 10^{-pD} + k_B \times 10^{pD-pK_D} + k_W \qquad \text{(Equation 4)}$$

This exchange rate is a function of pD in the solution, thus we could control the exchange-in and exchange-off reactions. During the LC separation, we set the pH of mobile phase at around 2.5 to minimize the back exchange, preserving more useful information. Both kinetics could occur in the protein while EX2 is more prevalent than EX1 kinetics. This two-step model is widely applied to the studies of protein folding, conformation and dynamics.[80]

Water molecules play an essential role in hydrogen bonding that is a fundamental intramolecular interaction in maintaining the higher order structure of proteins. Moreover, the significance of water molecule in protein-ligand binding is increasingly recognized in recent years. The displacement of water molecules can influence the binding affinity of protein and ligand.[81] As a result, using water to probe the structure of protein seems to be reasonable. Deuterated water (D_2O) is becoming a successful structural probe because it has little effects on protein structure and conformation.

HDX can be used to examine the solvent accessibility of a protein. The obtained structural information reflects the status of protein folding. It is not easy to interpret HDX into structural context without the complementation of other techniques, like X-ray crystallography and NMR. Another feature of HDX is that the HDX kinetics in different states cannot direct to the binding interface or hotspots of protein-protein interactions.

Over the years, hydrogen exchange has been widely utilized in many striking applications, as summarized in numerous perspectives.[78, 82-83] The most exciting application is related to the folding pathway of biologically essential proteins.[84-86] Conventionally, it is a great challenge to capture the conformations of disordered or misfolded proteins. Hydrogen exchange provides an alternative way to record the dynamics of these proteins.[87] This should be attributed to the advancing sub-second labeling technique with the arising of microfluidic chips[88-89] and ultra-sensitive detector, exemplified by up-to-date mass spectrometers.[90] Moreover, hydrogen exchange has been developed as a tool for quality control of biopharmaceutics.[91-92] Thus, we have seen that hydrogen exchange is playing a potent role in protein analysis and will continue to contribute to drug research and development.

1.3.2 Mass spectrometry for structural studies of proteins

Mass spectrometer was invented in the early twentieth century and primarily contributed to the discovery of isotopes.[93] Mass spectrometers were commercially available since 1940s while not until late 1980s, chemists were able to make the macromolecules fly in an elegant way through two significant ionization methods, electrospray ionization (ESI) and matrix assisted laser desorption ionization (MALDI). Both techniques were awarded the Nobel Prize in 2002, together with the application of nuclear magnetic resonance (NMR) in macromolecule investigation. From a historical perspective, it took a long while to bring mass spectrometry into biological study. Importantly, since the introduction of ESI and MALDI, they have been extraordinarily played around in every corner of the scientific field, especially in the field of protein study.

In 1990, for the first time ESI-MS was used to probe the protein conformational changes by Chait et al.[94] They exposed bovine cytochrome c into aqueous solutions with differed composition of acetic acid, resulting in various degrees of denaturation. This could be reflected in the patterns of charge distribution of the molecule ions due to the unfolding of the ternary structure in acidic solution (Figure 1.7). Their work demonstrated a new and easy method to probe the conformational change of proteins. Besides, other factors, like organic solvent and temperature,[95-96] have been taken into consideration to induce conformational change. Various charge distribution was observed after the protein samples were encountered disulfide reduction.[97] Insights into charge state envelopes in ESI mass spectra were lightened by Kaltashov group upon deconvolution of charge state distributions.[98] Different conformers coexisting in protein samples could be identified by linear combinations of limited basis Gaussian distribution. Two model proteins, α-helical myoglobin (Mb) and β-sheet cellular retinoic acid binding protein I (CRABP I), were determined as a function of solution pH.

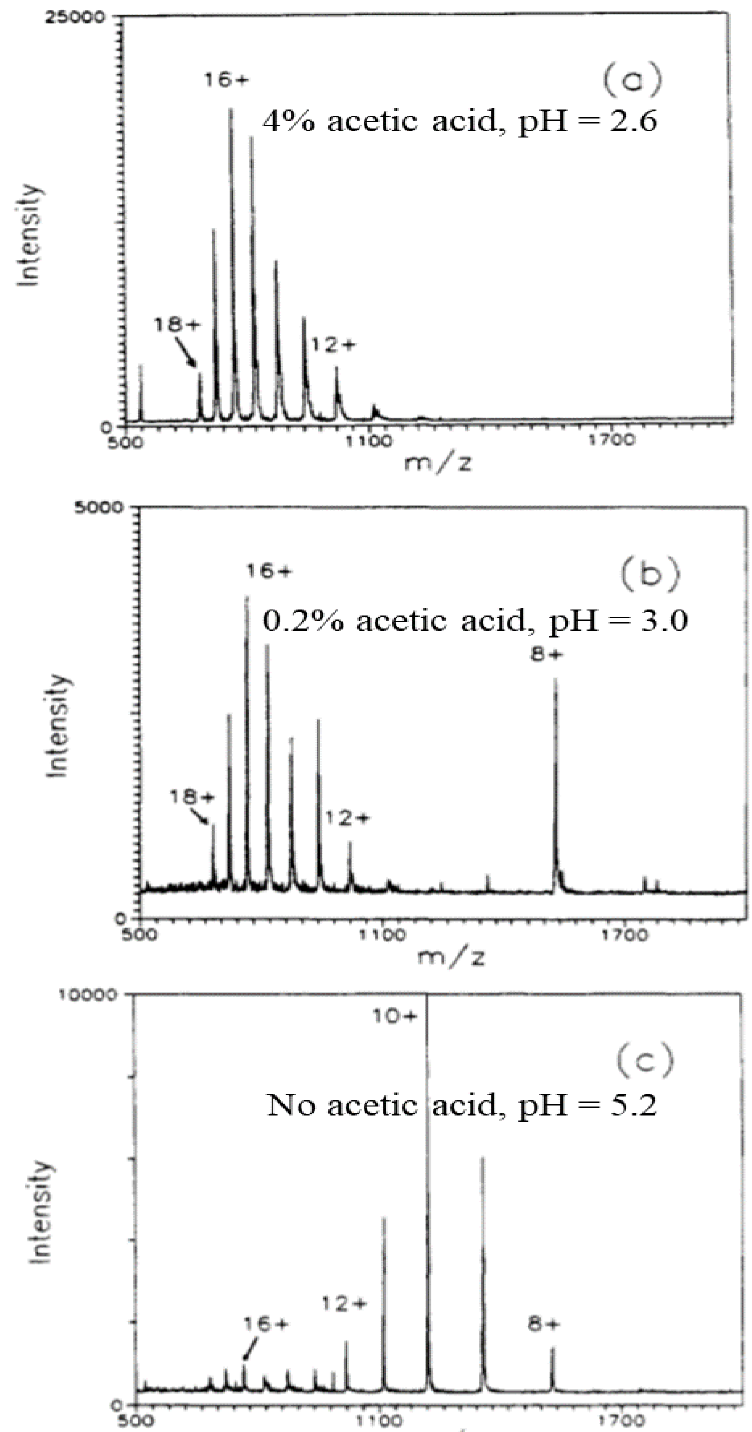

Figure 1.7 ESI-MS of bovine cytochrome c under different conditions. (Reprint from the literature[99])

ESI-MS has also been applied to probe protein-ligand noncovalent complexes since the beginning of 1990s.[99] ESI normally introduces samples from aqueous phase to gaseous ions after desolvation, which could preserve the dynamic conformation and the noncovalent complexes under native states. The aqueous condition should be soft and instrumental parameters should be gentle to avoid dissociation. Following the ESI-MS, the Nano electrospray (nanoESI) ion source invented by Wilm and Mann introduced the protein investigation into a new era.[100-101] This technique improved the desolvation in source and reduced the sample consumption, resulting in higher sensitivity and accuracy. The tolerance to buffer solution of protein samples also is improved by nanoESI, which could enable a rapid analysis in vitro.

In the end of 1990s, ion mobility mass spectrometry (IM-MS) was brought into research of structural properties of proteins and protein assemblies (Figure 1.8).[102-103] Waters Incorporation introduced the first commercial mass spectrometer equipped with IM-MS in the early 2000s. Since IM-MS added another dimension to characterize the protein structure, it was widely used to obtain the information related to protein stability and multiple conformations.[104-106] Rapid analysis and real-time reaction could be more easily achieved by IM-MS separation without forefront sample treatment, like size-exclusion separation.

At the beginning of this century, another breakthrough in protein analysis was the application of ESI-MS into large macromolecule assemblies by improving the transmission in a Q-TOF mass analyzer.[107] Nowadays a commercially available quadrupole could be extended to analyze m/z at 32,000 for mega-Dalton molecule machines (Figure 1.9). This methodology was used to probe the stoichiometry and topology of protein assemblies by many research groups in recent years.[108-110]

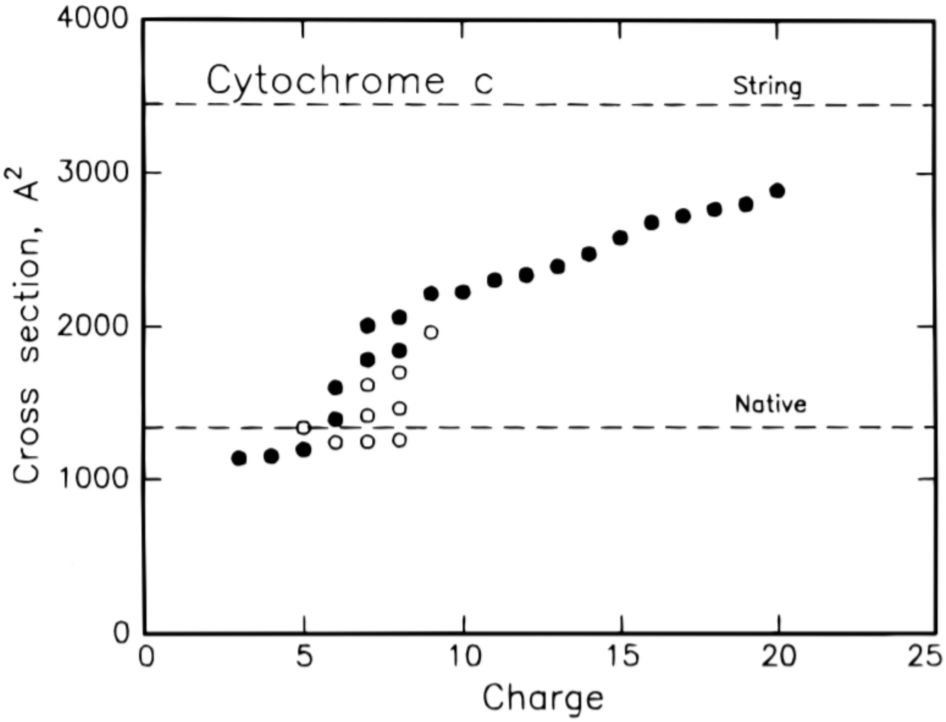

Figure 1.8 Plot of the collision cross-sections for the main features observed in the drift time distributions for the 3+ to 20+ charge states of cytochrome c. (Reprint from the literature[102])

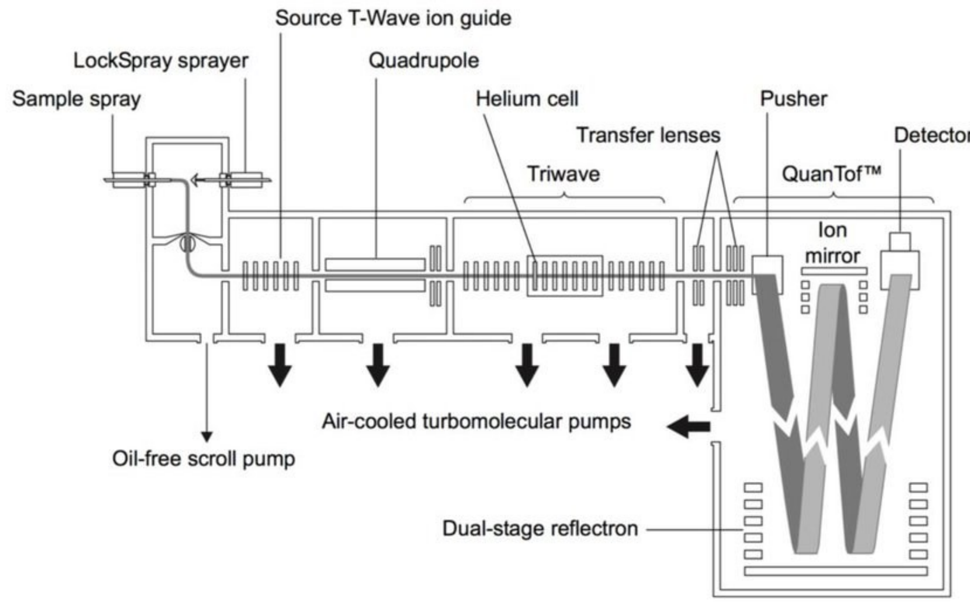

Figure 1.9 Schematic of the ion optics in the SNAPT G2-Si HDMS system. A commercially available ESI-Q-IM-TOF mass spectrometer. (Reprint from Waters Corporation, 2009)

New fragmentation methods also bring protein complex study into a new stage. In 1998, McLafferty et al introduced a distinct fragmentation method for intact protein and peptide ion, which was termed as electron capture dissociation (ECD).[111] Ten years later, this technique was firstly applied to solve the scrambling issue caused by collision induced dissociation (CID) when using MS to acquire HDX data.[112] Jørgensen et al found this problem could be reduced to a very low level by ECD and minimized scrambling could be realized by appropriate UPLC separation and source setting. Later, the same research group proposed another fragmentation method, electron transfer dissociation (ETD),[113] which was introduced by Hunt et al in 2004,[114] bringing the HDX study into residue level. The improvement in resolution made MS a more potent technique for protein study.

To summarize, mass spectrometry can be used for both qualification and quantification in protein studies. It can explore the high order of protein structure when coupled with other methods, such as chemical crosslinking, fast photochemical oxidation and hydrogen deuterium exchange.[115]

1.3.3 Hydrogen deuterium exchange mass spectrometry

After ESI was introduced into macromolecule study, Katta and Chait combined hydrogen exchange and mass spectrometry to investigate the conformational change of bovine ubiquitin under different conditions (Figure 1.10).[116-117] Compared to other detectors for HDX, such as IR, UV, NMR, mass spectrometry seems to be a more natural choice because it was invented to observe isotopic phenomenon in 1910s. Detecting mass shift is a direct way to monitor the exchange. Besides, local HDX could be obtained after the protein is digested by acidic proteases and separated by liquid chromatography. With the ion-mobility separation and non-scrambling dissociation methods,[118] high-resolution mass spectrometry alone can provide HDX information of some proteins in detail. Thus, the combination of HDX and MS is playing a vital role in protein studies. HDX-MS system has been developed for more than 20 years and great improvements have been achieved over these years. Several aspects for the experimental part are highlighted below (Figure 1.11).

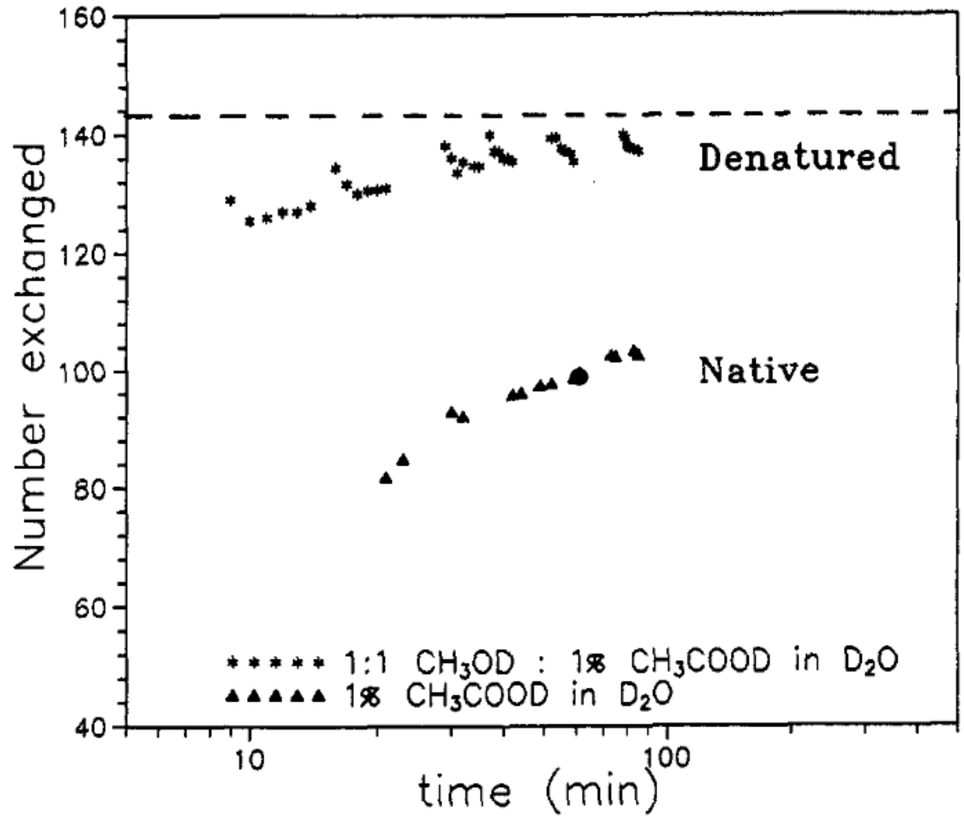

Figure 1.10 The number of hydrogens exchanged as a function of time under different conditions. (Reprint from the literature[117])

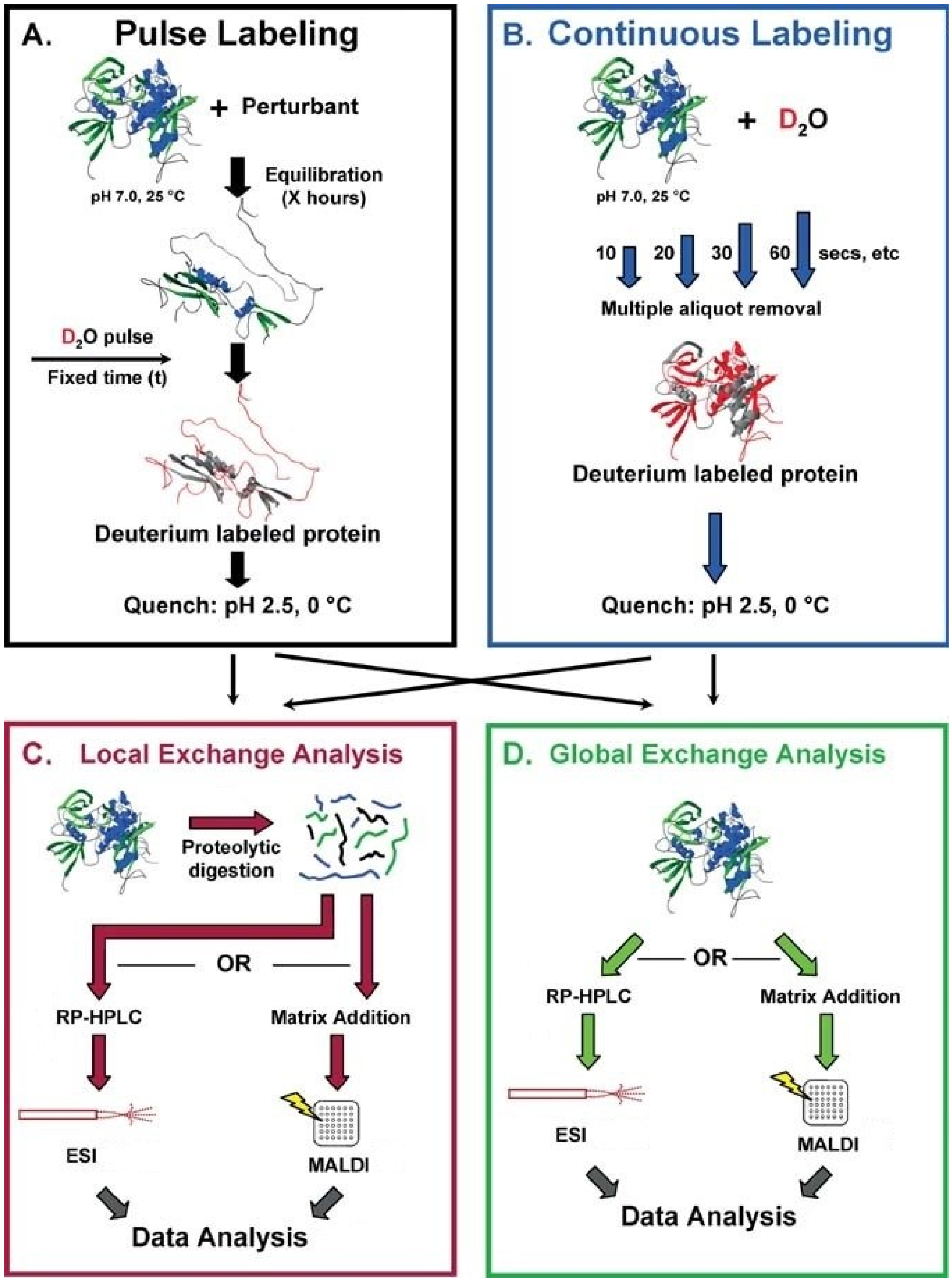

Figure 1.11 Overall scheme for hydrogen exchange mass spectrometry experiments. (Reproduced from the literature[119])

Pulse-labeling and continuous-labeling

Pulse-labeling and continuous-labeling methods are quite commonly adopted (Figure 1.11 A & B). The former is more suitable for study of protein folding and intermediate state, while the other is widely used for protein-ligand interaction. Since protein folding undergoes relatively short period, the labeling should be performed in milliseconds. Thus, special devices are required to do the labeling. There are many microfluidics applied to control the mixing, labeling and quenching.[120] More details of microfluidics coupling MS can be referred to a recent review.[121] Briefly, three steps are included in pulse labeling. Firstly, the protein sample fully unfolded by denaturants is exposed to refolding buffer for certain intervals. Then the protein refolded to various degrees is incubated with deuterated solvent for isotropic pulse-labelling followed by quenching. Finally, the mixture is introduced into a mass spectrometer. On the other hand, continuous labeling is more popular and commonly used. The time scale normally ranges from seconds to hours. Briefly, the protein sample is primarily diluted with deuterated buffer and undergoes exchange for certain time intervals. After labeling, quench buffer is added before the deuterated protein is analyzed. The labelling intervals cover 10s, 60s, 600s, 3600s and 4h, which are exponentially increasing. Protein may take days to be completely deuterated while this is not necessary since the difference is usually shown in hours.

Global and local HDX

After labelling, the protein or protein-ligand complex can be analyzed by chromatographic and mass spectrometric approaches (Figure 1.11 C & D). As beforementioned, mass spectrometry can reflect the structural information of intact protein. The feature obtained by MS identifies thousands of proteins. When the protein intakes deuterium, there will be a mass shift for the protein. We can plot the intact mass as a function of labeling period and different uptake plot indicates conformational changes induced by ligand binding or other factors. Within hours, this result can provide us with the conformational change under different conditions. Such evidence will direct to further investigation. Global exchange is favorable in high-throughput screening of potent targets in protein-ligand interaction. Global strategy only allows us an overview of protein dynamics without detailed structural information. Thus, it is quite necessary to get localized information of the protein structure. Local exchange could be read from fragmented proteins. The resultant fragments can be further sequenced by MS and aligned to the protein sequence. Structural information from local exchange depends on the sequence coverage, which is attributed to fragmentation efficiency and detection sensitivity. Residual resolution might be achieved by subtracting the overlapped fragments and gas-phase dissociation.

Top-down and bottom-up

It is known that two methodologies could be applied in protein study. Bottom-up is more prevalent than top-down due to the more established software packages for spectrum interpretation. Pepsin is one of the dominant enzymes that are used for digestion. It can keep good performance in acidic quench buffer under low temperature which are very critical conditions to avoid back exchange. Other functional proteases can also be used or combined to produce higher sequence coverage and more overlapped peptides.[122] For instance, a dual column packed with both protease type XIII and pepsin can significantly enhance digestion efficiency.[123] We can set up a house-build library for matching targeted protein sequences. Top-down is also rapidly developed these years due to the improvement of gas-phase dissociation in mass spectrometer. Collision-induced dissociation (CID) was widely used for tandem mass spectrometry. However, the high energy of CID might cause deuterium scrambling, thus resulting in disrupted exchange pattern.[124] Electron-activated dissociation, ECD and ETD, can reduce the scrambling to a minimal extent thus provide meaningful exchange information.[112, 125] These dissociation methods fragment protein molecules in a different pattern from CID, producing c and z ions rather than b and y ions, and require lower energy to break the peptides.[111, 114] Taking advantage of the fragmentation mechanism, ECD can be used to improve the detection of HDX of disulfide-rich proteins.[126] Ultraviolet photodissociation (UVPD) is a recently developed technique that can also be applied to the top-down analysis in HDX-MS.[127]

ESI and MALDI

Both ESI and MALDI are soft ionization methods for macromolecules. One of the distinctions between them is their compatibility to LC system. ESI could be connected to the flow of LC while MALDI seems not applicable. The LC system helps to separate the digested proteins and results in better sequence coverage. However, back exchange during LC separation reduces exchange to some extent. Fragmented protein can be directly introduced into both ESI and MALDI source without separation. In this case, nanoESI is more sensitive and optimal than normal ESI. MALDI produces singly charged peptides with less signal suppression compared to ESI. Considering the back exchange during LC separation, methods should be optimized to achieve a balance of better separation and less back exchange. Herein, UPLC is much more efficient than HPLC and can preserve high performance under low temperature. Compared to

LC-ESI-MS, MALDI-MS is less prevalent yet still a good choice due to the elimination of LC separation. Peptides are mixed with matrix (e.g., sinapinic acid and Dihydroxybenzoic acid) and are rapidly dried before MS analysis. Due to simplified mass spectrum, it would be more feasible for manual data processing.[128]

Manual and automated data processing

Data processing is one of the most important part in HDX-MS. However, with the evolution of computing techniques, automated software or algorithms appear to promote the data processing.[129-131] Essentially, two steps are involved to analyze HDX data. One is to identify and assign peptides corresponding to protein sequence and the other is to extract deuterium content for each peptide. Software-based extraction might fail due to mismatch thus still need to be manually checked. The algorithms might align the mass envelops incorrectly due to overlapping signals or noises and semi-automated tools may be useful for binomial fitting and bimodal deconvolution[70]. Software could be more applicable when epitope mapping is investigated with high throughput screening. Manual data processing is required for further verification. Targeted extraction by manual operation should be more reliable.

1.3.5 Data interpretation

Data interpretation is the last step of HDX-MS analysis and is quite empirical. That means different investigators could make distinct conclusions based on the identical results. It was comprehensively reviewed by Jürgen Claisen and Tomasz Burzykowski recently.[132] There are several aspects that need special attentions from my point of view.

Local and residual resolution

HDX data acquired by NMR normally achieve higher resolution than MS. There are also several methods that enable us to approach near-residual resolution with HDX-MS.[133] The overlapping digested peptides can be used to deduce the residue-level information. With additive proteases, resolution could be enhanced to some extent.[134] A redundant and overlapped peptide dataset tends to be a necessary constraint for enhanced resolution. A residue averaging algorithm was developed by D. D. Weis et al to approach residual resolution.[135] On the other hand, the introduction of electron-induced dissociations, including ECD[112, 126, 136] and ETD[137-139], can also improve the resolution to residue level. However, the detection sensitivity might be compromised upon dissociation. Gas-phase fragmentation results in numerous peptides, especially for large proteins. As a result, signals could also be overlapped and suppressed. This might be overcome with a high-resolution mass spectrometer, like Fourier Transform Ion Cyclotron Resonance Mass Spectrometry (FT-ICR).[123, 140] An alternative way is to separate the fragments by ion mobility which differentiates the molecular ions in the gas-phase electronic field. S. W. Englander et al reported that isotopic envelop rather than mass centroid included more useful structural information with higher resolution.[141] This is intriguing that same centroid mass might result from different isotope distribution patterns. After all, residual resolution might not always be necessary. Pros and cons should be weighed before putting more efforts to approach higher resolution since this can probably reduce the accuracy and reliability of HDX results.

Secondary and tertiary structure

HDX is aimed to probe the conformational dynamics of high ordered structure. Secondary structure is featured by hydrogen bonding,[75-76] which could apparently reduce the hydrogen exchange rate. In contrast, tertiary structure is contributed by several types of sidechain-sidechain interactions, including hydrogen bonding, hydrophobic interactions, salt bridge and disulfide bridge. Compared to hydrogen bonding, other interactions are less likely to alter the hydrogen exchange rate while they can fold the protein and expose the backbone to solvent, i.e., change the solvent accessibility. Thus, tertiary structure might have less impact on the apparent exchange rate compared to secondary structure. Allosteric effects of conformational change have drawn growing attention in protein-ligand interaction.[142-143] Tertiary structure is quite complicated and cannot be directly relate to hydrogen exchange rate. However, the conformational changes caused by ligand-binding can be related to the changes in hydrogen exchange. Such comparative studies could facilitate interpretation of tertiary structures.

Free and bound states

It is more complicated to conduct the HDX experiment of the bound state than the free state of protein. The primary reason is that the equilibrium of both states will defocus the HDX measurement of the bound state. In other context, the HDX information will be compromised to the free state, leading to less differentiation of these states. Though we could try to saturate the protein by its ligands, problem still arises when the ligand is also a protein. Because the excess ligand, usually 5-10 folds than protein samples, will make the signal of protein suppressed by the ligand. Additional separation by size-exclusion chromatography of the sample would reduce the noise of unrelated molecules. For those amount limited samples, this process might not be favorable. With the development of ion mobility mass spectrometry, it gives a preferred option to separate different conformers inside mass spectrometry.[144] Recent kinetic study on the intermediates of holo- and apo-Myoglobin showed different conformers with various charges, undergo different HDX patterns.[145] Coupling with gas-phase dissociation, it might be possible to differentiate local exchange rate between varying conformers.

1.3.6 Challenges and Chances

With the updates of hardware and software, better resolution could be achieved, thus more structural details could be obtained soon. The coupling of novel fragmentation and separation methods inside mass spectrometer provides huge data that can be used by powerful processing approaches. Given that enzymatic digestion and LC separation are substituted by gas-phase dissociation and ion mobility differentiation,[118] respectively, back exchange is no longer a disturbing issue. Meanwhile, automated sample preparation is changing the way to do the experiment and enables us to put more efforts into data interpretation. Also, mass spectrometer is evolving all the time and its performance is out of imagine in the following decades. The data processing software is playing an important role in saving time for data interpretation.[131] All these improvements together can achieve high-throughput HDX experiment and reliable HDX data. HDX-MS is able to probe the interaction of protein and its ligand, which could be small molecules, macromolecules and even nanoparticles.[146] Both covalent and non-covalent bindings with proteins are suitable for HDX-MS analysis. This technique is complementary to X-ray crystallography and NMR while it can also be useful with model-free method. Furthermore, with the assistance of computational methods, e.g., molecular dynamics simulation, we can elucidate the structure with constrains from HDX data. To make a step further, it is promising that single cellular analysis will be available one day using HDX-MS equipped with ultra-high sensitivity and intracellular labeling method. A big challenge is to transform HDX data to structural information. McAllister and Konermann showed their worry on the bias of HDX protection factors that may cause misleading results.[147] They proposed that backbone and sidechain hydrogen bonding as well as solvent accessibility might not be sufficient constraints to determine hydrogen exchange behaviors. This study emphasized the complexity of HDX that might not be fully understood.

1.4 Molecular dynamics simulation for elucidation of HDX

MD simulation can provide information at the atomic level for protein dynamics[148] and serve as a powerful tool to evaluate HDX behaviour.[149-150] Garcia and Hummer investigated the conformational dynamics of cytochrome *c* in an effort to correlate HDX kinetics and MD simulation in aqueous solution.[151] The system was examined at four temperatures ranging from 300 K to 550 K for at least 1.5 ns during which global motions and local hydrogen bonds were analyzed. The hydration and hydrogen bonding of backbone were evaluated, and multi-basin dynamics of cytochrome *c* was characterized to reflect a hierarchical energy landscape where multiple energy-minima basins were sampled within a few hundred picoseconds. It was interesting to see that large fluctuations were not adequate to break the hydrogen bonds in some rigid regions to allow HDX. More recently, Craig et al. simulated the HDX for ubiquitin, chymotrypsin inhibitor 2 and staphylococcal nuclease, based on funneled structure models.[152] The simulation using coarse-grained model examined large-scale unfolding transitions or unfolded sub-states which could dominate the HDX process. The open and closed conformations were characterized and protection factors at the residual level were predicted and compared with experimental values obtained by NMR. Their results emphasized the dominant role of the local environment in disturbing the stability of hydrogen bonds. Alternatively, the topology of protein could be determinant for the exchange of amide hydrogens. The coarse-grained conformational sampling method was also found to improve the correlation of predictive and experimental HDX data.[153]

Further understanding of HDX could be achieved by increasing the computing power for MD simulation.[154] A 100-ns-long simulation of ERK2 in four different states was conducted by Petrik et al. to obtain insight into the HDX experiments.[155] The averaged solvent accessible surface area (SASA) and the number of water molecules in the first NH solvation shell correlated well with the number of amide hydrogens exchanged per peptide, suggesting that the measured exchanges due to solvent exposure or fluctuations involved the breaking of hydrogen bonds. Persson and Halle analyzed a 0.262-ms MD trajectory of the protein BPTI to define the open state (O state) where more than two water molecules coordinated with exchangeable hydrogens for all the tested amide hydrogens.[150] Interestingly, a mean 100-ps residence of O state was identified, indicating that the HDX rates were governed by the reclosed frequency. The protection factors computed from this model correlated well with the experiments. Despite the demonstrated simulating power of molecular dynamics, a comparison with the HDX experiment for this protein G variant showed the demand for improvement of force fields to describe hydrogen bonding and backbone hydration.[156]

A recent study by McAllister and Konermann discussed the challenges in interpreting HDX data due to the complexity and ambiguous understanding of the HDX process.[147] From a 1-μs all-atom simulation, they explored the correlation between experiments and simulation where 57 out of 72 amide exchanges could be explained based on hydrogen bonding and solvent accessibility. Discrepancies were observed for some exposed amide hydrogens. These amide hydrogens are not hydrogen bonded and are solvent-accessible but were found to be protected from exchange. Electrostatic factors could account for this unexpected surface amide protection, but it is challenging to include this into consideration for simulations. The authors also criticized the idea that hydrogen bonding to crystallographic waters might protect surface amide hydrogens from exchange because these solvent molecules immobilize not only in the crystal, but also in bulk solution. Nevertheless, it should not be neglected that water molecules play an essential role in maintaining the high-ordered structures of proteins as hydrogen-bonding is a fundamental intramolecular interaction.[157] The importance of water molecules in protein-ligand binding is increasingly recognized in recent years because displacement of water molecules would influence the binding affinity between proteins and ligands.[81]

HDX, which makes use of water to examine the dynamics of protein molecules, could be an important technique in studying the effect of water molecules on protein structure/dynamics and functions.

Essentially, normal all-atom MD simulation does not directly compute hydrogen exchange or proton transfer at ambient conditions, so it is not surprising that discrepancies can be found between these two techniques. Basically, HDX monitors the interplay between proteins and water molecules while MD simulation captures the intrinsic motions of atoms in proteins. There are correlations between them, but such correlations would be in a complicated manner, leading to the difficulties encountered in interpreting these data. Perhaps further modifications of the force fields and criteria for defining hydrogen bonding and solvent accessibility are required for more reliable and comprehensive simulation of HDX behaviors.

1.5 The scope of this thesis

This project aims to employ HDX-MS, IM-MS and MD simulation to investigate the structural and particularly dynamic aspects of the interactions between BLIP and β-lactamases, to reveal the factors responsible for the differences in binding between BLIP and different β-lactamases and the enhanced inhibition efficiency of various BLIP mutant variants. The outcomes of this project will allow us a more comprehensive understanding on the interactions between BLIP and β-lactamases and provide new insight into the design of evolutionary BLIP mutant variants, which are important for us to combat the β-lactamase-mediated antibiotic resistance in long term.

The objectives are:

1.To investigate the structural and dynamic factors for the large difference in inhibition potency of BLIP towards three Class A β-lactamases, TEM1, SHV1 and PC1.

2.To investigate the structural and dynamic factors for the enhanced inhibition potency of three BLIP mutant variants, Y50A, E73M and K74G, towards β-lactamases.

In chapter two, the materials and methods used in this study are described. Three class A β-lactamases, TEM1, SHV1 and PC1 were expressed and purified. Protein characterization was performed by SDS-PAGE and mass spectrometry. Mass spectrometric approaches including HDX-MS and IM-MS are performed with a state-of-the-art high-resolution mass spectrometer (Synapt G2Si, Waters Corporation). MD simulation was performed with GROMACS.

In chapter three, the conformational dynamics of three class A β-lactamases, TEM1, SHV1 and PC1 was studied by HDX-MS and MD simulation in their free states and bound states. Individual class A β-lactamase bound with wild-type BLIP and an enhanced BLIP mutant were analyzed by HDX-MS.

In chapter four, the conformational dynamics of BLIP upon inhibitory binding to three class A β-lactamases (TEM1, SHV1 and PC1) was examined by HDX-MS and MD simulation. BLIP mutants with enhanced inhibition potency were compared with the wild type using HDX-MS and MD simulation. Furthermore, these mutants (Y50A, E73M and K74G) upon binding with TEM1, SHV1 and PC1, respectively, were analyzed by HDX-MS.

In chapter five, the conclusions and major findings of this study are summarized. Perspectives on the current study and future applications are provided.

2. Materials and Methods

2.1 Protein expression and purification

2.1.1 Expression and purification of TEM1, SHV1 and PC1.

His-tagged TEM1, SHV1 and PC1 were expressed in *E. coli* BL21 (DE3) through modified plasmids containing the kanamycin-resistant gene (pRSET-K), as previously described.[158] Briefly, transformed *E. coli* BL21 was pre-cultured at 37 °C in Luria Broth (LB) containing 50 µg/mL kanamycin for 16 hours with shaking at 250 rpm. 1 mL of preculture was transferred into 250 mL LB where protein overexpression was induced with 0.3 mM isopropyl β-D-1thiogalactopyranoside (IPTG) when OD_{600} of the culture reached 0.8. After 4-hour incubation at 30 °C, the cells were harvested by centrifugation (4,500 rpm, 15 min) at 4 °C followed by resuspension in lysis buffer (50 mM phosphate buffer, pH 7.4). Suspensions were lysed with sonication for 5 min at 15-sec intervals and insoluble debris was removed by centrifugation (10,000 rpm, 15 min). The supernatant was filtered through a 0.45 µg filter before purification.

The crude extract was applied to a HisTrap™ HP affinity column (5 mL, GE Healthcare) pre-charged with nickel for purification. HisTrap chelating column was applied to ÄKTA purifier and washed by Deionized water (5 mL/min) followed by charging with Ni^{2+}. The column was pre-equilibrated by 5-10 CV starting buffer (50 mM phosphate buffer, pH 7.4). Then 20 mL crude extract was injected into the sample loop and washed by starting buffer for 5-10 CV. Then eluted with elution buffer (0.5 M imidazole in 50 mM phosphate buffer) with linear gradients. Fractions were collected in the test tubes and analyzed by SDS-PAGE. The purified proteins were buffer exchanged into 50 mM phosphate buffer and sterilized through a 0.22 μm filter.

2.1.2 Expression and purification of BLIP and BLIP mutants Y50A, E73M and K74G

Wild-type BLIP was a kind gift from our collaborator Prof. Yun-Chung Leung. A detailed method for efficient production of secretory *Streptomyces clavuligerus* β-lactamase inhibitory protein (BLIP) in *Pichia pastoris* was described in the literature.[158]

Three BLIP mutants were purchased from Genscript Biotech Corporation. A general procedure was described as below. The target DNA sequences of BLIP mutants were optimized and synthesized followed by cloning into vector pET30a with His tag for protein expression in *E. coli*. The cloning strategy can be found in Figure 2.1. *E. coli* strain BL21 Star (DE3) was transformed with recombinant plasmid. A single colony was inoculated into LB medium containing kanamycin culture was incubated in 37 °C at 200 rpm and then induced with IPTG. SDS-PAGE and Western blot were used to monitor the expression. BL21 Star (DE3) stored in glycerol was inoculated into TB medium containing kanamycin and cultured at 37 °C. When the OD600 reached to about 1.2, cell culture was induced with IPTG at 15 °C for 16 hours. Cells were harvested by centrifugation. Cell pellets were resuspended with lysis buffer followed by sonication. The precipitate after centrifugation was dissolved using urea. Denatured supernatant after centrifugation was kept for future purification. Target protein was obtained by two-step purification by Ni column, TEV protease cleavage and Ni column. Target protein were refolded and analyzed by SDS-PAGE. The purified proteins were sterilized by 0.22 μm filter before stored in aliquots.

Full length protein:

> C0787DF040-5 (BLIP K74G)

NdeI--ATG--His tag -- TEV protease site --**BLIP K74G**--Stop codon--HindIII

Protein Length=185 MW=19886.3 Predicted pI=6.00 vector: pET30a

MHHHHHHENLVFQG**EAEAEFAGVMTGAKFTQIQFGMTRQQVLDIAGAENCETGGSFGDSIHCRGHAAGDYYAYATFGFTSAAADA**
KVDSKSQEGLLAPSAPTLTLAKFNQVTVGMTRAQVLATVGQGSCTTWSEYYPAYPSTAGVTLSLSCFDVDGYSSTGFYRGSAHLW
FTDGVLQGKRQWDLV..

DNA sequence: 570bp

CATATGCATCACCACCACCACCACGAAAATCTAGTATTTCAAGGA**GAAGCAGAGGCGGAGTTTGCGGGCGTTATGACCGGCGCGA**
AGTTCACCCAGATCCAATTCGGTATGACCCGTCAGCAAGTGCTGGACATTGCGGGTGCGGAGAACTGCGAAACCGGTGGCAGCTT
TGGTGACAGCATTCACTGCCGTGGTCATGCGGCGGGTGATTACTATGCGTACGCGACCTTTGGTTTTACCAGCGCGGCGGCGGAC
GCGAAGGTTGATAGCAAAAGCCAAGAAGGACTGCTGGCGCCGAGCGCGCCGACCCTGACCCTGGCGAAATTCAACCAAGTGACCG
TTGGTATGACCCGTGCGCAGGTGCTGGCGACCGTTGGTCAAGGCAGCTGCACCACCTGGAGCGAGTACTATCCGGCGTATCCGAG
CACCGCGGGCGTGACCCTGAGCCTGAGCTGCTTCGACGTTGATGGTTACAGCAGCACCGGTTTTTATCGTGGCAGCGCGCACCTG
TGGTTCACCGATGGTGTTCTGCAAGGCAAACGTCAATGGGATCTGGTTTAATGAAAGCTT

Derived Protein Sequence (after enzyme digestion):

Protein Length=172 MW=18201.7 Predicted pI=4.64 vector: pET30a

G**EAEAEFAGVMTGAKFTQIQFGMTRQQVLDIAGAENCETGGSFGDSIHCRGHAAGDYYAYATFGFTSAAADAKVDSKSQEGLLAP**
SAPTLTLAKFNQVTVGMTRAQVLATVGQGSCTTWSEYYPAYPSTAGVTLSLSCFDVDGYSSTGFYRGSAHLWFTDGVLQGKRQWD
LV..

Figure 2.1 Cloning strategy of BLIP mutants, exemplified by K74G

2.2 Protein characterization and analysis

2.2.1 Determination of purity and molecular weights

The SDS-PAGE showed that the purity of all proteins was higher than 95% (Figure 2.2). Such homogeneity is good enough to perform further experiment in this study. The molecular weights of TEM1, SHV1 and PC1 were around 30 kDa and those of BLIP and its mutants were around 20 kDa. ESI-MS further confirmed the molecular weight of BLIP mutants (Figure 2.3).

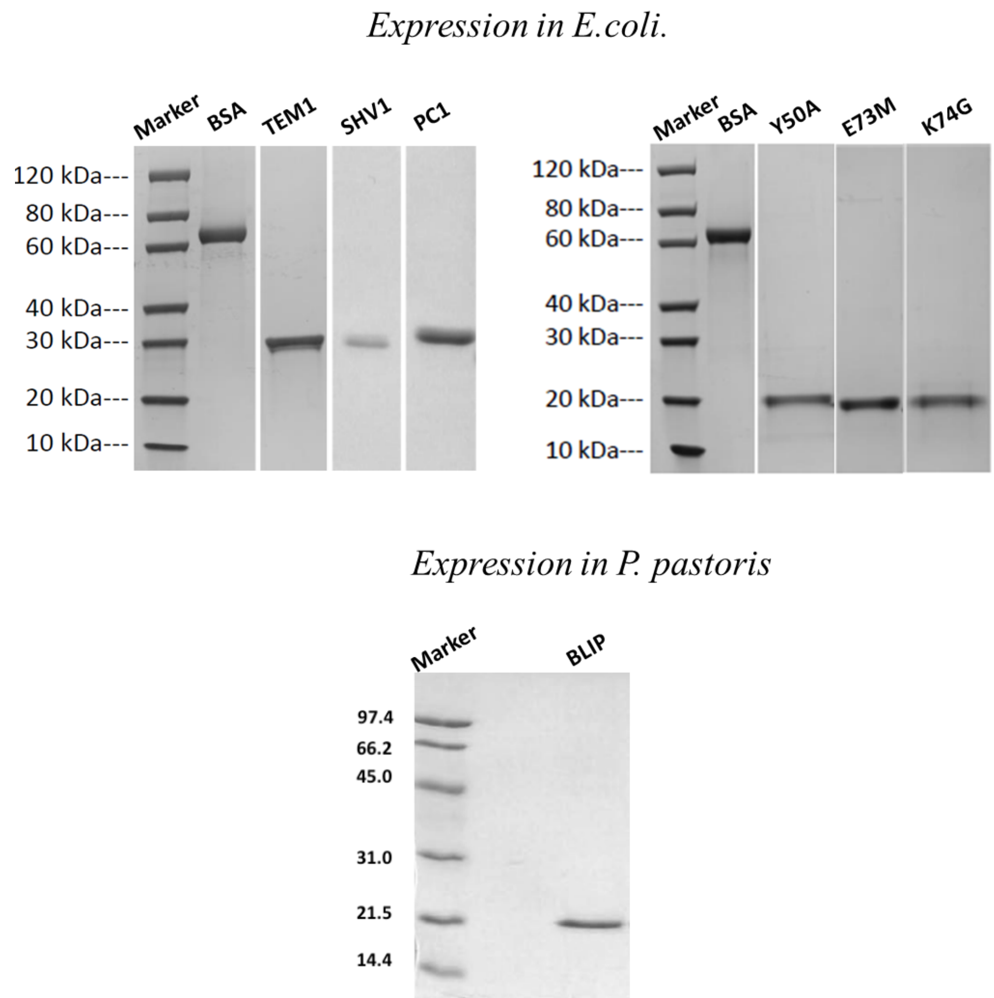

Figure 2.2 SDS-PAGE of purified proteins.

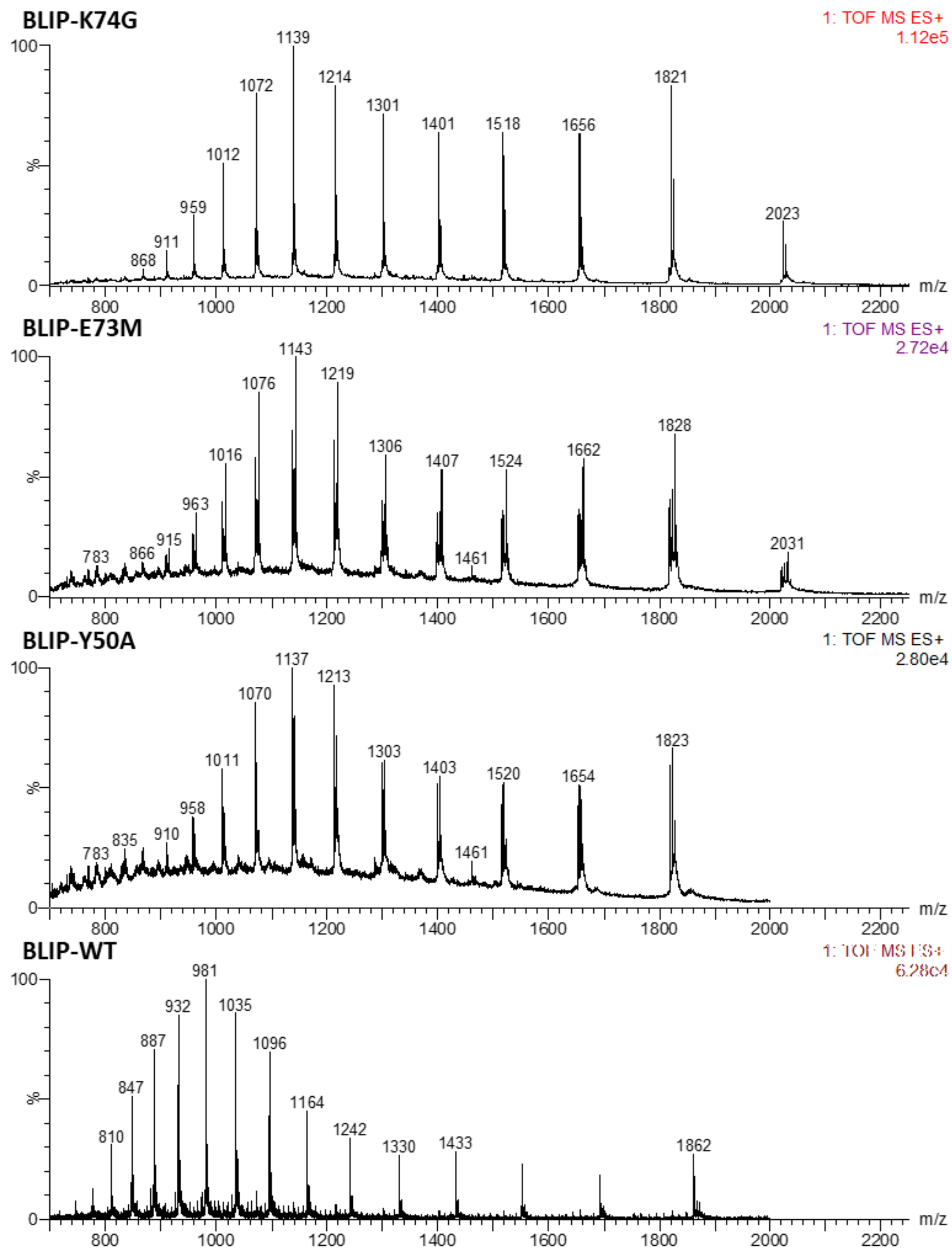

Figure 2.3 Mass spectra of BLIP wild type and its mutants.

2.2.2 Sequence confirmation for BLIP mutants

The BLIP mutants, Y50A, E73M and K74G were subjected to LC-MS analysis following trypsin digestion. The peptides containing the mutation sites were identified, thus the correct mutants could be confirmed. A LC-MS report can be found in the Figure 2.4.

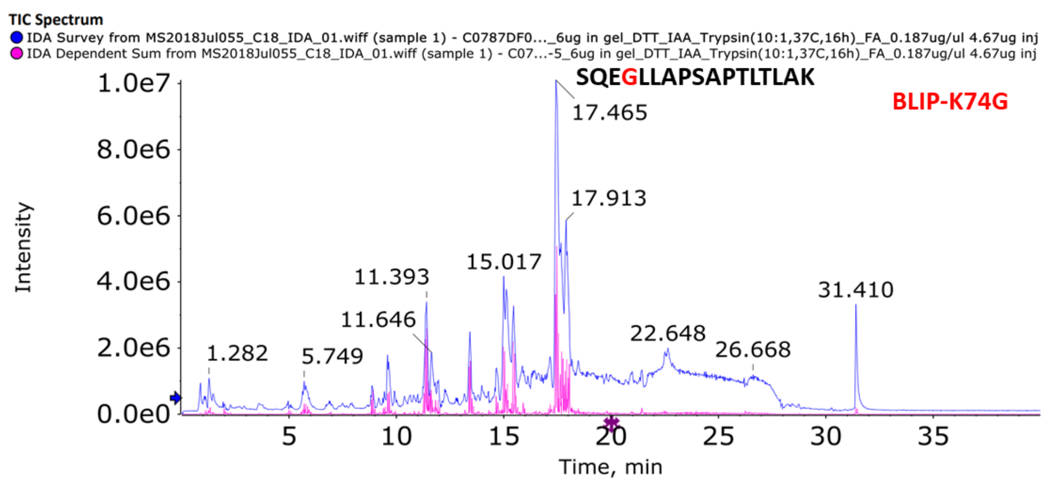

Figure 2.4 LC-MS report on the BLIP mutants, exemplified by K74G.

2.2.3 Enzyme activity assay for β-lactamases

The enzyme activity was measured by the rate of degradation of nitrocefin, whose absorbance at 486 nm appears upon catalysis by β-lactamases.[158] The initial slope of the reaction at steady state was determined using 2-10 nM enzyme and 200 μM nitrocefin. Reaction rates were plotted as a function of substrate concentration. The data were fitted to the Michaelis-Menten equation to determine k_{cat} and K_m.[159]

2.2.4 Inhibition assay for BLIP

2 nM TEM1 β-lactamase was pre-incubated with varying concentrations of BLIP (0-20 nM) in 100 mM potassium phosphate buffer at pH 7.4 for 2 hours at room temperature. Nitrocefin was added to the mixture at final concentration of 200 μM. The reaction was monitored by the absorbance at 486 nm and the inhibition constant (K_i) was calculated by fitting the plot of the fraction of unbound β-lactamase versus BLIP to the Equation 5. The fraction can be calculated by dividing the reaction rates by the one without adding BLIP.[158]

$$\text{Fraction} = 1 - \frac{[E] + [I] + K_i - \sqrt{([E]+[I]+K_i)^2 - 4[E]\times[I]}}{2[E]} \quad \text{(Equation 5)}$$

Where [E] and [I] are the concentrations of the β-lactamase and BLIP.

2.3 Native ion mobility mass spectrometry

For preparation of β-lactamases in complex with BLIP, 30 μM of β-lactamases was incubated with BLIP at a molar ratio of 1:1 for 30 min at room temperature. The protein samples were buffer exchanged into 50 mM ammonium acetate by an Amicon 10K centrifugal filter before MS analysis. Native mass spectra were acquired on a Synapt G2Si mass spectrometer (Waters, Milford, MA) with house-made nanoESI emitters. 3-5 μL protein solution (5-30 μM in 100 mM ammonia acetate) was loaded into the nanoESI emitter and sprayed under the following conditions: capillary voltage 1.5-2.0 kV, source temperature 25 °C, trap and transfer collision energy 15 V, trap DC bias 45, trap gas flow 4-8 ml/min.

Ion mobility measurements were carried out using the same equipment under the following conditions: IMS gas flow 30 ml/min, wave velocity 650 m/s, wave height 40 V and pressure 0.9 mbar. Collision-induced unfolding profiles were measured with trap collision energy from 15 to 120 V. Drift time was extracted manually using MassLynx 4.1 and Microsoft Excel.

2.4 Hydrogen deuterium exchange mass spectrometry

Hydrogen deuterium exchange

For preparation of β-lactamases in complex with BLIP, 20-30 μM of β-lactamases was incubated with BLIP or BLIP mutants at various ratios to saturate the protein under investigation (Table 2.1-Table 2.6). Percentage of the bound proteins was calculated based on the concentration of β-lactamase and BLIP and K_d (Equation 5). The mixtures were incubated on ice for at least 30 min before further experiments. Deuterium exchange was initiated by diluting 2 μL protein sample (20-30 μM) into 38 μL D_2O (99.9%, Cambridge Isotope Laboratories) buffer (100 mM phosphate buffer, pD 7.4) in room temperature (RT). The proteins were deuterated for 60 sec and 10 min for local HDX (optimized from global exchange for 10 sec, 60 sec, 10 min and 60 min) and the exchange reactions were quenched by addition of a quenching buffer (10 mM phosphate buffer, 6% formic acid) to reach a pH value of around 2.5.

LC-MS analysis of deuterated proteins or peptides

50 μL quenched solution was injected into a Waters nanoACQUITY UPLC system at 0 °C. After online pepsin digestion (Waters BEH pepsin column, 300Å, 5 μm, 2.1 mm x 30 mm) and desalting (Waters BEH C18 trap column,

130 Å, 5 μm, 300 μm x 50 mm) for 3 min at 100 μL/min, the peptides were eluted through a Waters UPLC BEH C18 column (130Å, 1.7 μm, 2.1 mm x 100 mm) with linear acetonitrile gradient (5-40%, 0.2% formic acid) at 60 μL/min. The pepsin column was washed 3 times between injections by a denaturing buffer (1.5M guanidine hydrochloride, 4% acetonitrile, 0.8% formic acid, pH 2.5). Global HDX was determined without pepsin digestion, and proteins were eluted with constant mobile phase (50% acetonitrile, 0.2% formic acid). The peptides were detected under the following conditions: capillary voltage 2.5-3.0 kV, source temperature 80 °C. Mass spectra were recorded using the Waters Synapt G2Si mass spectrometer in MS^e mode at 50-2000 Da.

Data analysis

Peptide lists including sequence, precursor ion, m/z, charge state, retention time, fragmentation profile and PLGS score, were generated by Waters ProteinLynx Global Server (version 3.0.2). The peptides fulfilled the following thresholds were processed with Waters DynamX (version 3.0) followed by manual inspection.

Maximum mass errors: 10 ppm

Maximum sequence length: 20

Minimum PLGS score: 7

The minimum intensity: 10000

Retention time root standard deviation: 5%

Peptides of interest were eluted from 3-8 min with an average length of 13. The average mass error for all the peptides for analysis was below 3 ppm. The average standard deviation of the deuterium uptake was below 0.1 Da. Back exchange was estimated to be an average rate of 35% by analyzing fully deuterated peptides from digested PC1 and BLIP (Equation 6). Back exchange was only corrected for the wild-type BLIP while it was not corrected when doing the comparison (Equation 7).

$$BE\% = 1 - \frac{HDX_{max}}{Exchangable\ NH} \times 100\% \qquad \text{(Equation 6)}$$

$$HDX_{BE\ corrected} = \frac{HDX_{observed}}{BE\%} \qquad \text{(Equation 7)}$$

The fractional deuterium uptake was calculated as reference to the maximum exchangeable amide hydrogens (Equation 8).

$$Fractional\ HDX = \frac{HDX_{observed}}{Exchangable\ NH} \times 100\% \qquad \text{(Equation 8)}$$

The deuterium uptake of BLIP wild type and mutants was normalized to allow quantitative comparison of the results obtained in different days (Equation 9).

$$Normalized\ HDX = \frac{HDX_{observed} - \mu_{HDX}}{\sigma_{HDX}} \qquad \text{(Equation 9)}$$

Where μ_{HDX} is the mean and σ_{HDX} is the standard deviation of all observed HDX for peptides in the datasets.

Difference in deuterium uptake with 95% confidence interval (CI) was considered as a significant change for differential HDX. Uptake figures were produced by Microsoft Excel and relative fractional deuterium uptake was mapped onto the crystal model of proteins using PyMol (version 2.0.7, Schrodinger, LLC). Experimental details have been summarized in Table 2.1-Table 2.6. HDX-MS datasets are provided in the Appendix.

Table 2.1 HDX summary for TEM1

Data Set	Free TEM1	TEM1 bound with BLIP-WT	TEM1 bound with BLIP-Y50A
HDX reaction details	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature
Molar ratio (inhibitor/enzyme)	0%	100%	100%
HDX time (min)	1, 10		
Number of peptides	84	83	85
Sequence coverage	90%	90%	94%
(Average peptide length / redundancy)	13.2 / 4.6	13.2 / 4.6	13.3 / 4.5
Replicates (biological or technical)	3	3	3
Repeatability	0.075 Da	0.037 Da	0.032 Da
Significant differences in HDX (ΔHDX > X D)	NA	0.29 Da (95% CI)	0.21 Da (95% CI)

Table 2.2 HDX summary for SHV1

Data Set	Free SHV1	SHV1 bound with BLIP-WT	SHV1 bound with BLIP-E73M
HDX reaction details	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature
Molar ratio (inhibitor/enzyme)	0%	430%	100%
HDX time (min)	1, 10		
Number of Peptides	61	28	61
Sequence coverage	89%	89%	90%
Average peptide length / Redundancy	12 / 3.7	12 / 1.2	12 / 3.2
Replicates (biological or technical)	3	3	2
Repeatability	0.057 Da	0.054 Da	0.028 Da
Significant differences in HDX (ΔHDX > X D)	NA	0.26 Da (95% CI)	0.24 Da (95% CI)

Table 2.3 HDX summary for PC1

Data Set	Free PC1	PC1 bound with BLIP-WT	PC1 bound with BLIP-K74G
HDX reaction details	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature
Molar ratio (inhibitor to enzyme)	0%	220%	100%
HDX time (min)	1, 10		
Number of Peptides	56	34	41
Sequence coverage	89%	89%	100%
Average peptide length / Redundancy	12 / 2.9	12 / 2.1	12 / 2.7
Replicates (biological or technical)	3	3	3
Repeatability	0.047 Da	0.076 Da	0.101 Da
Significant differences in HDX (ΔHDX > X D)	NA	0.46 Da (95% CI)	0.55 Da (95% CI)

Table 2.4 HDX summary for BLIPs in free states

Data Set	Free BLIP-WT	Free BLIP-Y50A	Free BLIP-E73M
HDX reaction details	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature
Molar ratio (enzyme/inhibitor)	0%	0%	0%
HDX time (min)	1, 10	1, 10	1, 10
Number of Peptides	17	45	19
Sequence coverage	48%	80%	58%
Average peptide length / Redundancy	10 / 2.34	11 / 3.97	9.2 / 2.23
Replicates (technical)	3	3	3
Repeatability	0.040 Da	0.074 Da	0.057 Da
Significant differences in HDX (ΔHDX > X D)	0.16 Da	0.30 Da	0.21 Da

Table 2.5 HDX summary for wild-type BLIP bound with β-lactamases

Data Set	BLIP-WT bound with TEM1	BLIP-WT bound with SHV1	BLIP-WT bound with PC1
HDX reaction details	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 50% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 50% D_2O, pH 7.4, room temperature
Molar ratio (enzyme/inhibitor)	100%	430%	220%
HDX time (min)	1, 10	1, 10	1, 10
Number of Peptides	17	18	17
Sequence coverage	48%	64%	64%
Average peptide length / Redundancy	10 / 2.34	11 / 1.99	11 / 1.91
Replicates (technical)	3	2	2
Repeatability	0.055 Da	0.035 Da	0.070 Da
Significant differences in HDX (ΔHDX > X D)	0.19 Da	0.22 Da	0.29 Da

Table 2.6 HDX summary for BLIP mutants bound with β-lactamases

Data Set	BLIP-Y50A bound with TEM1	BLIP-E73M bound with SHV1	BLIP-K74G bound with PC1
HDX reaction details	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature	100 mM Phosphate buffer in 90% D_2O, pH 7.4, room temperature
Molar ratio (enzyme/inhibitor)	100%	100%	100%
HDX time (min)	1, 10	1, 10	1, 10
Number of Peptides	45	19	16
Sequence coverage	80%	58%	57%
Average peptide length / Redundancy	11 / 3.97	9.2 / 2.23	10.5 / 1.91
Replicates (technical)	3	3	3
Repeatability	0.059 Da	0.067 Da	0.070 Da
Significant differences in HDX (ΔHDX > X D)	0.21 Da	0.22Da	0.21 Da

2.4 Molecular dynamics (MD) simulation

2.4.1 MD simulation

For the simulation of TEM1, SHV1 and PC1, the initial structures were directly obtained from protein data bank.[160-161] The free and bound TEM1 were constructed from PDB IDs, 1btl and 1jtg, while SHV1 from PDB IDs, 1shv and 2g2u. The crystal structure of PC1 (PDB ID: 3blm) was used for the protein-only simulation. The initial structure of the PC1/BLIP complex was prepared by homology modeling using Modeller.[162] The complex model was constructed using the crystal structures of apo-PC1 (PDB ID: 3blm) and the TEM1/BLIP (PDB ID: 1jtg), SHV1/BLIP (PDB ID: 2g2u) and KPC2/BLIP (PDB ID: 3e2l) complexes as templates. The starting structure for BLIP mutant E73M was obtained from the crystal structure of the protein complex BLIP-E73M/SHV1 (PDB ID: 3c4p). The structures of Y50A and K74G were prepared by mutagenesis in PyMol (Schrodinger, LLC) from wild-type BLIP (PDB ID: 3gmu).

The β-lactamase in its free and bound states were subjected to molecular dynamics simulation at consistent parameters using GROMACS[163-164] with the Charmm27 all-atom force field.[163] The protein atomic coordinates were encased in an octahedral or cubic box with a 1 nm-thick wall followed by solvation using the TIP3P water model. The charges on the protein were neutralized by adding Na^+ and Cl^- to 150 mM. Following energy minimization and 900 ps of restrained equilibration (100 ps under constant number, volume and temperature (NVT) and 800 ps under constant number, pressure and temperature (NPT), with progressively reducing restraints from 1000 to 0.1 kJ mol^{-1} nm^{-1}, 10 ns of unrestrained molecular dynamics simulations were performed. The time step was 2 fs and configurations were saved every 10 ps.

2.4.2 Trajectory analysis

For β-lactamases, trajectories were subjected to evaluation by calculating the root mean square deviation (RMSD) of Cα positions reference to the energy-minimized structure and further analyzed by calculating their root mean square fluctuation (RMSF) of backbone positions.

For BLIP and its mutants, all trajectories were subjected to evaluation by calculating the root mean square deviation (RMSD) of backbone positions referenced to the crystal structure of wild-type BLIP (PDB ID: 3gmu) and further analyzed by calculating their root mean square fluctuation (RMSF) of backbone positions. The distance between NTD and CTD was defined by the centre of mass of individual domains.[165] The angle was defined by two vectors pointing from the centre of mass of the three-residue linker (77-79) to that of individual domain. The hydrogen bonding was analyzed by VMD (version 1.9.3).

3. Conformational Dynamics of Helix 10 Region and Protruding Loop in Class A β-Lactamase Inhibitory Binding

3.1 Introduction

Among all classes of β-lactamases, class A β-lactamases are predominantly identified in clinical isolates for both Gram-negative and Gram-positive bacteria.[11] The TEM-type and SHV-type class A β-lactamases are most clinically prevalent and have evolved into hundreds of variants, some with abilities to confer resistance to inhibitors.[166] In this study, we focused on three parental class A β-lactamases, i.e., *Escherichia coli* TEM1, *Klebsiella pneumoniae* SHV1, and *Staphylococcus aureus* PC1. These enzymes share high sequence identity and structure similarity, especially between TEM1 and SHV1 (Figure 3.1). β-Lactamase inhibitory protein (BLIP), which is naturally produced by soil bacterium *Streptomyces clavuligerus*, is a unique inhibitor for β-lactamases and shows particularly high inhibition potency towards TEM1. Intriguingly, it inhibits SHV1 and PC1 with much lower efficiency (Table 3.1). In addition, the inhibition potencies of BLIP towards SHV1 and PC1 can be considerably improved by point mutations on BLIP, i.e., E73M and K74G, respectively.[167-168] Hence, BLIP can be a highly potential inhibitor towards various β-lactamases and offers an unusual and alternative way to overcome inhibitor resistance. In this regard, understanding the properties of interactions between BLIP and these β-lactamases is highly important for further development of novel BLIP-related therapeutic agents.

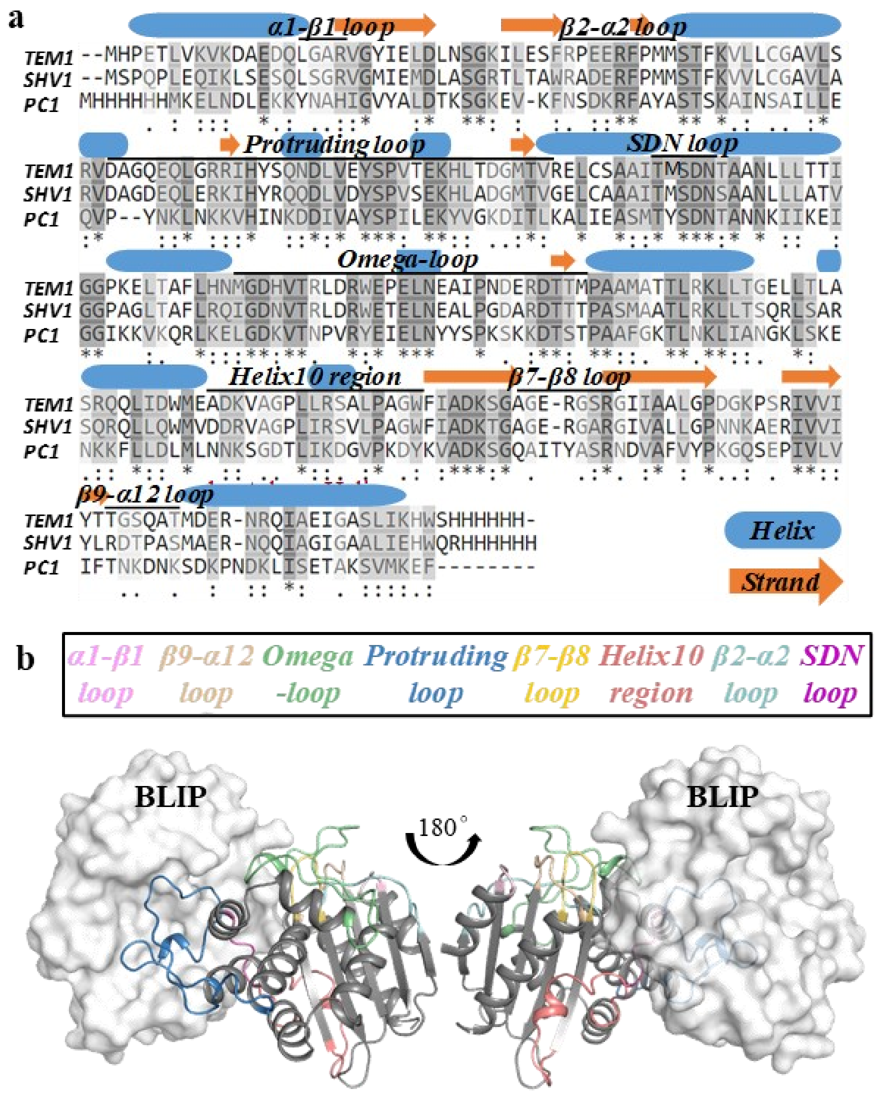

Figure 3.1 β-Lactamases and their complexes with BLIP.
(a) Sequence alignment of TEM1, SHV1 and PC1. Regions of helix and sheet are labelled.
(b) Representative complex of β-lactamases (e.g., TEM1) and BLIP (PDB ID: 1JTG). Loops of interest are labelled on TEM1.

Table 3.1 Characteristic properties of TEM1, SHV1 and PC1.

Type of class A β-lactamase	Inhibition constant by BLIP (nM)	Enhanced inhibition by BLIP mutants (nM)	Sequence identity to TEM1	Structural alignment to TEM1 before/after binding
TEM1	0.5	0.01	NA	NA
SHV1	1720	4.4	68%	1.5/1.1 Å
PC1	350	42	33%	3.0/2.7 Å

The crystal structures of β-lactamases TEM1, SHV1 and PC1 have been revealed.[169-171] A typical class A β-lactamase comprises of two domains, including a five-stranded antiparallel β-sheet domain, and an entirely α-helical domain that packs against the β-sheets from the first domain (Figure 3.1). The catalytic-site Ser70 serves as the nucleophilic amino acid, which is located at the interface of these two domains.[65] The catalytically important residues, including Ser70, Lys73, Ser130, Glu166 and Lys234, form an active pocket in catalysis. The crystal structure of BLIP (PDB ID: 3gmu) has been solved,[172-173] and its complexes with TEM1 (PDB ID: 1jtg) and SHV1 (PDB ID: 2g2u) have been extensively studied.[174-175] Particularly, eight β-strands from BLIP pack together to form a concave surface that accommodates the protruding loop on the β-lactamases and the β-hairpin loop on BLIP inserts into the active cavity of β-lactamases (Figure 3.1). Structural alignment of TEM1 (PDB ID: 1btl) and SHV1 (PDB ID: 1shv) showed highly similar structures, with a root mean square deviation (RMSD) in all-atom positions of 1.5 Å, and the RMSD of

TEM1 (PDB ID: 1jtg) and SHV1 (PDB ID: 2g2u) after binding decreased to 1.1 Å, showing highly similar binding modes (Table 3.1).

The broad range of inhibition potencies of BLIP towards different β-lactamases has attracted considerable research interests in studying the factors governing these inhibitory bindings and developing evolutionary BLIP variants with improved inhibition efficiency.[71, 73, 176-178] Regardless of extensive studies on this field, our understandings on the interactions between BLIP and β-lactamases have been mainly based on the static picture of the stable forms of the inhibitory interactions. The conformational dynamics,[161, 179] which could be determinant to differentiate the binding interactions of these "similar" β-lactamases, has been seldom explored. Ion mobility mass spectrometry (IM-MS) can sensitively probe the conformational changes of molecular ions in the gas phase,[21] while hydrogen deuterium exchange-mass spectrometry (HDX-MS) has been widely applied to study protein dynamics by examining the exchange of amide hydrogen with deuterium in solution.[105, 180-184] Molecular dynamics (MD) simulation is a powerful tool for exploring the complexity of protein molecules, especially in protein dynamics, thus providing complementary information to experimental studies.[182, 185]

In this chapter, we integrated IM-MS and HDX-MS with MD simulation to synergistically elucidate the mechanism of inhibitory interactions between three representative class A β-lactamases and BLIPs. Particularly, natural variants of class A β-lactamases here allowed investigation of the effects of extensive simultaneous mutations and large conformational rearrangements on inhibitory binding. Furthermore, BLIP mutants with enhanced binding affinity confirmed the changes in conformational dynamics regarding the inhibition of these β-lactamases.

3.2 Results and discussions

3.2.1 Global conformations of class A β-lactamases revealed by IM-MS and HDX-MS.

The β-lactamases were characterized by mass spectrometry under denatured (50% acetonitrile with 0.2% formic acid) and native (100 mM ammonia acetate) conditions (Figure 3.2). The mass spectra obtained for TEM1/BLIP, SHV1/BLIP and PC1/BLIP under native conditions confirmed the formation of the complexes with the binding ratio at 1:1 (Figure 3.3). The deconvoluted masses for the complexes TEM1/BLIP, SHV1/BLIP and PC1/BLIP were 48076.0±1.6 Da, 47910.3±0.5 Da, and 48094.8±0.4 Da, respectively, which were consistent to the expected values.

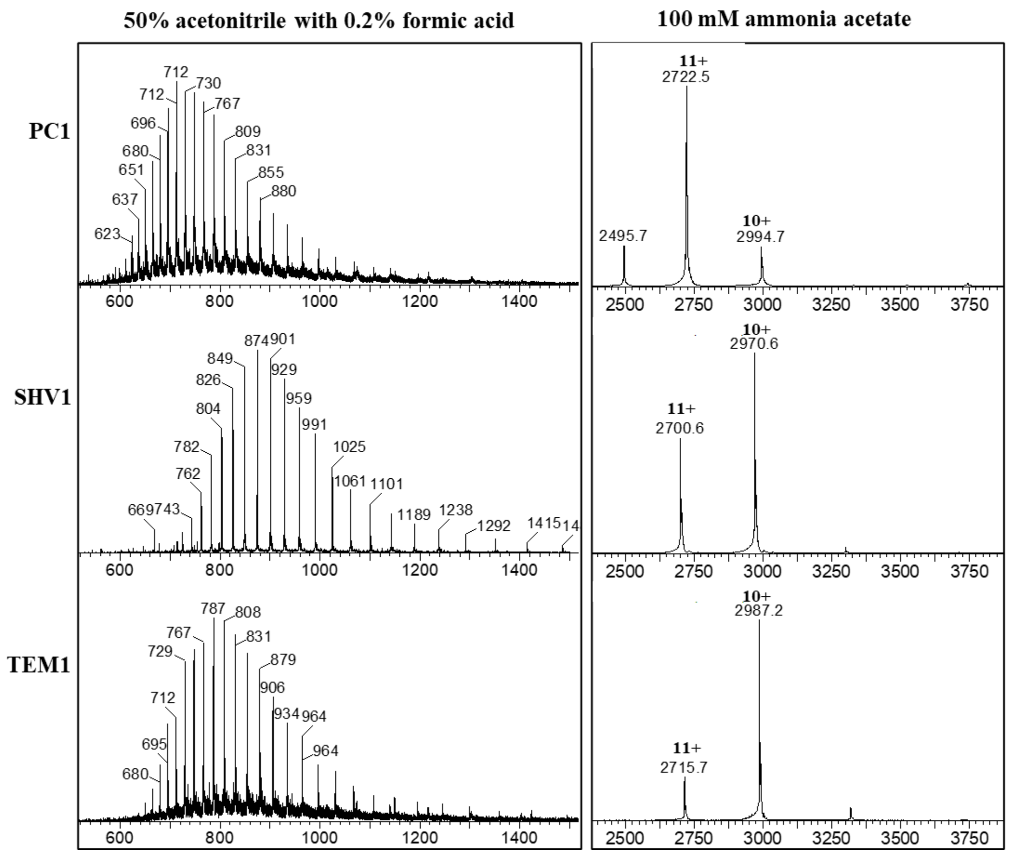

Figure 3.2 Mass spectra of β-lactamases under denatured conditions (50% acetonitrile with 0.2% formic acid, left) and native conditions (100 mM ammonia acetate, right).

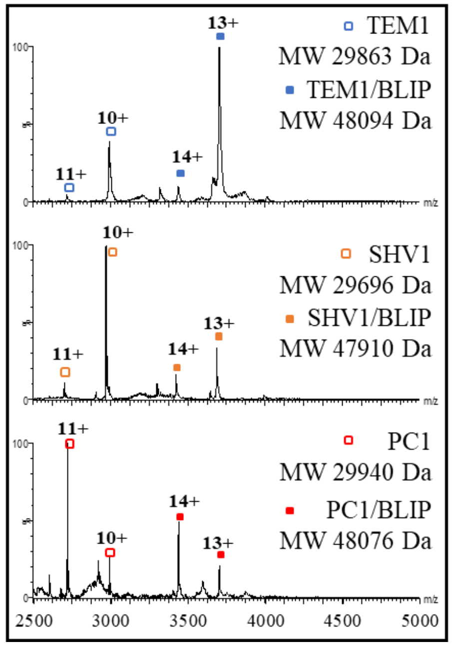

Figure 3.3 Native mass spectra of β-lactamases in their free (hollow) and BLIP-bound (filled) states.

Ion mobility of the free β-lactamases was performed to compare their native conformations. Their arrival time distributions at charge states 10+ were shown in Figure 3.4. The drift time of TEM1 and SHV1 was similar but shorter than that of PC1, indicating that PC1 displayed a more extended conformation than TEM1 and SHV1 in the gas phase. After the β-lactamases bound to BLIP, the complexes at charge state 14+ displayed little difference in drift time distribution. Such observation indicated similar conformations among the class A β-lactamases after binding BLIP. These results are consistent to the crystal structures (PDB IDs: 1jtg and 2g2u) that show little conformational differences among these homologous protein complexes.

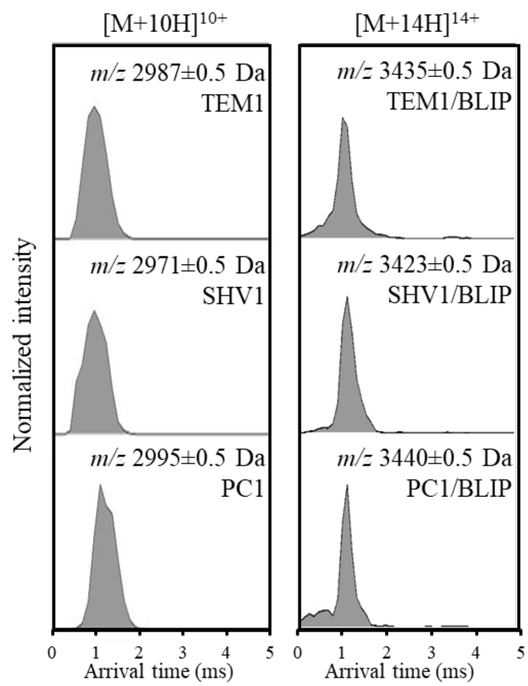

Figure 3.4 Arrival time distribution of SHV1, PC1 and TEM1 in their free states (left panel) and BLIP bound states (right panel).

The conformational dynamics was measured for the free and BLIP-bound states of intact β-lactamases. We compared the global HDX profiles spanning 10 sec, 60 sec, 10 min and 60 min for the three β-lactamases in solution (Figure 3.5). The deuterium uptake for TEM1 was higher than that for SHV1 and PC1. The difference in global HDX profiles indicated that the flexibility of backbone amide hydrogens for TEM1 was higher, reflecting that the overall structures of SHV1 and PC1 could be more rigid than that of TEM1. After inhibitory binding by BLIP, all β-lactamases showed similar HDX profiles that corresponded to similar flexibility of the bound states. Interestingly, the conformational changes for TEM1 induced less deuterium uptake, indicating stabilized conformations after the binding. These observations might also indicate that the higher affinity of BLIP for TEM1 than SHV1 and PC1 could result from a more dynamic unbound state.[186]

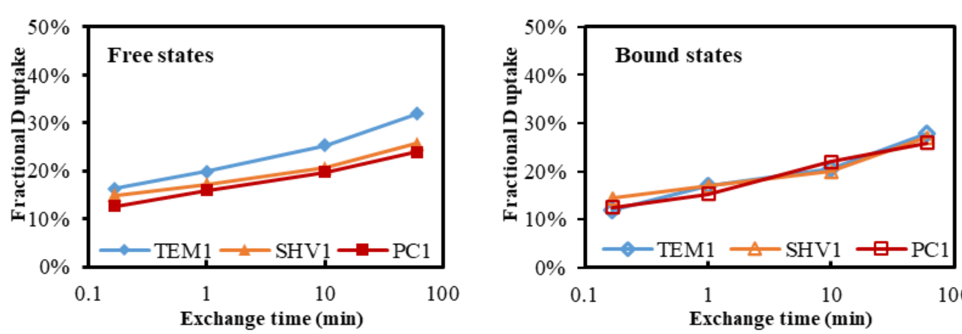

Figure 3.5 Global deuterium uptake plots of β-lactamases in their free (left) and BLIP bound (right) states.

3.2.2 Unravelling the local flexibility of key regions in class A β-lactamases.

Local HDX was performed by on-line pepsin digestion of the exchanged proteins followed by LC-MS analysis, with the sequence coverage as shown in Figure 3.6.

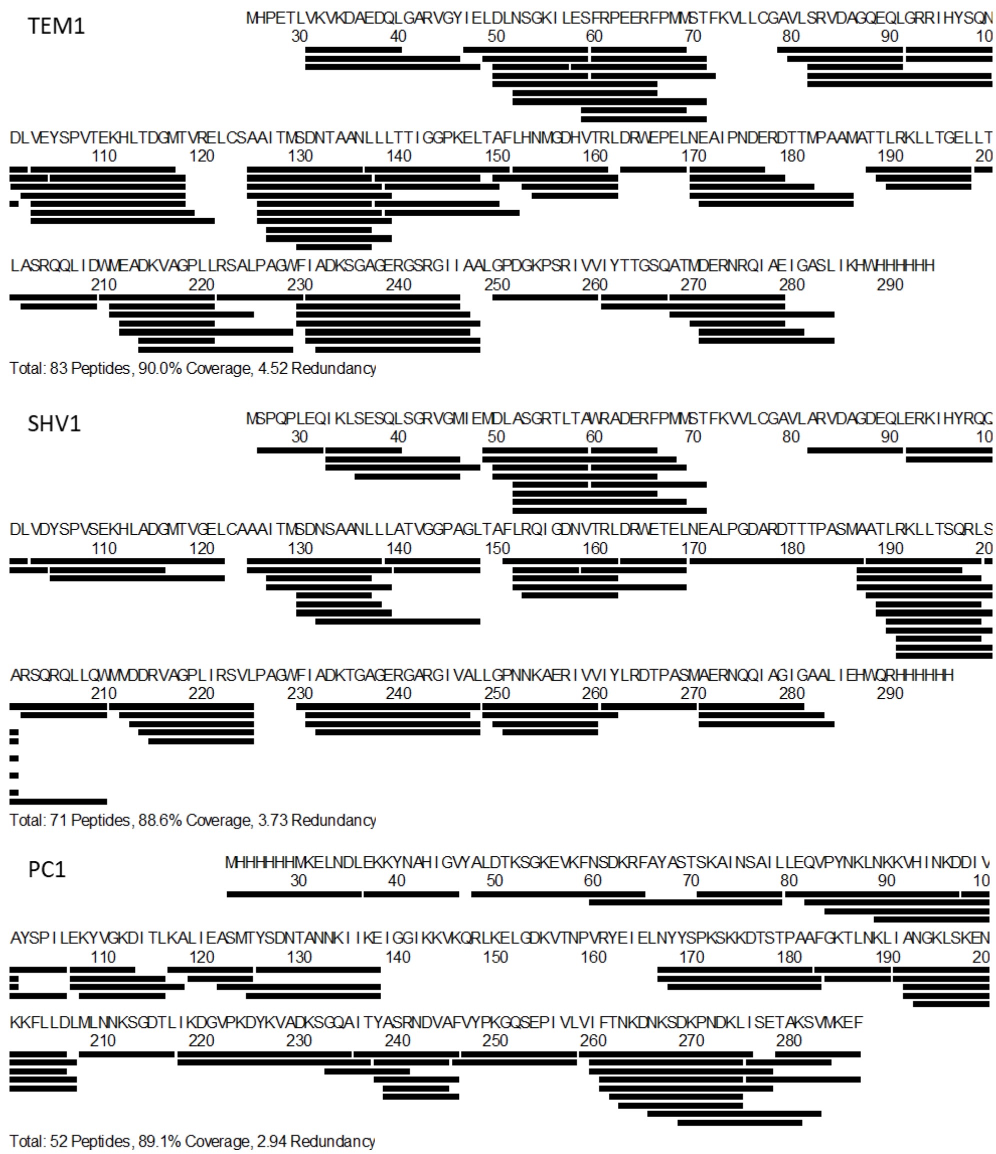

Figure 3.6 Sequence coverage of identified fragments from TEM1, SHV1 and PC1.

The H10 region showed fast deuterium exchange (Figure 3.7), reflecting very flexible structures for these class A β-lactamases. The fragment covering residues 211-225 from TEM1 underwent significantly faster deuterium uptake than that from SHV1 (Figure 3.7). From the available crystal structures (PDB IDs: 1btl and 1shv), it was observed that the H10 region of SHV1 showed the largest deviation (2.5 Å) from that of TEM1 compared to the other regions. Comparing the corresponding regions of the three β-lactamases, in TEM1 residues 211-225 comprise a loop between two helices, whereas in PC1 three residues 218-220 form a 3/10-helix (PDB ID: 3blm); and in SHV1, seven residues 218-224 form an α-helix. Helix, known as an ordered secondary structure, can protect amide hydrogen from deuteration.[187] The corresponding segment in SHV1 had the most helical structure and thus gave the slowest deuterium exchange, which was consistent with our HDX-MS results.

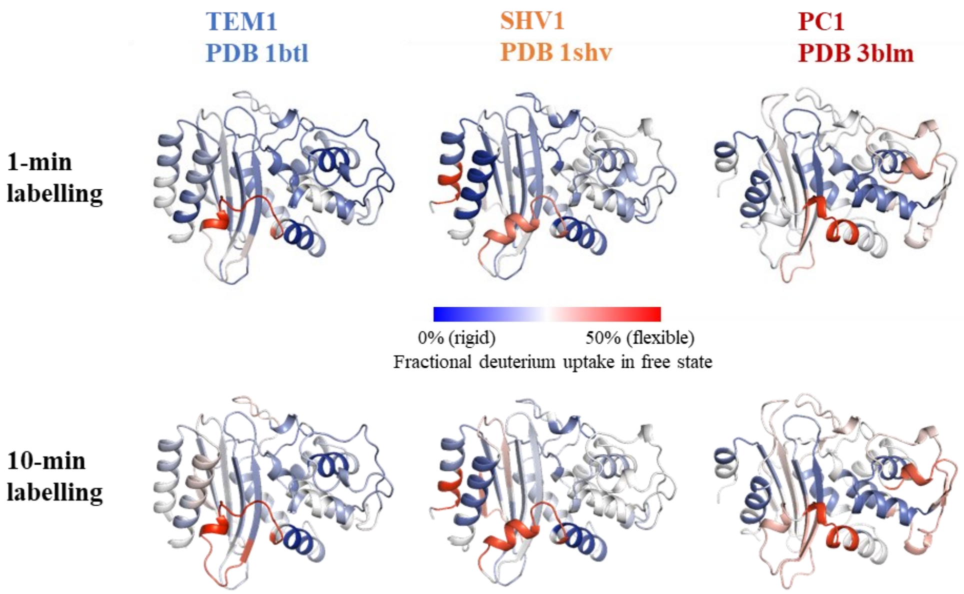

Figure 3.7 Local HDX of TEM1, SHV1 and PC1 in their free states.
Fractional deuterium uptake at 1 and 10 min is mapped onto the crystal models of TEM1 (PDB ID: 1btl), SHV1 (PDB ID: 1shv) and PC1 (PDB ID: 3blm). Deuterium uptake is labelled as the color bar indicated.

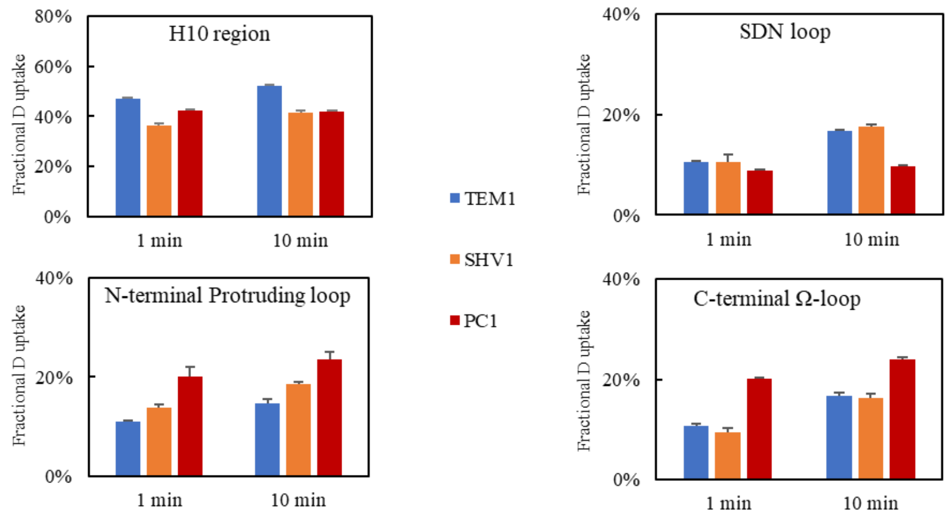

Figure 3.8 Fractional deuterium uptake of H10 region, SDN loop, N-terminal Protruding loop and C-terminal Ω-loop.

The deuterium uptake of the N-terminus of protruding loop (residues 92-104) for TEM1 was significantly less than that for SHV1 and PC1, reflecting a more ordered structure for TEM1 (Figure 3.8). By contrast, this loop on PC1 was the most flexible. It should be noted that the residue at position 104 for TEM1, SHV1, and PC1 is Glu, Asp, and Ala, respectively, with a decreasing size of side chain. Such variation might explain the difference in flexibility of this loop. Furthermore, it has been reported that the protruding loop comprises of the major binding interface with BLIP,[65] thus differentiated flexibility of this loop could be important for the accommodation onto the binding interface.

Interestingly, the HDX profiles of SDN loop and C-terminus of Ω loop for TEM1 and SHV1 were similar while those for PC1 were different from TEM1 and SHV1 (Figure 3.8). This is consistent with the similar substrate specificities for TEM1 and SHV1. In contrast, the results indicated that these loop regions on PC1 show different conformational dynamics from those on TEM1 and SHV1. Such observation supported that PC1 exhibited different substrate specificity from TEM1 and SHV1.[188-189] The difference might also contribute to the varying inhibition potencies by BLIP since the Ω loop plays an important role in the binding pocket of class A β-lactamases.

3.2.3 Cooperative binding interactions between the TEM1 β-lactamase and BLIP.

Briefly, we found that the protein was considerably protected from deuterium exchange after binding with BLIP (Figure 3.9). The protruding loop-covering fragments 82-101 and 101-118 from TEM1 showed around 15% decrement in deuterium uptake. Such obvious change strongly supported that this loop could be the predominant binding interface with BLIP, which has been proven by the crystal structure of TEM1-BLIP complex.[65] In this structure (PDB ID: 1jtg), the protruding loop of TEM1 and the concave sheet of the inhibitor form a concave/convex interface in the complex. Our study supported that this binding interface includes the protruding loop, and further analyzed the changes in dynamics at the residue level by comparing the overlapping fragments.[190] By subtracting the deuteration of these overlapped fragments, it was shown that residues 102-105 on TEM1 were unprotected from deuterium exchange in the free state while Val103, Glu104, and Tyr105 were under protection in the bound state (Figure 3.10). This protection corresponded to the hydrogen bonding network between TEM1 and BLIP as calculated from the crystal structure.[65] The protection of Val103 might be due to the reduced solvent accessibility since this residue interacted with Trp112 and Trp162 (both from BLIP) via van der Waals forces. By contrast, Glu104 and Tyr105 were protected mainly because of both the hydrogen bonding and van der Waals interaction with residues from BLIP. Hence, the protection of Glu104 and Tyr105 was more considerable than that of Val103. Previous studies revealed that a single mutation Asp104Glu for SHV1 could result in a 1000-fold enhancement in binding affinity with BLIP, indicating the important role of Glu104 in this inhibitory interaction. This enhancement was proposed to be explained by the restoration of a salt-bridge between Glu104 and BLIP Lys74.[178] Tyr105 was also investigated by mutagenesis, which showed that substitution of this position could cause inhibitor resistance due to oxyanion hole distortion.[191] Our observations from MD simulation showed that reduced fluctuation was observed for Glu104 and

Tyr105 but not for Leu102 and Val103 after the inhibition (Figure 3.10). Taking these together, our results revealed the important role of the dynamics of Glu104 and Tyr105 in the inhibitory binding.

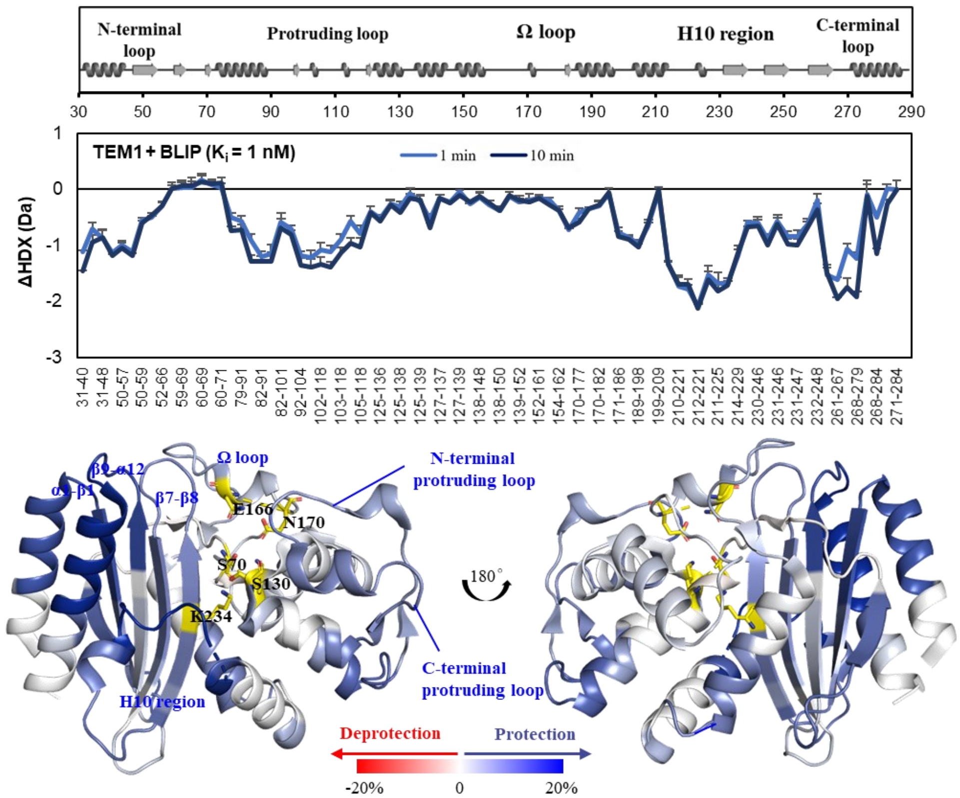

Figure 3.9 Dynamics change of TEM1 (PDB ID: 1btl) after inhibited by BLIP. HDX difference for all identified peptides from TEM1. Key secondary structures are shown on the top. Error bars indicate standard deviations at 1 and 10 min (n=3). Summed fractional HDX difference of 1- and 10-min labelling was labelled onto the crystal model of TEM1 (PDB ID: 1BTL), active sites are shown as yellow stick.

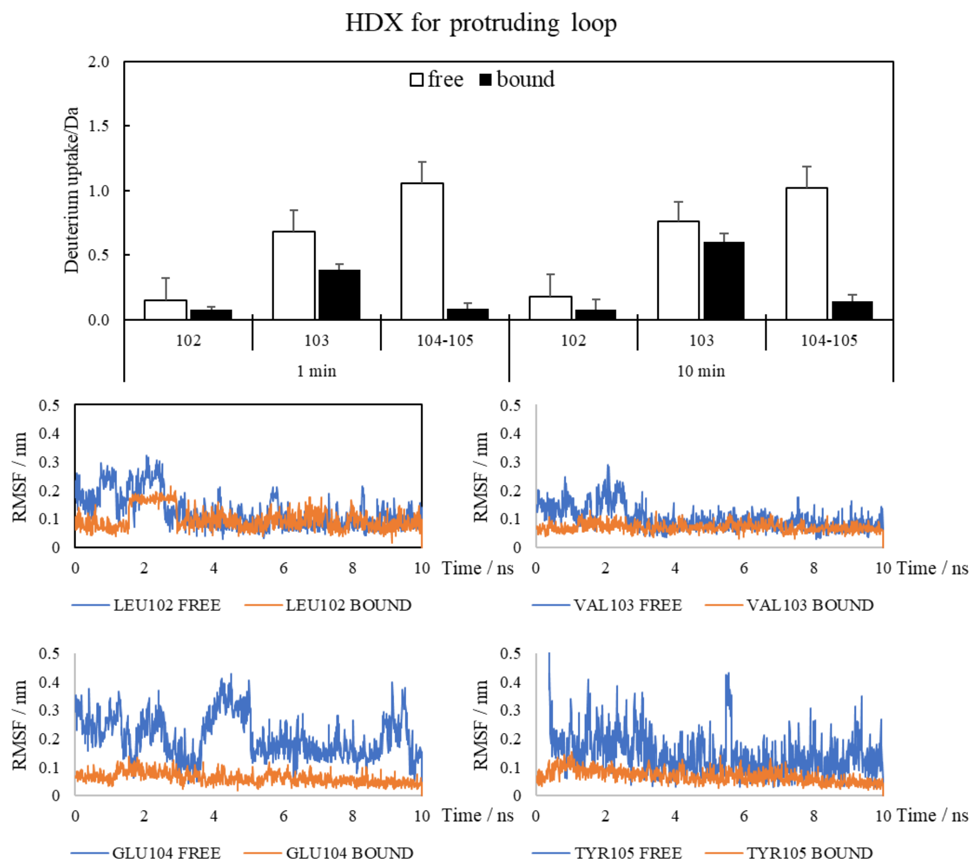

Figure 3.10 Residue-level dynamics for the protruding loop measured by HDX and MD simulation.

The H10 region, which was found to be the most flexible region in free TEM1, showed a remarkably reduction in deuterium uptake after the binding, indicating that this region was involved in the binding interaction (Figure 3.9). Fragment 214-221 showed more than 20% decrement for both 1 and 10-min exchange upon BLIP binding. This change was the most significant among all the identified fragments. Combining the exchange profile of fragments 210-221, 211-221, 212-221 and 214-221, we estimated the deuterium uptake for residue 211, 212 and 213-214, and found that residues 211-214 were not involved in the binding (Figure 3.11). In contrast, residues 215-221 were responsible for the inhibitory interaction. Trajectory analysis of these residues confirmed that Lys215, Val216, and Ala217 were less flexible upon the inhibition (Figure 3.11). Consistently, Lys215 was identified to form a hydrogen bond with Glu31 from BLIP in the crystal structure.[65, 177] This hydrogen bond stabilizes the whole loop and could account for the decreased exchange rate. Besides, Val216 contacts with Asp49 and Tyr50 in BLIP through van der Waals interactions, thus becoming less flexible after the binding. Since Ala217 has no contacts with any residues from BLIP, the protection of this residue is probably due to the conformational change induced by the neighbourhood. In contrast to our results, the crystal structure of the TEM1-BLIP complex (PDB ID: 1jtg) showed that TEM1 residues 214-217 (within the H10 region) are disordered compared to that of TEM1 (PDB ID: 1btl). We further investigated two newer crystal structures of TEM1 with BLIP-Y51A (PDB ID: 3c7u) and BLIP-Y150A (PDB ID: 3c7v). They both showed relatively low B-factors (Figure 3.12) and excellent electron densities in the H10 region.[177] The disorder observed in the TEM1-BLIP complex (PDB ID: 1jtg) was therefore more likely to be an experimental artefact under that particular crystallization conditions. Compared to B-factor that indicates protein dynamics in crystal state,[192] HDX-MS used in this study reflects protein dynamics in solution state and thus can reveal dynamics information that might not be obtained by using B-factors.

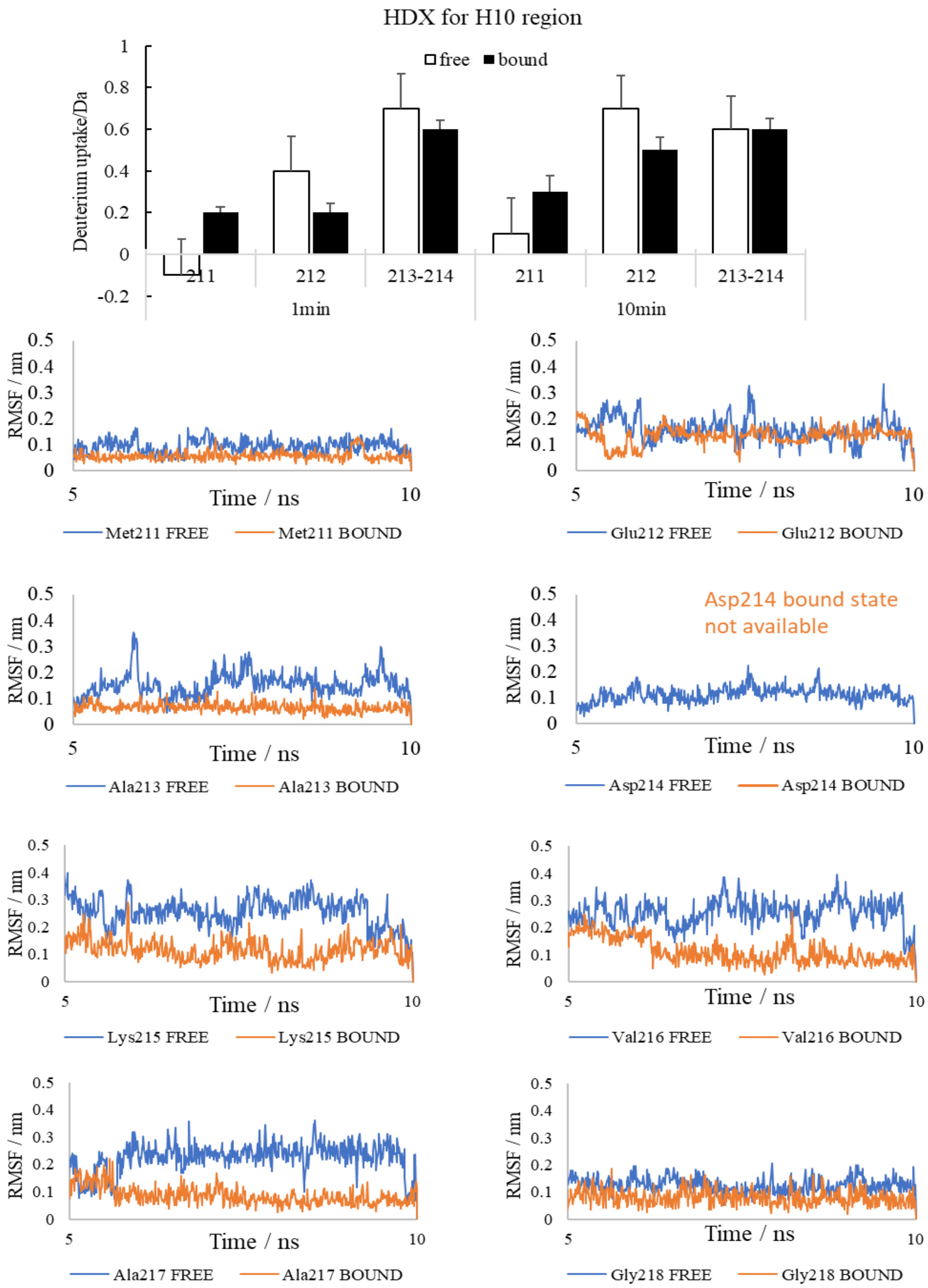

Figure 3.11 Residue-level dynamics for the H10 region by HDX and MD simulation.

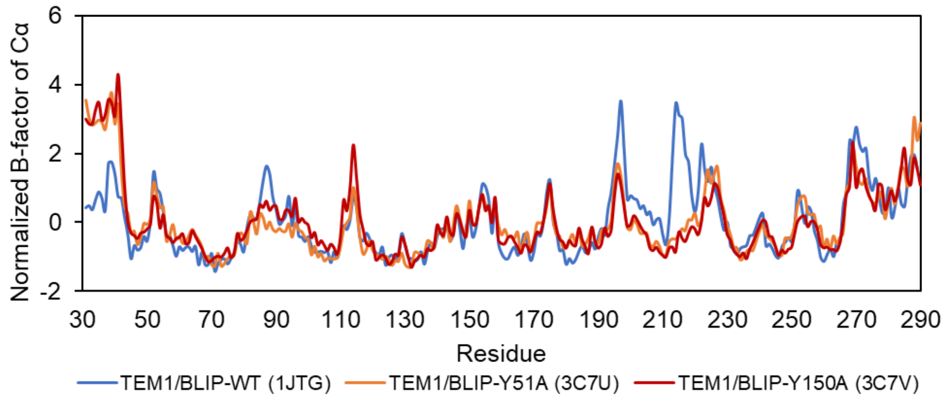

Figure 3.12 Normalized B-factors of α-carbon for the bound states of TEM1 with BLIPs. The PDB IDs for obtaining the B-factors are shown in the parentheses.

Another considerable difference was shown for residues 230-246, which covers the β7-β8 loop adjacent to the H10 region. This motif comprises part of the active cavity that interacts with the β-hairpin loop in BLIP. Three residues on this motif, i.e., Lys234, Ser235 and Arg243, forms essential hydrogen bonds with Asp49 in BLIP,[65] thus these residues could be more protected from deuterium exchange. Interestingly, we identified residues 170-177 (C-terminus of Ω loop), which showed slower exchange after formation of the complex, but there was no hydrogen bond formed in this region. This helical loop is quite flexible in the free state and is more rigid in the bound state as indicated by MD simulation (Figure 3.21). By comparison of the crystal structures of the free and the bound forms (PDB IDs: 1btl and 1jtg), it was found that the helix structure is extended in this loop after the binding, which would protect the amide hydrogen from exchange.

Regarding the catalytic sites (Ser70, Ser130, and Glu166), a fragment (residues 127-137) covering the SDN loop was found to show decreased HDX. The involvement of the Ser130 in this loop in the binding of substrate has been reported.[193] More importantly, hydrogen bonding between Ser130 (TEM1) and Asp49 (BLIP) indicated that the reduced flexibility of this loop could improve the inhibition potency of BLIP.[65] A fragment (residues 163-169) located at the N-terminus of end of the Ω loop covering Glu166 showed decreased HDX in 1 min, suggesting minor change in dynamics at this region. In contrast, several fragments (residues 58-71, 59-71, 60-71 and 60-72) covering Ser70 maintained low level of deuterium exchange upon binding, suggesting that this region was highly protected in both states of TEM1. The trajectory analysis of Ser70, Ser130 and Glu166 showed results that were consistent to HDX profiles of fragments covering these positions (Figure 3.13).

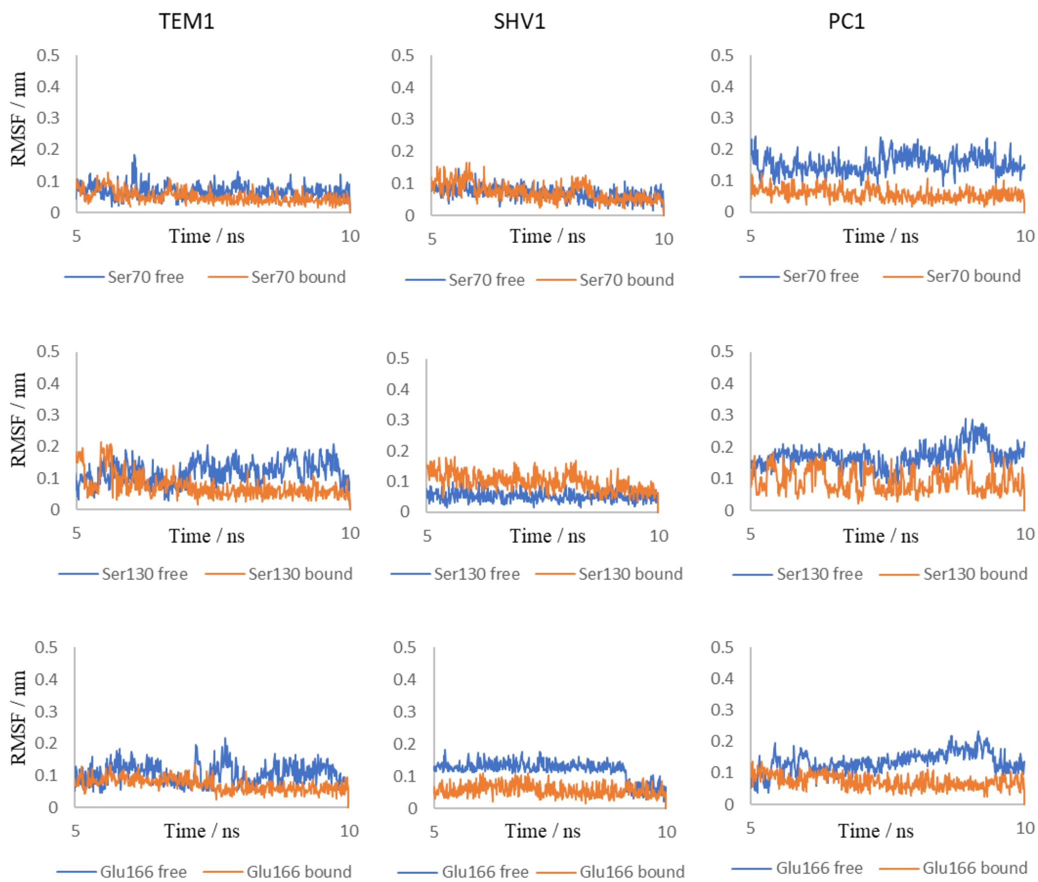

Figure 3.13 Residue-level RMSF (nm) for the catalytic sites of TEM1 (left panel), SHV1 (middle panel) and PC1 (right panel) measured by MD simulation.

3.2.4 Comparison of conformational changes among class A β-lactamases upon inhibition.

Intriguingly, we detected an increased exchange at 10-min labelling for the N-terminus of protruding loop on both SHV1 and PC1 (Figure 3.14 & Figure 3.15), which could be partially involved in the binding interface from the point of view by crystallography.[175] Our result indicated that the residues 92-104, which comprise the N-terminus of the protruding loop close to the interface, became deprotected upon the binding. This strongly suggested that the conformational change of this loop occurred in an unfavourable manner for tight binding with BLIP. Such unfavoured interactions lead to the loss of a salt-bridge between SHV1 (Asp104) or PC1 (Ala104) and BLIP (Lys74), which is preserved in the complex of TEM1 and BLIP. Furthermore, it has been demonstrated that the single mutation of SHV1 (D104E) and PC1 (A104E) would significantly enhance the binding affinity with BLIP,[168, 178] highlighting the important role of residue at position 104 for effective inhibition. As mentioned before, this loop in TEM1 is more rigid than that in SHV1 and PC1. The conformational dynamics of this loop could be crucial to the inhibitory binding, indicating that rigid and convex shape of this loop in class A β-lactamase could dock better onto the concave interface in BLIP. Assumably, the preference of rigid interface reduced the entropic penalty of unfavourable conformational change, thus enhancing the inhibitory binding.

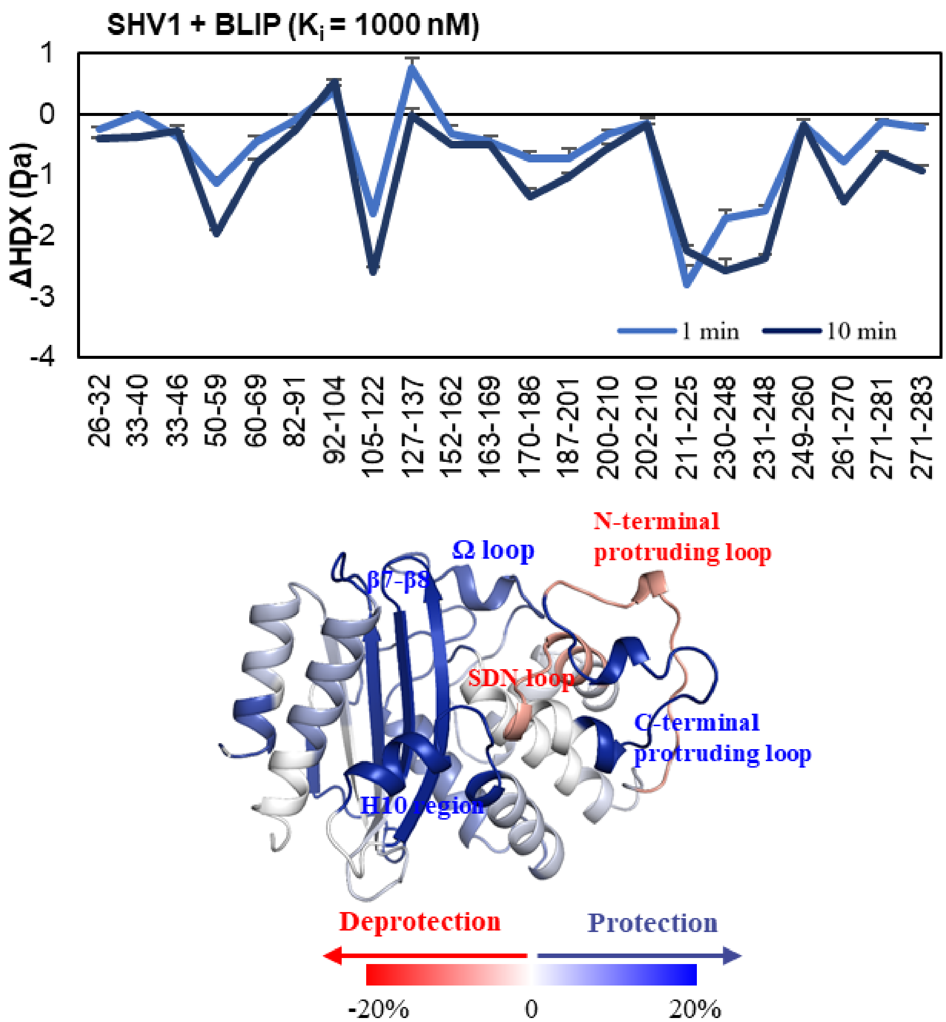

Figure 3.14 HDX-MS results of SHV1 after inhibited by BLIP.

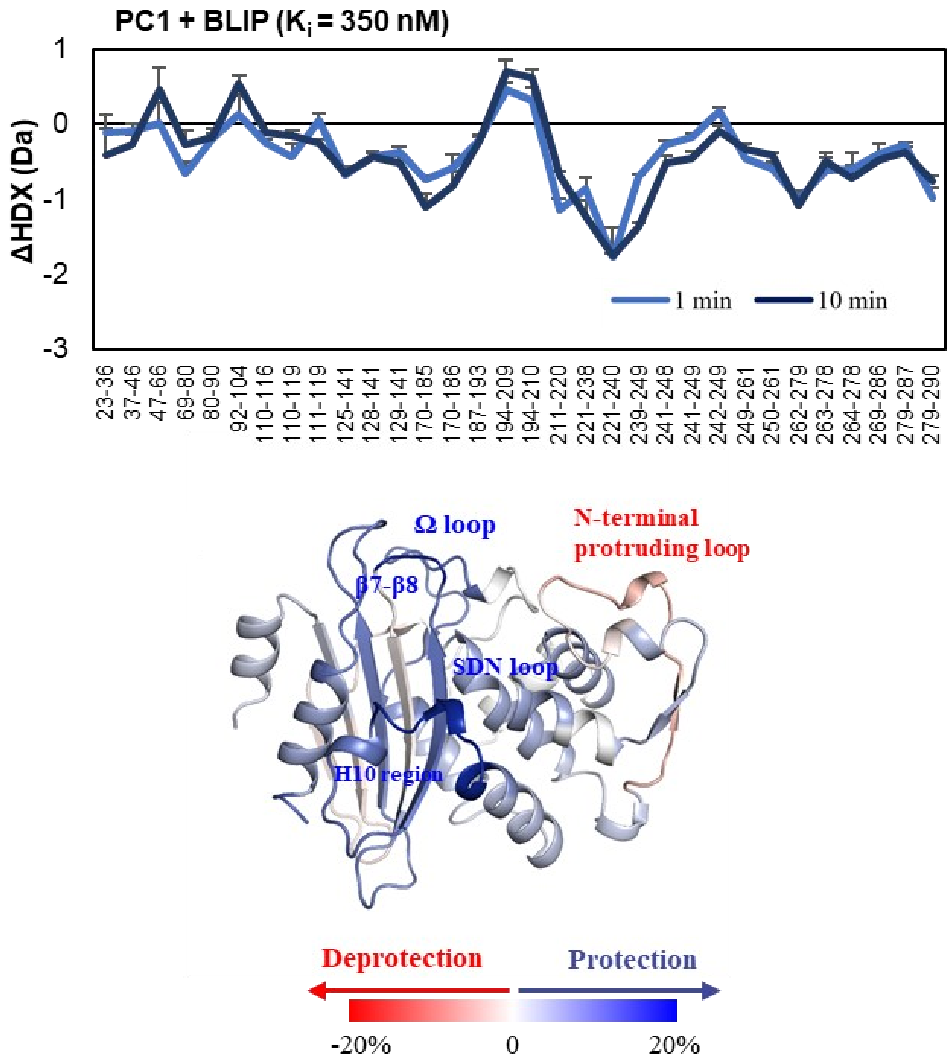

Figure 3.15 HDX-MS results of PC1 after inhibited by BLIP

H10 region in SHV1 (211-225) and PC1 (211-228) showed the most significantly decreased deuterium exchange as observed in TEM1, indicating that this region was stabilized in all the inhibitory interactions under investigation (Figure 3.9 & Figure 3.14 & Figure 3.15). Based on the crystal structures of SHV1 and BLIP complex (PDB IDs: 1shv and 2g2u), the H10 region was displaced to the core of active pocket in the complex.[175] Such conformational change was consistent with our observation. Besides, it has been reported that two novel inhibitors allosterically inactivated TEM1 through destabilization of the H10 region,[194] which was recently further proven to be an evolutionarily conserved allosteric site unique to class A β-lactamases.[195] Our results by HDX-MS revealed the conformational dynamics of this cryptic site among class A β-lactamases and the considerable changes induced by BLIP binding. A recent study on the dynamics of TEM1 upon substrate binding showed a similar HDX change in the H10 region during catalytic processes.[196] By contrast, our work involved study of different class A β-lactamases and the inhibitor protein, extending the observation to a broader and more general scope. From the 10-ns MD trajectory of these enzymes, H10 showed flapping motion to modulate the binding pocket. This strongly suggested that the flexibility of the H10 region could be determinant for the effective binding with BLIP. Higher flexibility of H10 on TEM1 might facilitate the insertion of BLIP into the binding pocket, causing higher binding affinity towards the inhibitor. Together, disrupting H10 region could be an alternative strategy to inactivate the class A β-lactamases.

As one of the highly conserved loops in class A β-lactamases, the SDN loop has been proven to be important for the structural stability and function by site-directed mutagenesis.[192] After the inhibitory binding, residues 127-137 which refer to the SDN loop in SHV1 showed increased deuterium uptake at 1 min of exchange (Figure 3.14). On the contrary, the SDN loop in TEM1 and PC1 showed both decreased exchange level at 1 and 10 min upon the inhibition. MD simulation of Ser130 showed that this region on SHV1 was more flexible after the binding while it was less flexible on TEM1 and PC1 (Figure 3.13). Besides, no difference in deuterium exchange was found for the SDN loops of TEM1 and SHV1 in their free states (Figure 3.8). Comparison of the sequences spanning residues 127-137 of SHV1 (ITMSDNTAANL) with TEM1 (ITMSDNSAANL) indicated that only the residue at position 133, a threonine for TEM1 and a serine for SHV1, is different. Since threonine has a bulkier sidechain than serine, it is reasonable that after the binding the SDN loop of TEM1 could be less flexible than that of SHV1. Furthermore, according to the crystal models of TEM1 and SHV1 (PDB IDs: 1btl and 1shv), both Thr133 and Ser133 have contacts with residues Val103 and Ser106 located on the protruding loop. However, due to the orientation of the side chain at position 133, the distance of the contact towards Ser106 in TEM1 (2.7Å) is shorter than that in SHV1 (3.1Å), indicating that TEM1 has a tighter structure than SHV1 in this region. As a result, this region was less protected after the inhibitory binding for SHV1 while the protection was enhanced for TEM1. Perturbation of SDN loop could be important to suppress the activity of the class A β-lactamases.

Specifically, PC1 shares lower sequence and structural similarity with TEM1 and SHV1, and the crystal model of the complex PC1/BLIP has not been available yet. We constructed the complex by homology modeling (see details in Methods) and equilibrated the structure by MD simulation. Our HDX data showed that the binding with BLIP induced some unique changes in PC1 covering the catalytic sites, Ser70 and Lys73. Decreased deuterium exchange was observed in fragment 69-80. Together with the results from the MD simulation, we confirmed that the fluctuation of Ser70 was obviously reduced after binding with BLIP (Figure 3.8).

3.2.5 Enhanced inhibition by BLIP mutants.

To further explore the role of protruding loop on the binding interface, we introduced two mutations at E73 and K74 on BLIP, which specifically couple with the residues 104-106 upon binding with SHV1 and PC1, respectively.[65] Strikingly, the inhibition towards SHV1 and PC1 could be significantly enhanced with K_i at nanomolar level by the single mutation of E73M and K74G on BLIP, respectively.[167-168] Intriguingly, our results showed that $BLIP_{E73M}$ and $BLIP_{K74G}$ reversed the deprotection effect on the N-terminal of the protruding loop on SHV1 and PC1 caused by wild-type BLIP, indicating a more stabilized binding interface of the complexes (Figure 3.16 & Figure 3.17). The stabilization of the protruding loop could majorly contribute to the enhanced inhibition potency of BLIP mutants towards SHV1 and PC1 β-lactamases. Previous studies reported that single mutation on the protruding loop at position 104 could also enhance the binding affinity of SHV1 and PC1 towards wild-type BLIP.[4,5] Together, it can be concluded that the interactions between the protruding loop and BLIP is crucial for novel design of BLIP derived inhibitors.

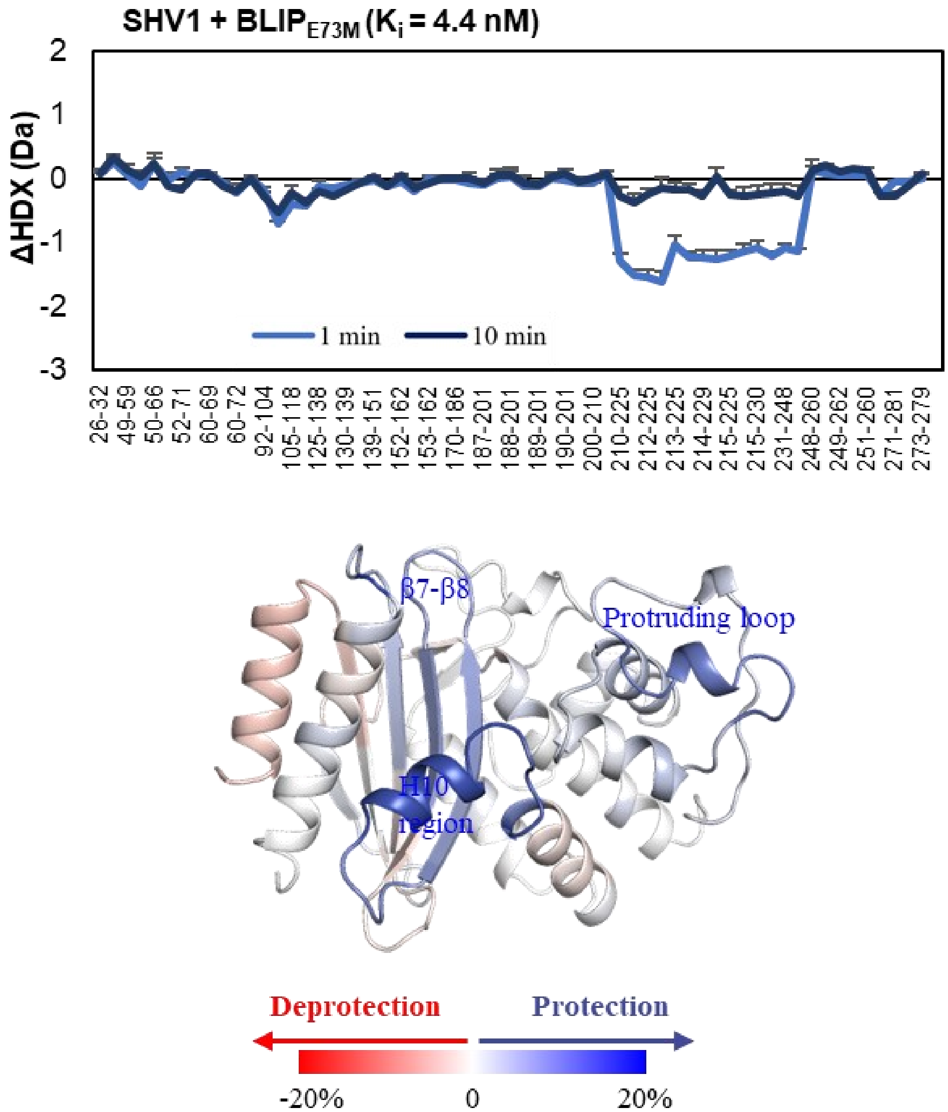

Figure 3.16 HDX-MS results of SHV1 after inhibited by $BLIP_{E73M}$

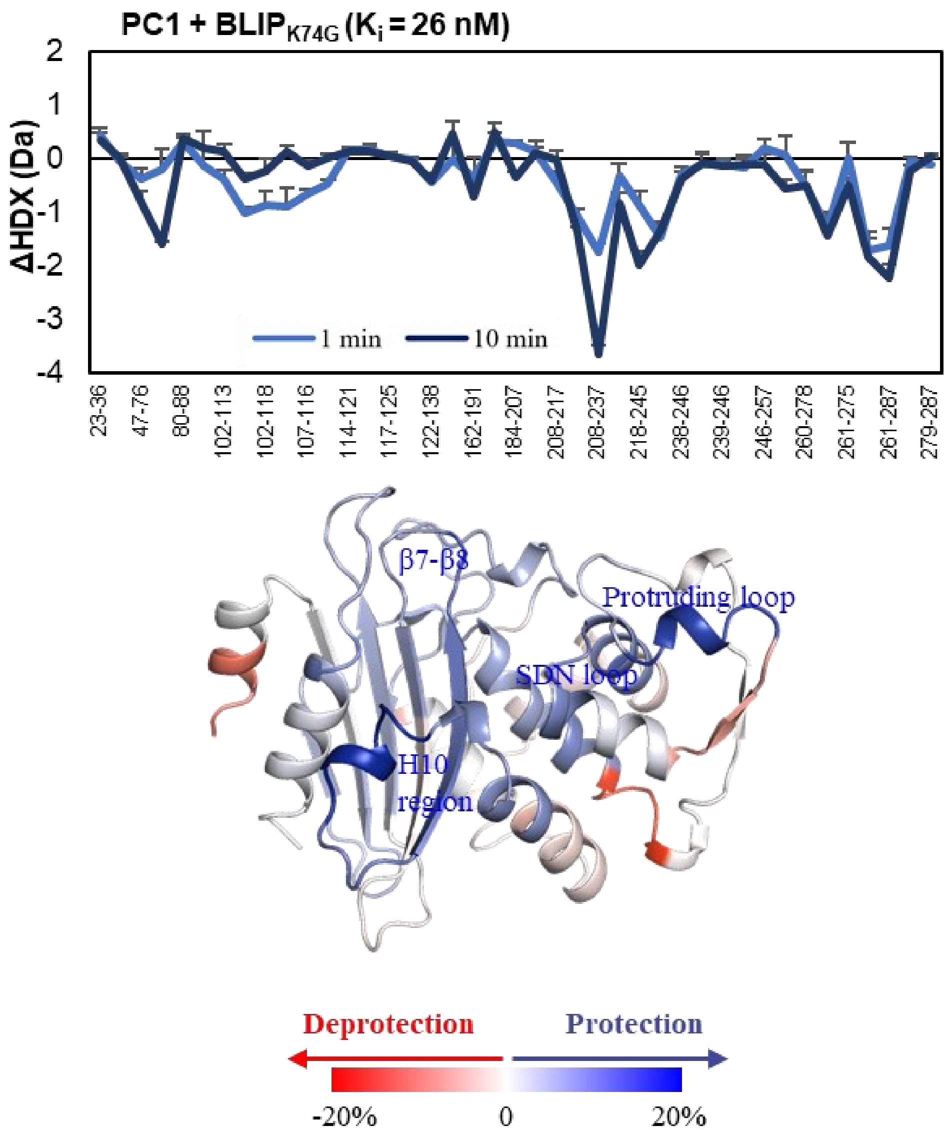

Figure 3.17 HDX-MS results of PC1 after inhibited by BLIP$_{K74G}$

BLIP$_{E73M}$ also eliminated the deprotection effect from wild-type BLIP towards the SDN region of SHV1, which could facilitate more effective inhibition

(Figure 3.16). In contrast, $BLIP_{K74G}$ showed more protection of residues 50-79 in PC1 including active sites Ser70 and Lys73, which could severely inactivate the function of these sites (Figure 3.17). Both BLIP mutants demonstrated predominant protection in the H10 region of SHV1 and PC1, similarly reflected in the inhibition by the wild-type BLIP. Such conformational changes indicated a common mechanism of the bindings between these β-lactamases and BLIP as discussed above.

Tyr50 on BLIP forms favorable interactions with several key regions on β-lactamases, including Pro107 on protruding loop, Met129 on SDN region and Val216 on H10 region.[65] Strangely, alanine substitution of this residue can further promote the potency towards TEM1 with K_i at picomolar level.[197] Although the Ala50 would lose several contacts with TEM1, presumably it could eliminate the colliding effect caused by the side-chain of Tyr50 and allow a more favorable conformation of Asp49.[197] More significant protections from HDX were observed in SDN loop and β7-β8 in TEM1 upon inhibition by $BLIP_{Y50A}$ (Figure 3.18), suggesting more stabilized conformations of these regions. Our results could support that Asp49 might show a more favorable conformation, which interacted more intensively with SDN loop (Ser130) and β7-β8 (Lys233, Ser234 and Arg244) in the complex.[65] This might be the major cause for the enhanced inhibition of TEM1.

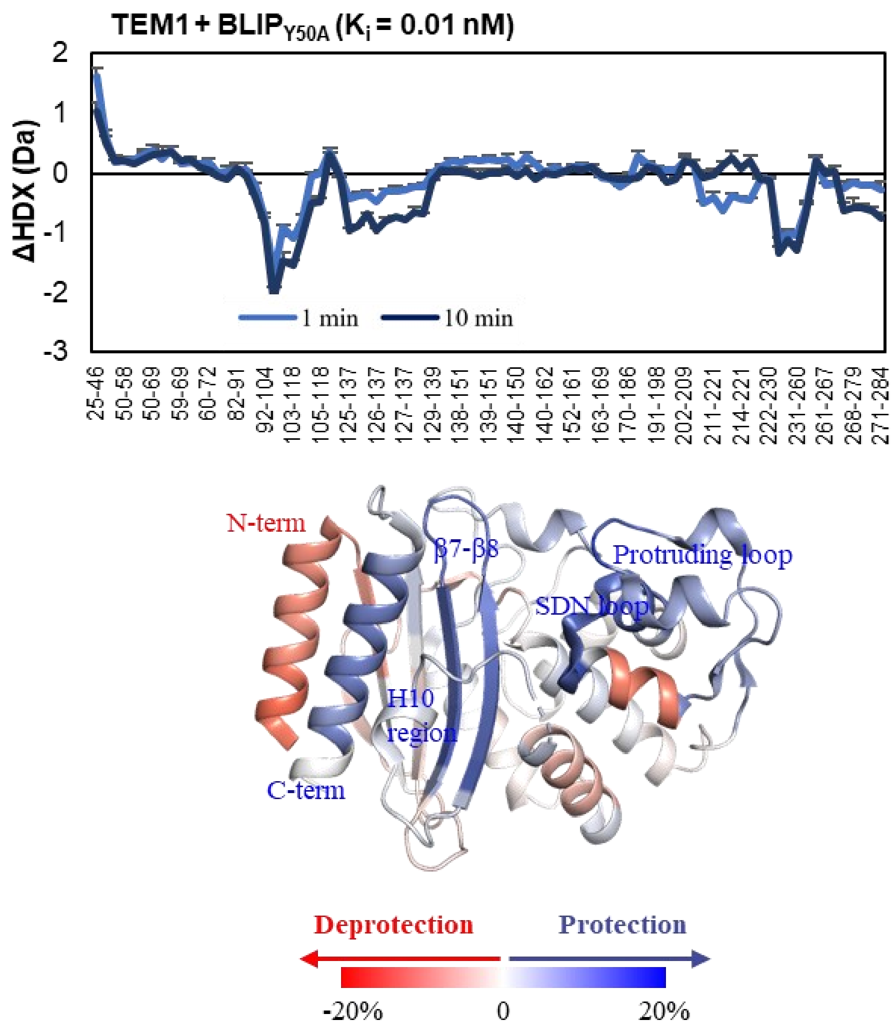

Figure 3.18 HDX-MS results of TEM1 after inhibited by $BLIP_{Y50A}$

Interestingly, obvious deprotection of the N-terminal helix, which is far away from the active binding domain, indicated that relaxation of this region might be favorable in the cases of strong inhibition (Figure 3.18). A recent study showed that the mutation Q39K on the N-terminal helix of TEM1 destabilized the protein, revealing that interactions involved in this region could significantly influence the binding and catalytic properties of the protein.[198] Such observation, which might not be obtained by other techniques, could support that deprotection of this region might suppress the activity of β-lactamases.

3.2.6 MD simulation revealed protein dynamics at complementary time scale to HDX.

Obviously, the deviation of TEM1 was reduced upon binding BLIP while it was maintained for SHV1 and PC1 (Figure 3.19). This result showed consistency with that obtained with global HDX (Figure 3.5). Our results confirmed that the global stabilization of TEM1 upon binding was more obvious than that of the others during the period of simulation. It should be noted that the simulation was examined at ns time scale, and thus it might not necessarily be consistent but complementary to some HDX results that reflected protein dynamics at μs to ms scale.[147] However, integration of these two techniques enabled us to dig into some unique dynamic features of the proteins.

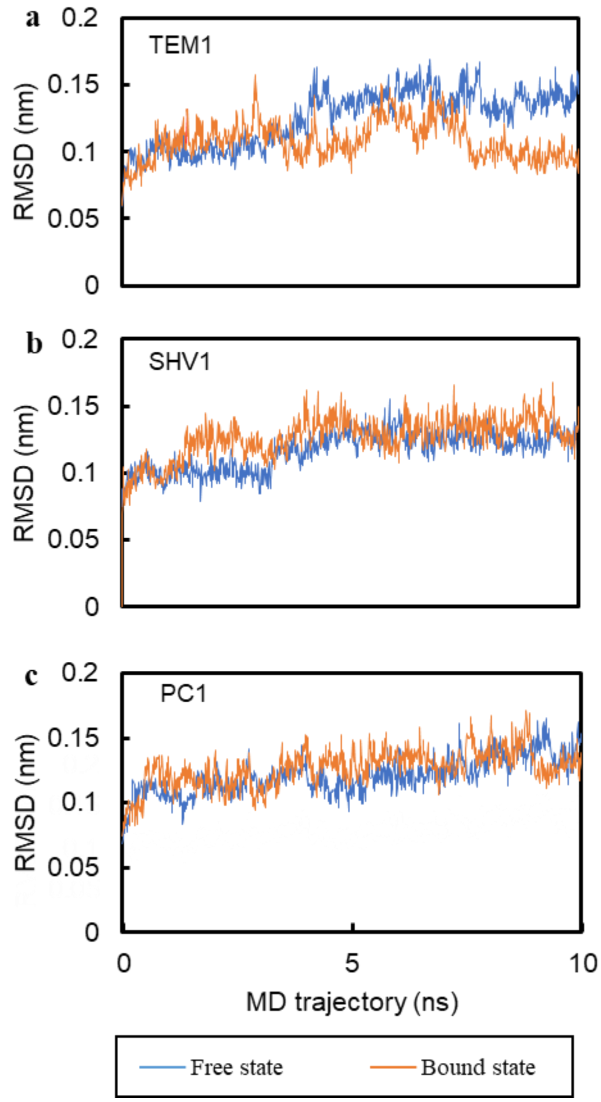

Figure 3.19 Root mean square deviation (RMSD) as a function of time (ns) for the free (blue) and BLIP-bound (orange) (a)TEM1, (b)SHV1 and (c)PC1 β-lactamases.

The last 5-ns simulation showed that most regions across these β-lactamases had similar fluctuations, strongly indicating similar folding of the proteins. Interestingly, variation was found in the Ω loop, where TEM1 showed the much higher fluctuation than PC1 and SHV1 (Figure 3.20). This result was complementary to the result obtained by HDX-MS, which showed that this loop in PC1 was more flexible than that in TEM1 and SHV1 (Figure 3.8). This might be explained by the different timescales of the loop motion reflected by the MD simulation and HDX. A previous study by NMR demonstrated that the Ω loop of TEM1 was highly ordered on the ps-to-ns time scale while slow motions were observed on the μs-to-ms time scale,[161, 179] supporting our assumption. Such dynamic duality plays an important role in the substrate binding and catalytic reaction. Taking advantage of our results from MD simulation and HDX-MS, we infer that the fast motion of Ω loop on TEM1 is faster than that on SHV1 and PC1, and in contrast, the slow motion of this loop on PC1 is faster than that on TEM1 and SHV1.

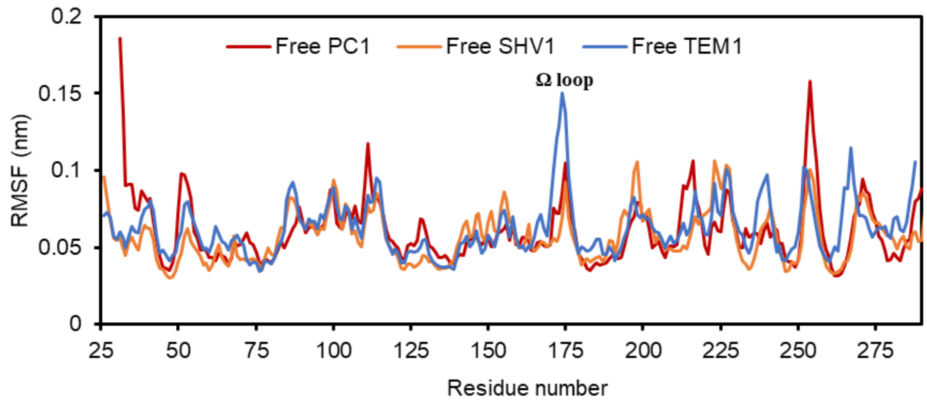

Figure 3.20 Root mean square fluctuation (RMSF) of individual residue of free β-lactamases during 5-10 ns simulation.

More interestingly, significant protection from HDX was detected in the C-terminus of Ω loop on all these class A β-lactamases after bound with BLIP (Figure 3.9 & Figure 3.14 & Figure 3.15), indicating that the slow motion of this loop was suppressed. By contrast, the fluctuation of this loop was escalated on SHV1 and PC1 while it was attenuated on TEM1 (Figure 3.21). This observation suggested that the loop might be catalytically more active on SHV1 and PC1 than on TEM1 upon bound with BLIP. Such difference could explain the lower inhibition potency of BLIP towards SHV1 and PC1. Together, it was strongly suggested that the motion of Ω loop was affected in both fast and slow fashions upon the inhibition.

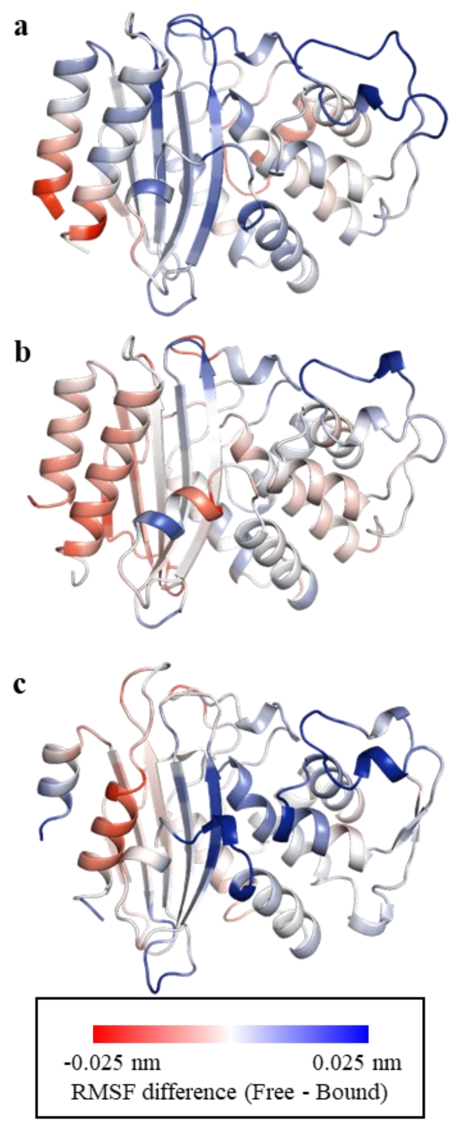

Figure 3.21 Difference in root mean square fluctuation (RMSF) of individual residue upon BLIP binding during 5-10 ns simulation was mapped onto the crystal model of (a)TEM1, (b)SHV1 and (c)PC1.

3.3 Summary

Although class A β-lactamases are highly similar with respect to their sequences and structures, our results revealed that SHV1 exhibited the most compact and rigid conformation, while PC1 was most extended and flexible. After inhibition by BLIP, TEM1 was globally more rigid while SHV1 and PC1 showed no significant change in conformational dynamics. The protruding loop on TEM1 showed remarkable change in conformational dynamics upon BLIP binding. However, the N-terminus of the protruding loop on SHV1 and PC1 were both deprotected from HDX upon the binding, indicating the unfavorable interactions in this region. BLIP mutants, $BLIP_{E73M}$ and $BLIP_{K74G}$, reversed this deprotecting effects on SHV1 and PC1 and stabilized the interfaces of these bindings. Rigid conformation of this interface might be favored to reduce the entropic penalty of the binding. Unexpectedly, marked conformational changes in the H10 region were observed upon the inhibition. Although this region is far away from the binding interface, it could modulate the plasticity of the binding pocket and enhance the inhibition of the enzymes. Conformational dynamics of several functional loops, i.e., SDN loop, β7-β8 loop and Ω loop, provided novel and unique insights into the mechanistic process of these inhibitory interactions. Cooperative interaction of these loops with BLIP could improve the inhibition potency of BLIP towards the class A β-lactamases.

Potent inhibitors such as engineered BLIPs can be rationally designed to bind β-lactamases by disrupting the flexibility of the loop regions, especially H10 region and SDN loop. To conclude, our study is the first to provide comprehensive insights into the conformational dynamics of class A β-lactamases upon inhibitory binding with BLIP, which implies alternative and unique strategies for the study of β-lactamases and the design of novel inhibitors.

4. The Role of Interdomain Flexibility of BLIP

4.1 Introduction

BLIP naturally produced by *Streptomyces clavuligerus* is a protein consisting of a tandem repeat of αβ domains conjugated by a flexible interdomain loop (Figure 4.1).[172] It can effectively bind to several types of class A β-lactamases, including *Escherichia coli* TEM1, *Klebsiella pneumoniae* SHV1 and *Staphylococcus aureus* PC1. However, the inhibition potency of BLIP towards different β-lactamases varies, with measured K_i values in subnanomolar to micromolar range. BLIP can bind TEM1 around 1000-fold tighter than SHV1 and PC1. The crystal structures of BLIP/TEM1 (PDB ID: 1jtg) and BLIP/SHV1 (PDB ID: 2g2u) exhibit high structural homology, with an all C_α root mean square deviation (RMSD) of 0.6 Å, and the bound BLIPs show a C_α RMSD of 1.0 Å (Figure 4.2).[175] Such structural similarity could hardly reveal the mechanism underlying different binding affinities and specificities between BLIP and β-lactamases.

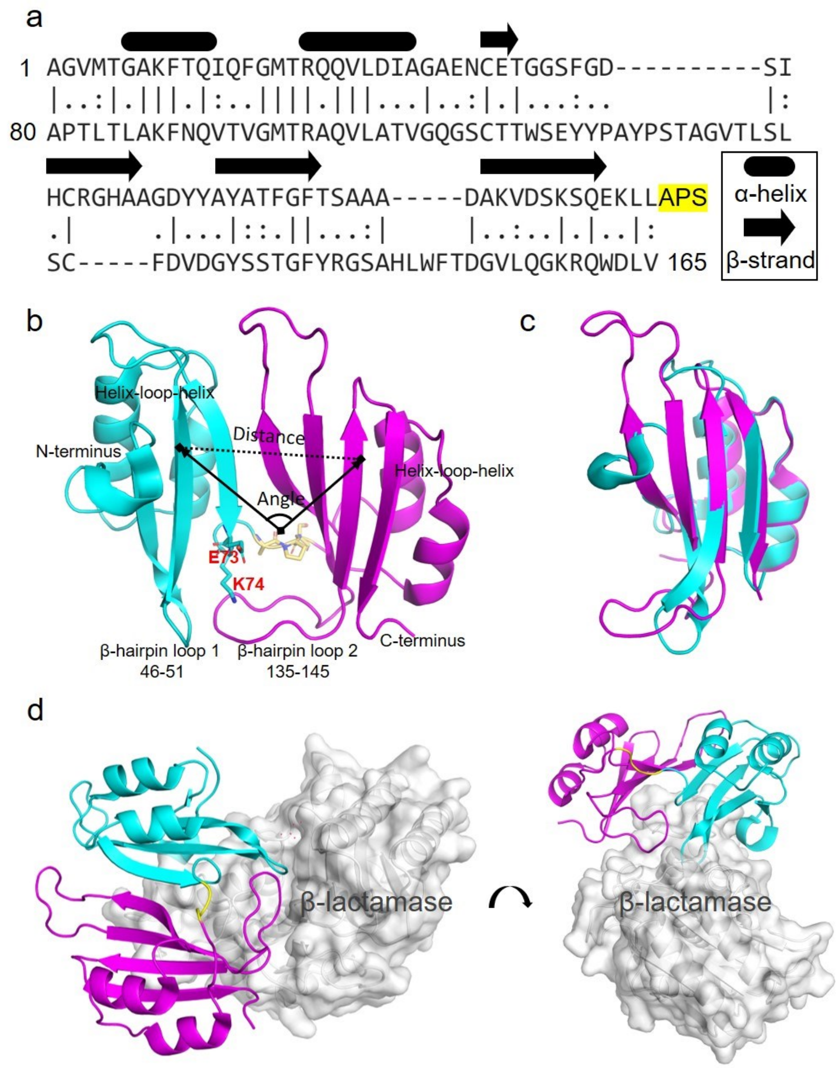

Figure 4.1 Sequence and structure of BLIP.
(a) Sequence alignment of the N-terminal (1-76) and C-terminal (80-176) domains. Secondary structures are indicated above the sequence and the three-residue linker is highlighted in yellow. (b) Full-length crystal structure of BLIP (PDB ID: 3gmu). The three-residue linker at the interdomain loop is labelled as sticks in cyan. The distance and angle between the N-terminal (in cyan) and the C-terminal (in magenta) domains are indicated. (c) The structural alignment of the N-terminal (in cyan) and C-terminal (in magenta) domains. (d) Representative crystal structure of BLIP and TEM1 β-lactamase (PDB ID: 1jtg).

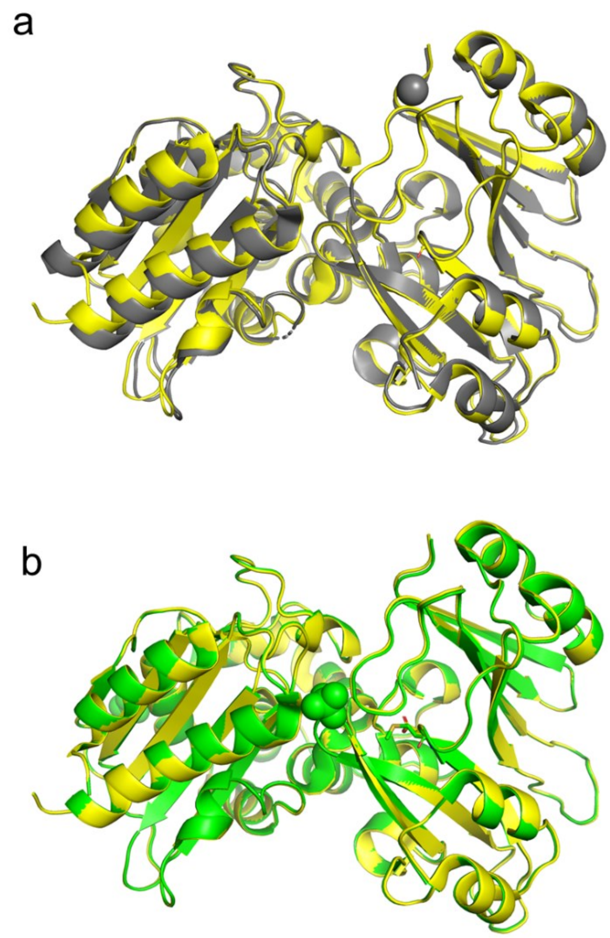

Figure 4.2 Structural alignment of the complexes of BLIP/SHV1 (PDB ID: 2g2u, yellow) with (a) BLIP/TEM1 (PDB ID: 1jtg, grey), and with (b) BLIP-E73M/SHV1 (PDB ID: 3c4p, green).

Researchers have attempted to enhance the inhibitory activities of BLIP towards these β-lactamases using phage display,[168] computational design program[167, 178] and site-directed mutagenesis.[197, 199] These studies strongly suggested BLIP as a potential template for inhibitor design. Reynolds et al redesigned the interface of BLIP and SHV1 in an attempt to enhance the inhibitory binding.[167] A point mutation at E73M of BLIP improved the binding affinity to 1000-fold higher (with $K_d \approx 28$ nM) than that of the wild-type BLIP with SHV1 (with $K_d \approx 1800$ nM). Such improvement was hypotesized to be due to the restoration of a salt bridge between SHV1 (D104) and BLIP (K74), which was present in TEM1 (E104) and BLIP (K74).[167] Hanes et al performed alanine mutagenesis on K74 (BLIP) and D104 (SHV1) based on E73M (BLIP) to eliminate the salt bridge formation. BLIP-E73M/SHV1-D104A and BLIP-E73M-K74A/SHV1-D104A showed significantly improved binding affinities (with $K_d \approx 48$ nM and 130 nM, respectively) compared with that of BLIP/SHV1 (with $K_d \approx 1800$ nM).[178] This study suggested that the role of the D104 (SHV1)-K74 (BLIP) salt bridge could be important, and that other unknown factors that could account for the enhanced inhibitory binding. The elimination of an electrostatic repulsion between SHV1 D104 and BLIP E73 at the interface was one possible reason.[178] Structural alignment of SHV1-BLIP (PDB ID: 2g2u) and SHV1/BLIP-E73M (PDB ID: 3c4p) illustrated no major changes in the overall structures of BLIP (Figure 4.2).[167] Yuan et al have identified a potent BLIP variant, K74G, that inhibits PC1 β-lactamase with binding affinity (with $K_d \approx 26$ nM) increased by 10 folds compared with the wild-type BLIP (with $K_d \approx 380$ nM).[168] Alanine at position 104 on PC1 differs from aspartate on SHV1 and glutamate on TEM1, and lacks the ability to form any salt bridges with K74 from BLIP. The mutant

A104E of PC1 binds 15-fold more tightly with BLIP presumably by restoring the salt bridge between E104 and K74. However, BLIP variant K74G also lacks the ability to form a salt bridge with A104 from PC1 while the inhibitory binding between BLIP-K74G and PC1 (with $K_d \approx 26$ nM) is significantly enhanced to an equivalent level with that between BLIP and PC1-A104E (with $K_d \approx 20$ nM). The underlying contribution of this mutation is still unclear. Charge matching of residue 104 (PC1) and residue 74 (BLIP) dissected by the thermodynamic cycle showed that this pair of residues with either opposite charges or nonpolar sidechains could be favourable for the bindings between PC1 and BLIPs.[168] Unfortunately, no crystal structures of the complexes of PC1 and BLIPs has been available yet.

It was proposed that the ability of BLIP to adapt to various class A β-lactamases is probably due to the flexibility between the two domains of BLIP.[65] However, little work has been done on the role of flexibility or conformational dynamics of BLIP upon binding β-lactamases. Following our previous chapter on β-lactamases regarding their binding with BLIP, in this chapter, we focused on the conformational dynamics of BLIP and its mutants upon binding class A β-lactamases using hydrogen deuterium exchange mass spectrometry (HDX-MS) complemented with molecular dynamics (MD) simulation.[181-182, 200-201] Analysis of the molecular dynamics produced by longer time scale simulations (i.e., μs to ms) can examine the conformational changes under equilibrium and dynamic conditions.[154, 202] MD simulation combined with HDX-MS have been demonstrated to provide complementary information about protein flexibility.[147, 152, 156, 203] In this study, these methods together were employed to investigate the conformational dynamics of BLIP and its mutants upon their binding to β-lactamases TEM1, SHV1 and PC1. The results have enabled us to draw a dynamic picture of BLIP at different time scales and provide unique insights into its inhibitory binding with β-lactamases.

4.2 Results and discussion

4.2.1 Interdomain region tended to be more flexible in the BLIP mutants.

A characteristic property of BLIP is that this protein folds with a repeat of a 76 amino-acid αβ domain which is bridged by three interdomain residues 77-79 (Figure 4.1).[172] The fragments 74-85 spanning the interdomain region of wild-type BLIP were identified after pepsin digestion and their HDX profiles were compared upon binding to three class A β-lactamases (Figure 4.3). Our results showed that the protection of this region was significant upon binding to TEM1 while it was not obvious upon binding to SHV1 or PC1. Our observation suggested that the interdomain region of BLIP was significantly less flexible upon binding to TEM1 β-lactamases while the flexibility of this region remained unchanged upon the relatively weak bindings to SHV1 and PC1. Structural alignment of the crystallized complexes TEM1/BLIP (PDB ID: 1jtg) and SHV1/BLIP (PDB ID: 2g2u) illustrated that the backbones are nearly identical in these regions (Figure 4.2). Thus, it was of great interest to observe these differentiated conformational dynamics between these topologically similar bindings. Such changes supported the presence of a salt bridge between K74

(BLIP) and E104 (TEM1) that is absent between K74 (BLIP) and D104 (SHV1) or A104 (PC1).[167-168]

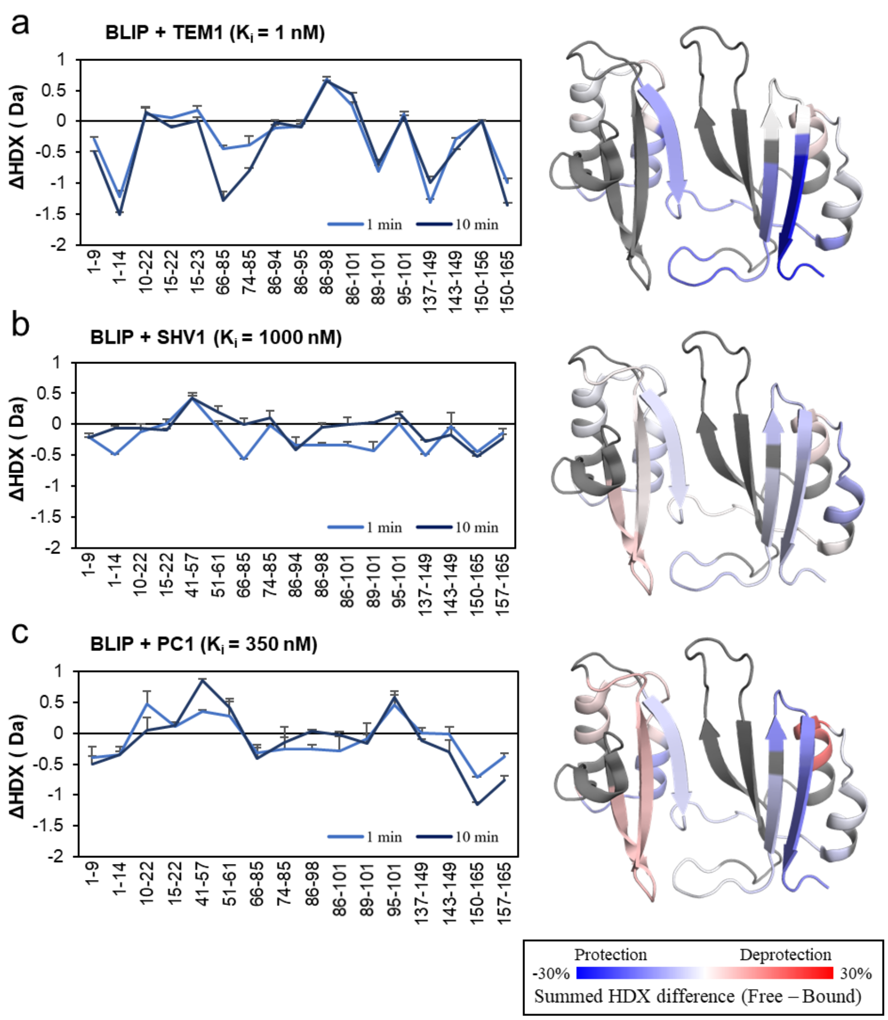

Figure 4.3 HDX-MS profiles of BLIP upon binding β-lactamases
(a) TEM1, (b) SHV1 and (c) PC1. ΔHDX represents the difference between the free and bound states. Positive values reflect deprotection from HDX while negative values reflect protection. Error bars indicate standard deviations for the time points 1 and 10 min (n=3). ΔHDX values larger than 0.19 Da, 0.17 Da and 0.29 Da were considered as significant changes upon binding TEM1, SHV1 and PC1, respectively. The summed fractional differences in 1- and 10-min labelling are mapped onto the crystal model of BLIP (PDB ID: 3gmu) for better visualization.

To investigate the roles of the interdomain region and the salt bridge, a BLIP mutant E73M with the ability to form a salt bridge between K74 (BLIP-E73M) and D104 was examined by HDX-MS in the free state and the results were compared with wild-type BLIP (Figure 4.4). The normalized HDX level of fragment 66-85 was significantly higher after the mutation (Figure 4.4), indicating that the BLIP E73M mutant has a more flexible interdomain linker than the wild-type BLIP.. Moreover, the interdomain region (residues 74-85) was more protected from HDX after binding to SHV1 (Figure 4.3 vs. Figure 4.5). Such observation might be explained by the restoration of the salt bridge between K74 (BLIP-E73M) and D104 (SHV1), which could reduce the flexibility of the interdomain loop. However, the 1-μs MD trajectory of the complex SHV1/BLIP-E73M showed that the restored salt bridge in SHV1/BLIP-E73M was broken after 58 ns. Simultaneously, an intramolecular hydrogen bond between K74 and S138 was formed. These results were not visible in the crystal structure while reflected the dynamics of the interactions close to the binding interface. It was suggested that the enhancement of the binding might be due to not only the intermolecular salt bridge but also the integrity between the interdomain region and the large interfacial loop on the BLIP side. Our finding demonstrated that the enhanced binding affinity is not only contributed by the interactions between the binding partners but also stabilization of either side.

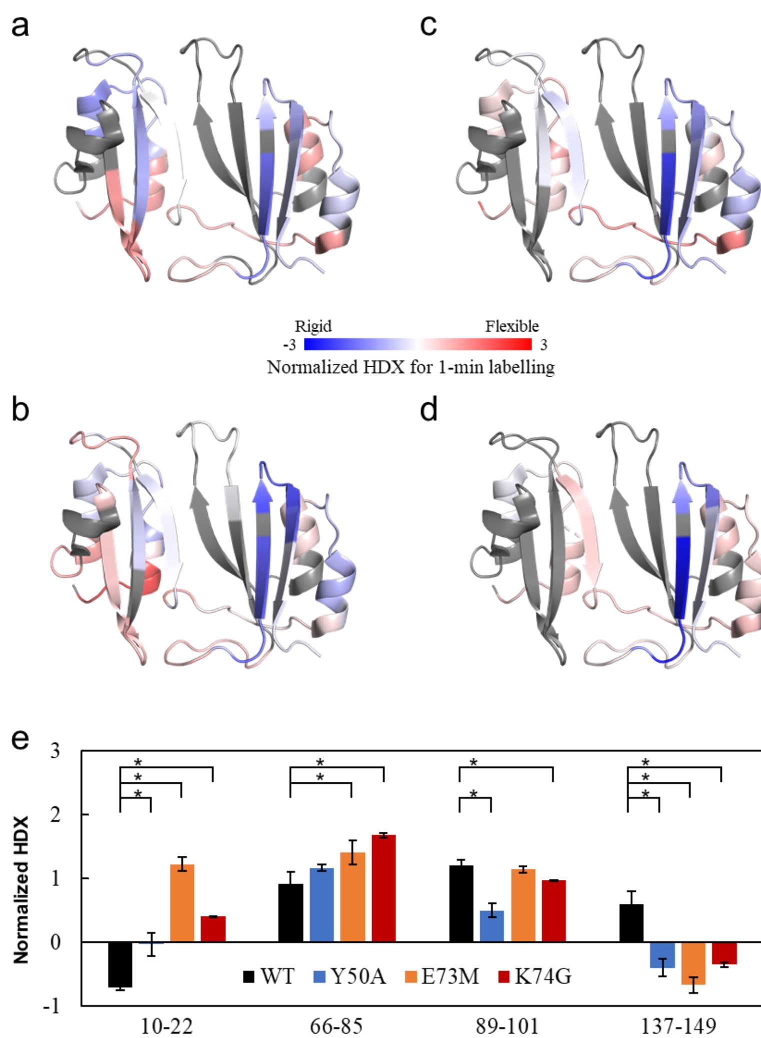

Figure 4.4 HDX-MS profiles of unbound BLIP and its mutants.
Normalized deuterium uptakes of (a) BLIP-WT, (b) BLIP-Y50A, (c) BLIP-E73M and (d) BLIP-K74G 1-min labelling are mapped onto the crystal structure of BLIP (PDB ID: 3gmu) for visualization. (e) The normalized HDX of some key regions, including helix-loop-helix motif from the N-terminal domain (10-22), interdomain region (66-85), helix-loop-helix motif (89-101) and the β-hairpin loop (137-149) from the C-terminal domain, are compared before and after the mutations. Error bars indicate standard deviations for the time points 1 min (n=3). Asterisks indicate p-value < 0.01 with one-tailed student's t-test.

To further validate the role of the interdomain region, another mutation BLIP K74G that could enhance the binding between BLIP and PC1 was investigated.[168] The HDX results for K74G suggested that the flexibility of fragment 66-85 was significantly increased compared to that of the wild type (Figure 4.4). Moreover, the interdomain region of BLIP-K74G showed significant protection from HDX after binding with PC1, indicating reduced interdomain flexibility compared with that of the wild-type complex (Figure 4.3 vs. Figure 4.5). In contrast to BLIP-E73M, K74G is disabled to form the salt bridge between G74 (BLIP) and A104 (PC1). Thus, the enhanced inhibition must be caused by other factors, i.e., the reduced side chain of residue 74 could allow better adaption of the interdomain region in the binding interface. However, details have not been revealed yet.

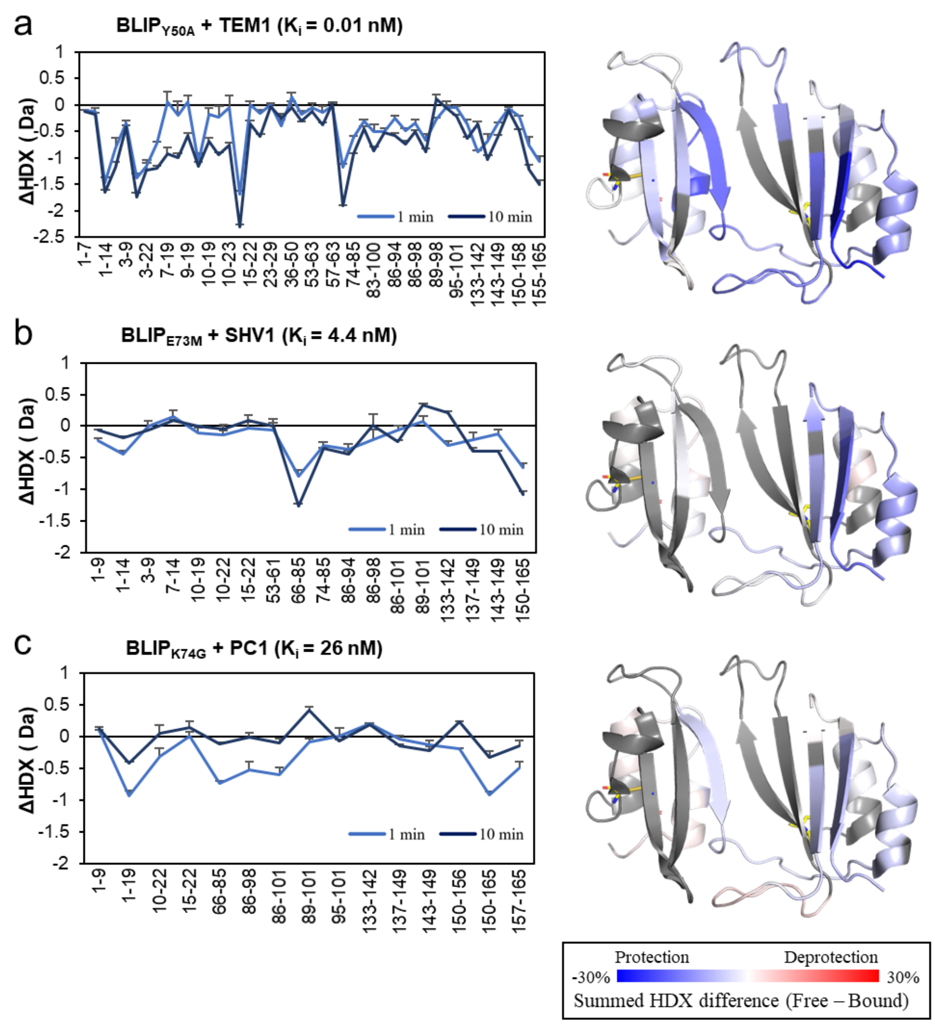

Figure 4.5 HDX-MS profiles of enhanced BLIP mutants upon binding β-lactamases (a) BLIP-Y50A with TEM1, (b) BLIP-E73M with SHV1 and (c) BLIP-K74G with PC1. ΔHDX represents the difference between the free and the bound state. Positive values reflect deprotection from HDX while negative values reflect protection. Error bars indicate standard deviations for the time points 1 and 10 min (n=3). ΔHDX larger than 0.21 Da, 0.22 Da and 0.21 Da was significant for BLIP mutants upon binding TEM1, SHV1 and PC1, respectively. The summed fractional difference in 1- and 10-min labelling are mapped onto the crystal model of BLIP (PDB ID: 3gmu) for better visualization.

The abilities to widely adapt to several class A β-lactamases might be contributed by the flexibility of the two domains of BLIP, but no solid evidence has been reported yet. In this study, compared to the wild-type BLIP, both of the BLIP mutations E73M and K74G were significantly more flexible in the interdomain region, where significant protection from HDX was observed upon binding to SHV1 and PC1, respectively. An intramolecular salt bridge E73-K74 present in the wild type BLIP might integrate the structure of the interdomain region and further enhanced upon binding to SHV1 (). However, BLIP mutants E73M and K74G were unable to form this salt bridge and allowed enhanced flexibility of the interdomain region. Our results indicated that the conformational dynamics of the interdomain region for mutations E73M and K74G could be important to enhance the inhibitory binding towards class A β-lactamases, suggesting that the relative orientation of the two domains of BLIP could be determined by the flexibility of the interdomain linker. Such interdomain dynamics allowed the ability of BLIP to bind a variety of β-lactamases which adapt onto the concave interface cooperatively formed by the two domains from BLIP. Such cooperativity was also reflected in the binding interface of the complex $SHV1_{D104K}$/BLIP.[174]

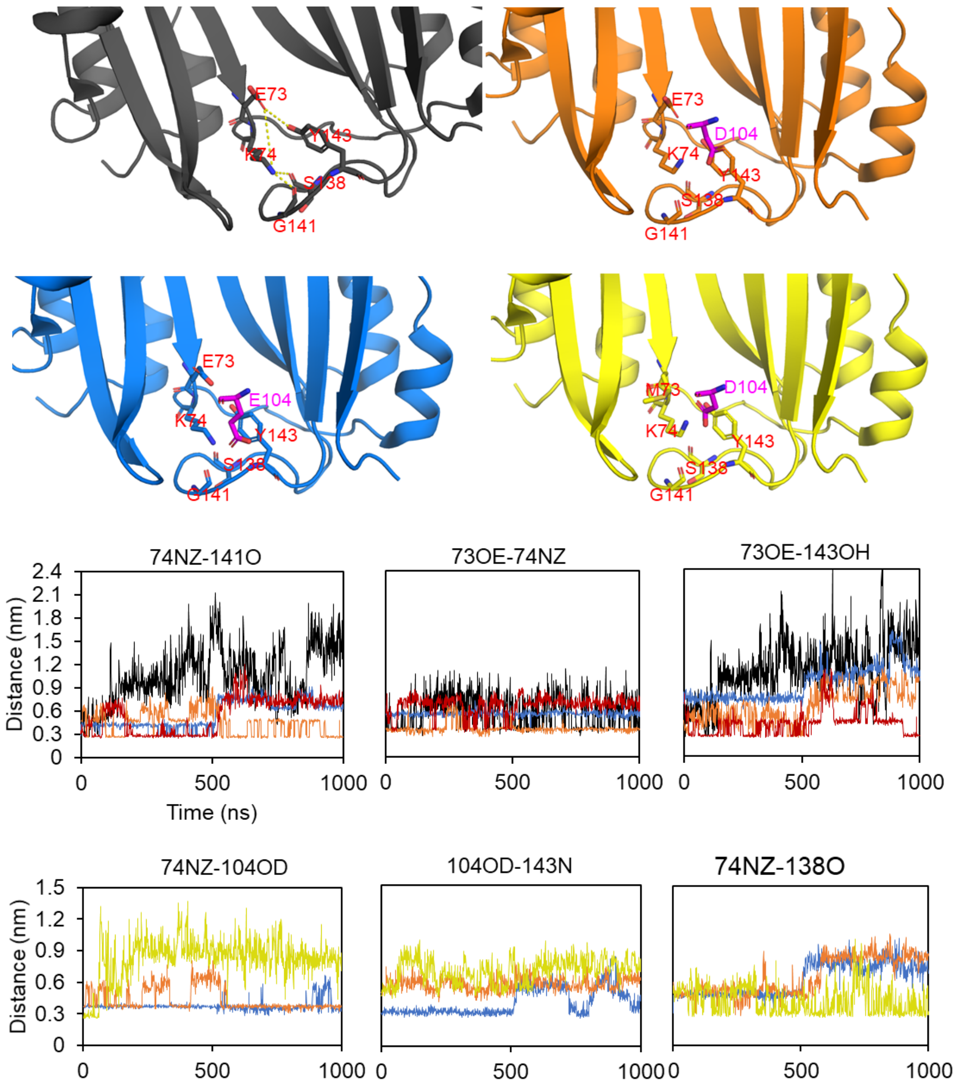

Figure 4.6 Hydrogen bonding network in BLIP (PDB ID: 3gmu).
E73 and K74 stabilize loop 135-146 via hydrogen bond with Y143, G141 and S138.

The flexibility of the interdomain linker of multidomain proteins has been proven to provide the ability to modulate the orientation of each domain, a feature that can be critical for the ligand binding[165] and functioning of chaperon protein.[207] For instance, the orientation of the PDZ domains connected by a conserved peptide linker facilitates the binding to multiple targets and the binding affinity can be modulated by the interdomain rearrangements.[164] A recent study on a chaperon protein SurA using HDX-MS also revealed the important role of the interdomain dynamics in multiple binding to protein clients.[207] In this study, by using HDX-MS and site-directed mutagenesis, for the first time, we revealed that the increased flexibility of the interdomain region in BLIP could allow better adaption of the β-lactamases into the binding pocket of BLIP. This finding could lead to rational design of the interdomain linker on BLIP for enhanced inhibition efficacy towards the class A β-lactamases.

4.2.2 Cooperative effect of BLIP mutations on the structural integrity of interfacial loop 135-145

In the unbound structure of BLIP, one of the significant disordered regions was the β-hairpin loop 46-51 conjugating β2 and β3 in the N-terminal domain (NTD) (Figure 4.1). This region showed major interactions with the active sites of β-lactamases[65] and served as a template for designing analogous inhibitors[206]. Our results showed that this region on BLIP was deprotected after binding to SHV1 and PC1 (Figure 4.3). Combining with the HDX profiles of protruding loop and SND region (chapter 2) from the β-lactamases that intensively contact with the β-hairpin loop, we found that both sides of this interface were partially deprotected from HDX, indicating increased flexibility in the complexes. This could be due to unfavourable interactions compensated by the entropic benefit from the increased flexibility. This compensation would be favourable for the overall binding free energy.[208]

Residues 135-145 on the C-terminal domain (CTD) is a large loop analogous to the β-hairpin (residues 46-51) on the NTD, and was disordered to some extent (Figure 4.1). These topologically equivalent loops together were inserted into the active pocket of β-lactamase and blocked the function of the enzyme.[65] The HDX profiles of the fragments 137-149 show significant protection from HDX upon binding with TEM1 (i.e., reduced flexibility) while there was no significant changes upon binding SHV1 or PC1 (Figure 4.3). Unexpectedly, both mutants E73M and K74G on their own showed significantly reduced HDX for fragment 137-149 compared with wild-type BLIP in the free states (Figure 4.4). Yet, the HDX values of this large loop in E73M and K74G upon binding to SHV1 and PC1, respectively, are not significantly changed compared to those of the BLIP mutants in their free states (Figure 4.5). Together, these results reflected that this loop in the complexes of SHV1/BLIP-E73M and PC1/BLIP-K74G showed reduced HDX than that in the complexes of SHV1/BLIP and PC1/BLIP, respectively.

Notably, E73 and K74 could interact with the residues in the loop 135-145 (PDB ID: 3gmu), suggesting the coherence between the interdomain region and the large interfacial loop. This highly supported our observation that single mutation on the interdomain region would cooperatively alter the conformational dynamics of the large interfacial loop. Furthermore, the BLIP mutant E73M-F142A would totally lose high affinity to SHV1, resulting in a K_d value that is similar to that of SHV1/BLIP.} Such evidence confirmed that the cooperativity between the interdomain region and the large interfacial loop was essential for the enhanced inhibitory binding. Additionally, the F142A mutation that could lose the integrity of the interfacial loop was not favourable for the binding. On the contrary, more order (or less flexibility) of the large interfacial loop 135-145 in the E73M and K74G mutants in comparison with that of the wild type could be favourable for their enhanced bindings with SHV1 and PC1, respectively.

Previous chapter showed that the interfacial protruding loop preferred some rigidity for tight bindings. In the case of TEM1, its E104 on the protruding loop interacts with K74 (via salt bridge) and Y143 (via hydrogen bonding) on BLIP as observed from the crystal structure (PDB ID: 1jtg). Therefore, the loop 135-145 in the BLIP/TEM1 complex has significantly reduced flexibility compared to that in the apo-BLIP. In the case of SHV1, the interactive network K74-E104-Y143 is less stable than that in TEM1 (). Interestingly, the BLIP mutant E73M could contribute little to this network while enhanced the hydrogen bonding between K74 and S138 (). This enhanced bonding supported the change in flexibility of the loop 135-145 in the complex of SHV1/BLIP-E73M compared to that of SHV1/BLIP. Taken together, our results suggested that the rigidity of the interface between the interfacial loop 135-145 of the BLIP mutants and the protruding loop of β-lactamases could contribute to their inhibitory binding.

4.2.3 N-terminal helix-loop-helix motif showed structural heterogeneity for enhanced BLIP mutants

The helix-loop-helix motifs exhibit the highest internal sequence identity and structural similarity within the repeat domains (Figure 4.1). They pack against a four-stranded antiparallel β-sheet on the opposite side of the binding interface (Figure 4.1).[5] HDX profiles of these motifs in the NTD and CTD were analyzed upon binding the β-lactamases. The N-terminal motif spanning residues 10-22 on BLIP showed no differences in deuterium uptake upon binding to all tested class A β-lactamases, i.e., TEM1, SHV1 and PC1, regardless of their binding affinities. Interestingly, this motif on all BLIP mutants showed significantly improved flexibility compared to that of the wild type. More intriguingly, this region in both BLIP mutants exhibited bimodal mass distribution compared to that in the wild type (Figure 4.7). For the observed slow-exchange (blue lines) and fast-exchange (red lines) bimodal HDX, both BLIP mutants shared similar exchange levels and nearly constant subpopulations (Figure 4.7). Such HDX profiles were not significantly changed upon binding to SHV1 or PC1. Consistently, a shorter fragment spanning residues 15-22 on both mutants also exhibited bimodal HDX profiles compared to that on the wild type (Figure 4.7), further verifying the observation.

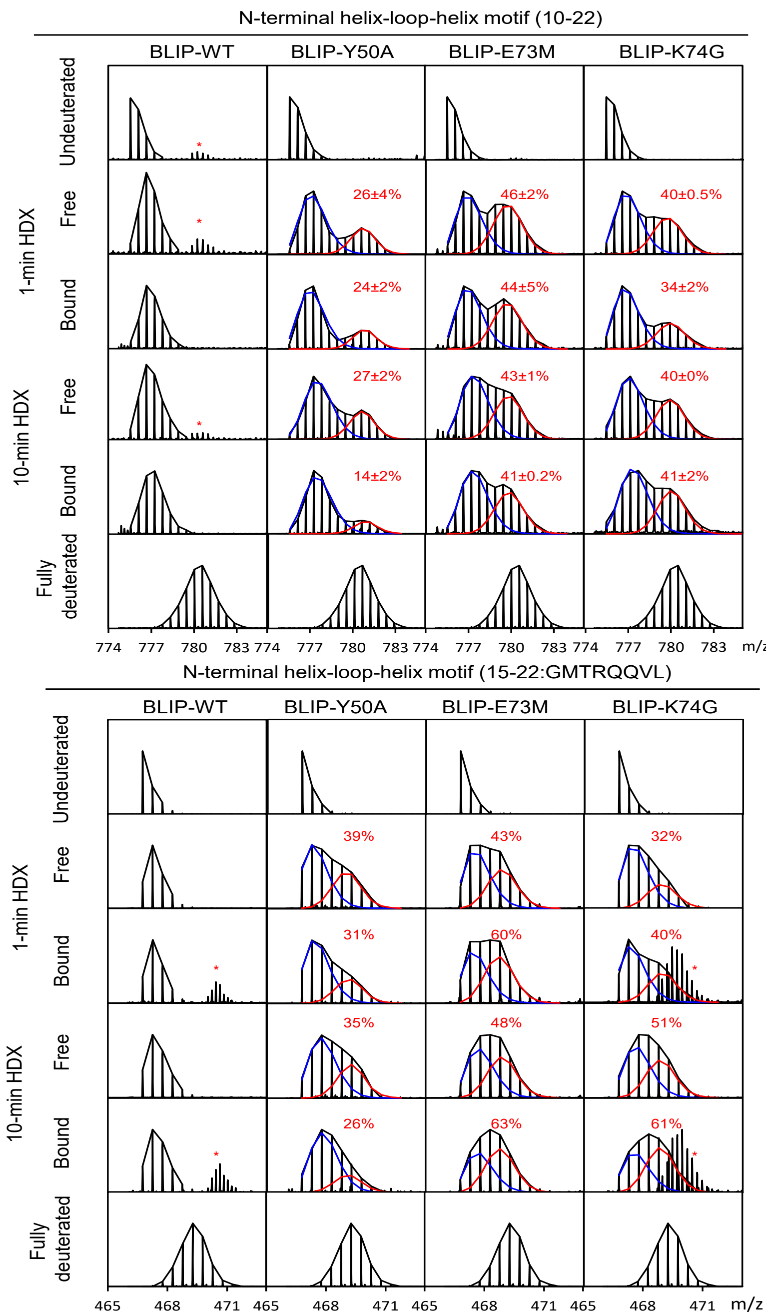

Figure 4.7 Bimodal HDX of the N-terminal helix-loop-helix motif on the BLIP mutants. Representative mass spectra are shown for this motif in the free and bound states. Fraction of the unprotectted is annotated above the corresponding peak. Asterisks indicate mass envelops of other peptides.

This observation was rare while particularly valuable for native proteins under physiological conditions. Our results suggested that the BLIP mutations induced an additional subpopulation of conformation with enhanced flexibility in the N-terminal helix-loop-helix motif compared to BLIP wild type. This change might relax this motif to facilitate a more favoured conformation upon binding to β-lactamases. Such observations might be due to the destabilizing effects of the mutations, which could alter the hydrogen exchange mechanism. For example, two mutants I106A and V108G of Ribonuclease A demonstrated strongly destabilized structures throughout the protein indicated by a shift from the EX2 to the EX1 mechanism.[209] Besides, Ye et al observed a bimodal distribution similar to our results for peptides in the noncanonical interface, revealing the heterogeneity of the hexameric Hsp104 structure.[210] For the first time, our results indicated that structural heterogeneity of the N-terminal helix-loop-helix motif was allosterically induced by the single mutations, i.e., Y50A, E73M and K74G. A more flexible conformation might be favoured in better binding with the class A β-lactamases. From a thermodynamic point of view, multiple conformations for the complex could provide favourable entropic energy of the binding, although further investigation for the binding kinetics has not been performed.

4.2.4 The C-terminus of BLIP was more protected from HDX upon the binding compared to the N-terminus.

Both termini of BLIP were protected from HDX after binding to TEM1 and PC1 (Figure 4.3). By contrast, the HDX of neither terminus was significantly changed upon binding to SHV1 (Figure 4.3). No significant changes in HDX were observed for both termini comparing BLIP mutants and wild type (Figure 4.4). Furthermore, no significant changes were found for the N-termini of BLIP mutants after the binding (Figure 4.5). In contrast, the C-termini of the BLIP mutants were all protected from HDX after the binding (Figure 4.5). Our results showed that the binding with β-lactamases induced significant conformational changes in the C-terminus rather than the N-terminus of BLIPs. The N-terminus of BLIP sits on the other side of the binding while the C-terminus is located on the edge of the concave interface (Figure 4.1). The deformation of the C-terminus of BLIP could led to a better adaptation of the protruding loop from β-lactamases onto the concave interface.

4.2.5 MD simulation of BLIPs revealed the flexibility of important loop regions.

MD simulation can provide explicit information on the dynamics of BLIPs, which is complementary to the HDX-MS results. 1-μs MD simulations of BLIP were performed in its free and bound states with TEM1, SHV1 and PC1 (Figure 4.8c). When the structural fluctuations of protein backbones were compared, the most obvious change was located at the loop 135-145 where the fluctuation was significantly decreased for all bound states (Figure 4.8c). This was in good agreement with the HDX-MS results, which all indicated that this loop, as a major binding interface, would become more rigid upon the tight binding to the β-lactamases. In addition, the analogous loop 46-51 on the NTD that was not covered in the HDX-MS study underwent changes to varying extents upon binding to different β-lactamases. The binding to SHV1 induced a moderate change in flexibility while the binding to PC1 caused the most significant change at this region. By contrast, this loop showed little change in the complex with TEM1. BLIP mutants Y50A, E73M and K74G were simulated in parallel with the wild-type BLIP (Figure 4.8d). The mutant E73M showed increased fluctuation at the interdomain linker in the vicinity where the mutation was introduced (Figure 4.8d). This observation agreed with our HDX-MS results that showed improved flexibility in the interdomain region. In addition, residues in the N-terminal helix-loop-helix motif also showed enhanced fluctuation and the fluctuation was further enhanced in the complex with SHV1. Again, this result strongly supported our finding for the additional more flexible conformation from HDX-MS. Besides, significantly enhanced flexibility was located at the β-hairpin 46-51, which is known to be the major active sites. Furthermore, the β-turn conjugating β1' and β2' (residues 115-125) were also significantly more flexible in all mutants. By contrast, the 3/10 helix (residues 34-36) in the NTD of the mutant K74G exhibited higher fluctuations while those of the mutant Y50A or E73M were unchanged compared to the wild type.

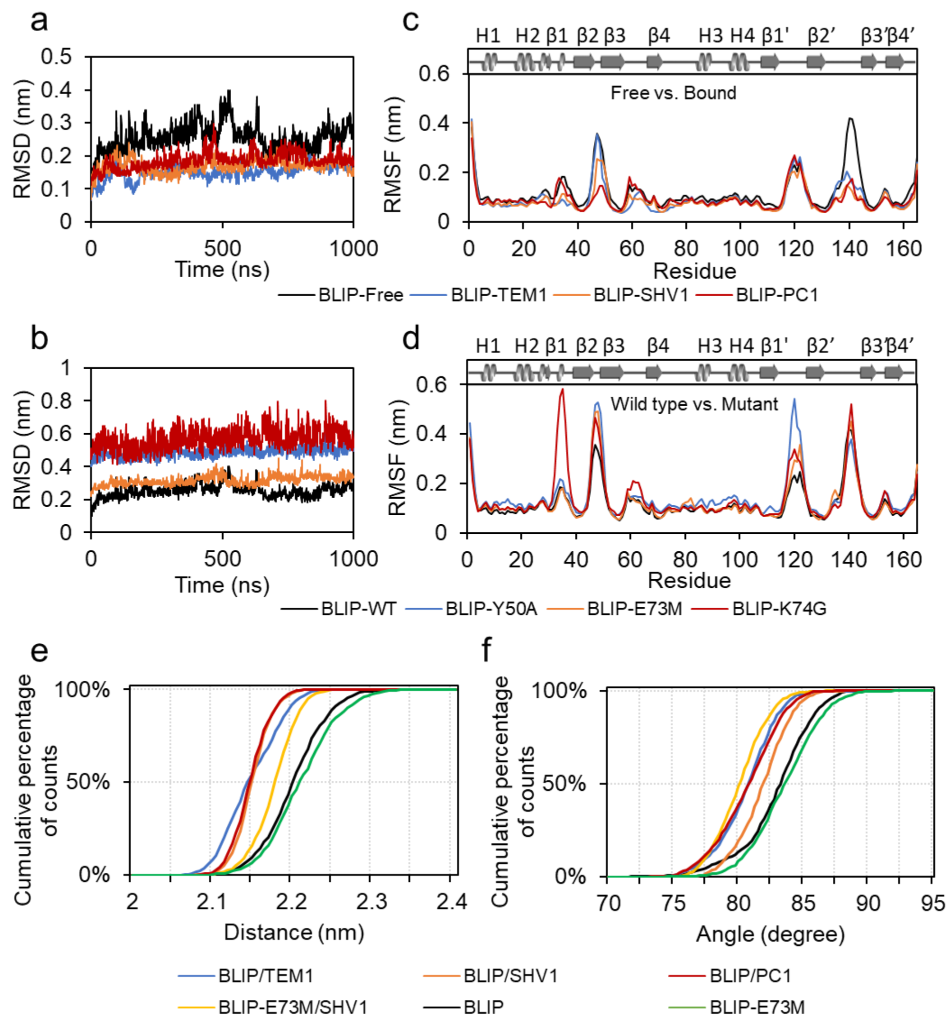

Figure 4.8 MD simulation of BLIPs and their complexes with β-lactamases.
(a-b) The RMSD and RMSF of BLIP in its free state (black line) and bound states with TEM1 (blue line), SHV1 (orange line) and PC1 (red line). (c-d) The RMSD and RMSF of BLIP for its wild type (black line), mutants Y50A (blue line), E73M (orange line) and K74G (red line). (e) Distribution of the distance and angle between the N-terminal and the C-terminal domains of BLIP in its free state and bound states with TEM1, SHV1 and PC1. (f) Distribution of the distance and angle between the N-terminal and the C-terminal domains of BLIP for its wild type and mutants Y50A, E73M and K74G. Statistical analysis presents the ranges of distance and angle at 99% confidence intervals.

The interdomain flexibility of BLIP upon the binding was analysed through the distance and angle between the NTD and CTD.[165] The distance is defined by the centres of mass (COM) of NTD and CTD and the angle is defined by two vectors, from the COM of the three-residue (Ala77-Ser79) interdomain linker to the COMs of the NTD and the CTD (Figure 4.1). Both the distance and the angle were significantly reduced upon binding to all the class A β-lactamases under investigation (Figure 4.8e-f), which was also reflected by the conformational dynamics in the interdomain region as observed by HDX-MS. Such observation was consistent with the results of the RMSD analysis (Figure 4.8a-b), suggesting that the overall conformation of BLIP was significantly changed. This finding supported the conformational change in the crystal structure where a 5.3° inward rotation of the two analogous domains was observed.[65]

The BLIP mutant E73M exhibited little changes in the orientation between NTD and CTD when compared to the wild type. Interestingly, the distance increased while the angle decreased when comparing the complexes of BLIP-E73M/SHV1 (PDB ID: 3c4p) and BLIP/SHV1 (PDB ID: 2g2u) (Figure 4.8f). Such changes supported a favoured inward rotation of the NTD and CTD upon binding to SHV1. These results completely agreed with our HDX-MS results, indicating that the affinity of BLIP towards these class A β-lactamases might be modulated by the interdomain orientation, i.e., a structural feature that relates to the flexibility of the interdomain region. A more flexible interdomain linker on BLIP could result in a more dynamic pocket that could adapt better with SHV1. Thus, the modulation of the interdomain orientation could be critical in the enhancement of the binding affinity and specificity of BLIP towards class A β-lactamases.

4.3 Summary

BLIP is a potent protein inhibitor towards a variety of class A β-lactamases and engineered BLIPs with single mutations can enhance the inhibition potency by up to 1000 folds. The protein dynamics of BLIPs in their free and bound states in complexation with three class A β-lactamases was examined by HDX-MS and MD simulation in this study.

Our results demonstrated for the first time that the flexibility of the interdomain region in wild-type BLIP was significantly suppressed by the high-affinity interactions with TEM1 while this suppression was not observed by the low-affinity interactions with SHV1 and PC1. Interestingly, the enhanced BLIP mutants E73M and K74G towards SHV1 and PC1, respectively, showed higher flexibility than the wild type in the interdomain region. The loops 135-145 of the mutants were significantly more rigid compared to that of the wild type. Such change might be more favoured for insertion to the active pockets of the class A β-lactamases. An additional conformation with higher flexibility for the N-terminal helix-loop-helix motif was observed in the enhanced mutants E73M and K74G. The relaxation of these motifs could promote the favourable interfacing towards β-lactamases. Both termini of BLIP showed reduced flexibility upon binding to the class A β-lactamases under investigation. In contrast, the C-terminus tended to be more protected while the N-terminus was not significantly changed as the inhibitory binding enhanced.

MD simulation exhibited the significant changes of BLIP upon binding to the β-lactamasees regardless of the binding affinity of the complexes. Particularly, the changes of the loop 135-145 was predominant throughout the sequence of BLIP. Such results agreed with the changes observed in the HDX-MS study. Single mutations could alter both the interdomain linker and remote loop regions in BLIP. Importantly, significant changes occurred with the β-hairpin loop 46-51. These findings suggested that such mutations could induce broad changes to the conformational dynamics of BLIP. Such changes improve the plasticity of the binding interface, i.e., surface loops and interdomain linker, between BLIP and the β-lactamases, thus enhancing the inhibitory interactions.

Our results presented novel insights into the mechanism underlying the broad and enhanced binding of BLIP mutants towards various β-lactamases. These findings revealed the role of the inhibitor protein from the perspective of protein dynamics. Engineered BLIPs with enhanced potency exhibited critical changes in the plasticity of interfacial loops and interdomain orientation. Such insights into the conformational dynamics of BLIPs can provide general guidelines towards the designing of novel inhibitors.

5. Conclusions

In this study, the conformational dynamics of the interaction between three class A β-lactamases and β-lactamase inhibitory proteins were investigated by native and ion mobility mass spectrometry (IM-MS), hydrogen deuterium exchange mass spectrometry (HDX-MS) complemented with molecular dynamics (MD) simulation. The major findings can be divided into two parts.

In the first part, we focused on the conformational dynamics of three class A β-lactamases in their free states and BLIP-bound states. In the free states, PC1 exhibited more extended conformation than TEM1 and SHV1 as revealed by IM-MS. In contrast to the results obtained in the gas phase, HDX-MS indicated that the structure of TEM1 was more flexible than SHV1 and PC1 in solution. The conformations of these β-lactamases could not be differentiated upon binding to BLIP as suggested by both IM-MS and HDX-MS. We further compared the local flexibility of TEM1, SHV1 and PC1. The SDN loop and Ω loop of PC1 showed different flexibility from those of TEM1 and SHV1. Both loops are involved in the catalysis of β-lactam antibiotics, thus different substrate specificity could be supported by our results. The protruding loop serving as the major binding interface from TEM1 was the most rigid, by contrast, the H10 region from TEM1 was the most flexible, among these β-lactamases. Upon binding to BLIP, the H10 region showed remarkable protection from HDX for all β-lactamases under investigation. By contrast, the protruding loop from TEM1 was significantly protected while those from SHV1 and PC1 were deprotected upon binding to BLIP. However, when bound with enhanced BLIP mutants, E73M and K74G, the protruding loops became more protected in SHV1 and PC1, respectively. These results indicated that the plasticity of major binding interface could be determinant for the affinity of inhibitory binding to BLIP. Furthermore, the flexibility of H10 region could be critical for the modulation of these interactions. Meanwhile, the flexibility of other loop regions, e.g., SDN loop and Ω loop, might cooperatively contribute to the inhibition of β-lactamases. The slow motion of Ω loop was suppressed upon binding to BLIP. By contrast, MD simulation revealed that the fast motion was attenuated in TEM1 while escalated in SHV1 and PC1 upon binding. Such results indicated that the BLIP binding to TEM1 caused a more negative effect on the catalytic efficiency compared to SHV1 and PC1.

In the second part, we focused on the conformational dynamics of the inhibitory protein and its enhanced mutants towards the three β-lactamases investigated in the first part. Our results were featured by the conformational changes in the interdomain region. This region in wild-type BLIP was significantly protected from HDX upon binding to TEM1 while was not changed upon binding to SHV1 or PC1. Both single mutations, E73M and K74G, improved the flexibility of the interdomain linker compared to the wild type. Significant protection was observed for the linker in these mutants, E73M and K74G, upon binding to SHV1 and PC1, respectively. These results indicated that the flexibility of the interdomain region could be determinant to the enhanced potency of BLIP mutants. The flexible motion of the two domains from BLIP could be supported for the cause of the wide ability BLIP to bind various β-lactamases. A highly disordered β-hairpin loop 135-145 from the N-terminal domain of BLIP exhibited more rigidness in all the enhanced mutants, indicating that such conformation might be favourable for the inhibitory binding. On the contrary, more flexibility could be desirable for the loop-helix-loop motifs from both domains. Both termini of BLIP showed reduced flexibility upon binding while the C-terminus showed more conformational change than the N-terminus as the inhibitory binding tightened. MD simulation provided consistent and complementary results to HDX-MS. The flexibility of the β-hairpin loop 135-145 was significantly reduced upon binding to β-lactamases. In contrast to the HDX results, the flexibility was not obviously changed upon mutation. It is possible that the slow motion of this disorder loop might not be sampled by the simulation within a nanosecond while such motion was sensitively detected by

HDX. By contrast, significant changes were identified in the β-hairpin loops 46-51 on the C-terminal domain and a distant β1'-β2' loop. These results indicated the possible allosteric effects induced by mutation E73M and K74G.

Together, this study provided a novel and unique understanding on the inhibitory bindings between these class A β-lactamases and BLIP. Such understanding provided general guideline for the design of inhibitors towards β-lactamases.

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7. Appendices

Supplementary Table S1. HDX-MS dataset for TEM1/BLIP

	Sequence	MaxUptake	MHP / Da	RT / min	Deuterium uptake / Da Free 1 min	Deuterium uptake / Da Free 10 min	Deuterium uptake / Da Bound 1 min	Deuterium uptake / Da Bound 10 min	Standard deviation / Da Free 1 min	Standard deviation / Da Free 10 min	Standard deviation / Da Bound 1 min	Standard deviation / Da Bound 10 min
31-40	VKVKDAEDQL	9	1144.621	3.80	1.45	1.88	0.42	0.45	0.02	0.00	0.02	0.01
31-46	VKVKDAEDQLGARVGY	15	1747.934	4.34	1.28	1.72	0.16	0.27	0.07	0.06	0.08	0.09
31-48	VKVKDAEDQLGARVGYIE	17	1990.060	4.70	0.94	1.31	0.23	0.38	0.14	0.14	0.05	0.01
47-59	IELDLNSGKILES	12	1430.774	5.60	1.86	2.12	0.98	1.27	0.01	0.02	0.03	0.03
50-57	DLNSGKIL	10	1188.647	5.01	2.14	2.66	1.01	1.47	0.06	0.06	0.04	0.05
49-59	LDLNSGKILES	7	859.488	4.86	2.03	2.45	1.04	1.41	0.04	0.04	0.01	0.03
50-59	DLNSGKILES	9	1075.563	4.52	2.21	2.64	1.10	1.45	0.03	0.03	0.03	0.02
50-66	DLNSGKILESFRPEERF	15	2037.040	5.67	2.08	2.57	1.52	1.97	0.02	0.02	0.02	0.04
52-66	NSGKILESFRPEERF	13	1808.929	5.07	1.81	2.31	1.31	1.84	0.02	0.02	0.07	0.04
52-71	NSGKILESFRPEERFPMMST	17	2356.142	5.53	1.46	1.88	1.17	1.59	0.03	0.01	0.05	0.11
59-69	SFRPEERFPMM	11	1743.783	5.32	1.20	1.64	1.22	1.66	0.09	0.09	0.04	0.04
58-71	ESFRPEERFPMMST	8	1426.661	5.55	0.98	1.34	1.00	1.40	0.09	0.09	0.06	0.09
60-69	FRPEERFPMM	10	1614.740	5.27	0.83	1.32	0.86	1.39	0.10	0.11	0.06	0.02
59-71	SFRPEERFPMMST	7	1339.629	5.48	0.94	1.34	1.10	1.48	0.07	0.07	0.05	0.05
60-71	FRPEERFPMMST	9	1527.708	5.22	0.71	1.15	0.79	1.24	0.07	0.07	0.06	0.05
60-72	FRPEERFPMMSTF	10	1674.777	6.13	0.66	0.96	0.68	1.07	0.07	0.07	0.04	0.03
79-91	AVLSRVDAGQEQL	12	1385.738	4.62	1.19	1.72	0.68	0.99	0.09	0.09	0.02	0.01
80-91	VLSRVDAGQEQL	11	1314.701	4.36	1.36	1.93	0.80	1.20	0.09	0.09	0.08	0.07
82-91	SRVDAGQEQL	9	1102.549	3.83	1.62	2.30	0.71	1.01	0.10	0.09	0.03	0.03
82-100	SRVDAGQEQLGRRIHYSQN	18	2214.112	3.64	1.87	2.29	0.67	1.00	0.15	0.16	0.06	0.03
82-101	SRVDAGQEQLGRRIHYSQND	19	2329.139	3.68	1.76	2.24	0.61	0.96	0.16	0.15	0.04	0.07
92-102	GRRIHYSQNDL	10	1358.692	3.51	0.80	1.22	0.21	0.53	0.06	0.05	0.03	0.04
92-104	GRRIHYSQNDLVE	12	1586.803	3.69	1.07	1.42	0.37	0.62	0.08	0.12	0.01	0.01
101-118	DLVEYSPVTEKHLTDGMT	16	2034.969	5.09	1.62	2.16	0.43	0.81	0.11	0.13	0.04	0.08
102-118	LVEYSPVTEKHLTDGMT	15	1919.942	4.73	1.63	2.20	0.42	0.81	0.12	0.12	0.06	0.05
103-117	VEYSPVTEKHLTDGM	13	1705.810	4.59	1.35	1.92	0.27	0.58	0.10	0.09	0.04	0.02
103-118	VEYSPVTEKHLTDGMT	14	1806.858	4.44	1.31	1.87	0.20	0.48	0.09	0.09	0.04	0.03
103-119	VEYSPVTEKHLTDGMTV	15	1905.926	4.89	1.18	1.67	0.28	0.54	0.09	0.09	0.02	0.01
105-118	YSPVTEKHLTDGMT	17	2191.070	4.41	0.74	1.41	0.17	0.46	0.14	0.14	0.05	0.08
103-121	VEYSPVTEKHLTDGMTVRE	12	1578.747	4.21	0.99	1.45	0.20	0.41	0.09	0.09	0.04	0.07
125-136	AAITMSDNTAAN	11	1179.531	3.94	0.90	1.40	0.50	0.99	0.02	0.04	0.05	0.06
125-137	AAITMSDNTAANL	12	1292.615	5.02	0.94	1.44	0.43	0.87	0.07	0.08	0.05	0.03
125-138	AAITMSDNTAANLL	13	1405.699	5.91	0.75	1.19	0.49	0.90	0.09	0.09	0.04	0.08
126-137	AITMSDNTAANL	14	1518.783	6.54	0.85	1.47	0.54	1.06	0.04	0.04	0.08	0.06
125-139	AAITMSDNTAANLLL	11	1221.578	4.98	0.56	0.94	0.49	0.79	0.09	0.09	0.06	0.04
126-138	AITMSDNTAANLL	12	1334.662	5.91	0.65	1.14	0.43	0.95	0.08	0.08	0.04	0.03
127-137	ITMSDNTAANL	13	1447.746	6.56	1.04	1.66	0.51	0.97	0.06	0.06	0.04	0.03
126-139	AITMSDNTAANLLL	10	1150.541	4.86	0.60	1.05	0.44	0.88	0.02	0.02	0.02	0.02
127-139	ITMSDNTAANLLL	12	1376.709	6.52	0.72	1.16	0.49	0.91	0.01	0.02	0.04	0.02
130-137	SDNTAANL	7	805.369	3.89	0.14	0.31	0.09	0.21	0.01	0.01	0.01	0.00
138-148	LLTTIGGPKEL	13	1573.920	6.88	1.47	1.82	1.21	1.59	0.07	0.07	0.01	0.03
137-151	LLLTTIGGPKELTAF	9	1141.683	5.20	0.90	1.38	0.76	1.25	0.06	0.06	0.02	0.01
138-150	LLTTIGGPKELTA	11	1313.768	5.05	1.21	1.76	0.94	1.51	0.08	0.07	0.00	0.02
139-150	LTTIGGPKELTA	10	1200.683	4.58	1.38	1.98	1.00	1.60	0.07	0.07	0.01	0.02
139-152	LTTIGGPKELTAFL	12	1460.836	6.63	0.99	1.48	0.87	1.37	0.07	0.07	0.00	0.03
150-162	AFLHNMGDHVTRL	12	1510.758	4.69	0.36	0.56	0.13	0.37	0.07	0.06	0.06	0.03
152-161	LHNMGDHVTR	9	1179.569	3.49	0.37	0.69	0.19	0.46	0.04	0.03	0.03	0.04
153-162	HNMGDHVTRL	9	1179.569	3.73	0.34	0.60	0.19	0.45	0.03	0.03	0.02	0.02
154-162	NMGDHVTRL	8	1042.510	3.73	0.32	0.65	0.12	0.38	0.02	0.07	0.01	0.03
163-169	DRWEPEL	5	944.447	5.52	1.03	1.19	0.66	0.80	0.03	0.02	0.01	0.02
170-177	NEAIPNDE	6	901.390	3.59	1.84	2.76	1.12	2.07	0.04	0.04	0.05	0.04
170-179	NEAIPNDERD	8	1172.518	3.28	1.19	2.23	0.78	1.65	0.00	0.02	0.02	0.00
170-182	NEAIPNDERDTTM	11	1505.654	3.97	0.85	1.57	0.50	1.22	0.08	0.08	0.01	0.01
170-186	NEAIPNDERDTTMPAAM	14	1875.821	4.64	1.03	1.62	0.74	1.35	0.08	0.08	0.03	0.04
171-186	EAIPNDERDTTMPAAM	13	1761.778	4.62	0.63	1.18	0.56	1.12	0.09	0.09	0.04	0.01
188-198	TTLRKLLTGEL	10	1244.757	5.83	1.38	1.56	0.61	0.72	0.04	0.04	0.02	0.02
189-198	TLRKLLTGEL	9	1143.710	5.88	1.44	1.60	0.58	0.70	0.06	0.06	0.01	0.02
190-198	LRKLLTGEL	8	1042.662	4.85	1.69	1.88	0.72	0.85	0.06	0.05	0.03	0.02
199-209	LTLASRQQLID	10	1257.716	4.85	1.21	1.42	0.56	0.82	0.02	0.01	0.07	0.02
202-209	ASRQQLID	7	930.500	3.85	0.01	0.02	-0.03	0.02	0.02	0.02	0.03	0.04
210-221	WMEADKVAGPLL	10	1329.687	6.12	3.63	3.92	2.31	2.59	0.15	0.15	0.02	0.06
211-221	MEADKVAGPLL	9	1143.608	5.37	4.12	4.34	2.40	2.64	0.08	0.08	0.02	0.05
212-221	EADKVAGPLL	13	1570.862	5.42	4.21	4.10	2.42	2.40	0.07	0.07	0.03	0.04
214-221	DKVAGPLL	8	1012.567	5.13	4.46	4.44	2.37	2.32	0.15	0.14	0.05	0.06
211-225	MEADKVAGPLLRSAL	15	1851.012	6.05	4.23	4.68	2.70	3.07	0.11	0.07	0.16	0.08
212-229	EADKVAGPLLRSALPAGW	6	812.488	5.07	4.10	4.72	2.42	2.90	0.08	0.08	0.02	0.03
214-229	DKVAGPLLRSALPAGW	13	1650.933	6.05	4.15	4.73	2.47	3.02	0.11	0.11	0.01	0.07
222-230	RSALPAGWF	7	1004.531	6.39	2.54	3.77	1.35	2.64	0.03	0.01	0.02	0.02
230-246	FIADKSGAGERGSRGII	16	1733.929	4.11	0.99	1.24	0.40	0.57	0.08	0.10	0.07	0.05
230-247	FIADKSGAGERGSRGIIA	17	1804.966	3.90	1.03	1.29	0.43	0.62	0.09	0.10	0.04	0.02
231-246	IADKSGAGERGSRGII	18	1876.004	3.94	1.08	1.32	0.18	0.32	0.09	0.09	0.05	0.01
230-248	FIADKSGAGERGSRGIIAA	15	1586.861	3.62	0.97	1.22	0.40	0.61	0.09	0.09	0.04	0.02
231-247	IADKSGAGERGSRGIIA	16	1657.898	3.41	1.09	1.34	0.25	0.36	0.10	0.11	0.04	0.04
231-248	IADKSGAGERGSRGIIAA	17	1728.935	3.49	1.04	1.30	0.20	0.31	0.09	0.09	0.04	0.03
232-248	ADKSGAGERGSRGIIAA	16	1615.851	3.94	1.03	1.28	0.41	0.61	0.10	0.11	0.03	0.01
250-260	GPDGKPSRIVV	8	1124.642	4.72	1.00	1.76	0.81	1.40	0.11	0.11	0.05	0.10
261-267	IYTTGSQ	6	769.373	3.20	2.07	2.09	0.55	0.59	0.04	0.04	0.02	0.01
261-279	IYTTGSQATMDERNRQIAE	18	2184.035	3.83	2.05	2.59	0.44	0.63	0.07	0.16	0.04	0.02
268-279	ATMDERNRQIAE	11	1433.680	3.24	1.33	2.26	0.27	0.51	0.10	0.10	0.03	0.03
270-279	MDERNRQIAE	16	1874.939	5.27	1.54	2.51	0.30	0.59	0.24	0.21	0.04	0.01
268-284	ATMDERNRQIAEIGASL	9	1261.595	3.24	0.90	1.60	0.82	1.48	0.09	0.08	0.01	0.03
271-281	DERNRQIAEIG	10	1300.660	3.95	0.93	1.89	0.43	0.74	0.08	0.07	0.03	0.01
271-284	DERNRQIAEIGASL	13	1571.814	5.18	0.58	1.44	0.59	1.18	0.13	0.14	0.08	0.08

Supplementary Table S2. HDX-MS dataset for SHV1/BLIP

	Sequence	MaxUptake	MHP	RT / min	Deuterium uptake / Da				Standard deviation / Da			
					Free		Bound		Free		Bound	
					1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
26-32	SPQPLEQ	4	798.3992	3.60	2.55	2.95	2.31	2.54	0.02	0.02	0.04	0.02
33-40	IKLSESQL	7	917.5302	4.66	0.59	1.29	0.59	0.93	0.03	0.01	0.01	0.03
33-46	IKLSESQLSGRVGM	13	1504.8152	4.66	1.04	1.77	0.67	1.51	0.06	0.05	0.07	0.07
50-59	DLASGRTLTA	9	1004.5371	4.18	2.46	3.69	1.32	1.72	0.09	0.05	0.04	0.02
60-69	WRADERFPMM	8	1338.6082	5.73	0.54	0.98	0.10	0.17	0.06	0.05	0.06	0.05
82-91	ARVDAGDEQL	9	1073.5222	4.01	1.12	1.58	1.01	1.32	0.02	0.03	0.09	0.04
92-104	ERKIHYRQQDLVD	12	1699.8874	3.25	1.80	2.40	2.18	2.94	0.05	0.04	0.09	0.03
105-122	YSPVSEKHLADGMTVGEL	16	1932.9372	5.24	2.55	4.05	0.92	1.44	0.14	0.08	0.08	0.02
127-137	ITMSDNSAANL	10	1136.5252	4.80	1.17	1.86	1.94	1.86	0.15	0.10	0.04	0.03
152-162	LRQIGDNVTRL	10	1284.7383	4.45	1.37	1.90	1.05	1.40	0.09	0.06	0.11	0.01
163-169	DRWETEL	6	948.4421	5.36	0.90	0.92	0.47	0.41	0.03	0.03	0.07	0.02
170-186	NEALPGDARDTTTPASM	14	1746.7963	4.31	1.43	2.45	0.71	1.11	0.11	0.11	0.06	0.07
187-201	AATLRKLLTSQRLSA	14	1628.9806	4.77	2.33	2.98	1.60	1.95	0.07	0.05	0.15	0.02
200-210	SARSQRQLLQW	10	1372.7444	5.43	0.45	0.76	0.12	0.20	0.04	0.04	0.05	0.07
202-210	RSQRQLLQW	8	1214.6753	5.44	0.13	0.13	-0.02	-0.05	0.01	0.01	0.10	0.01
211-225	MVDDRVAGPLIRSVL	13	1640.9152	5.8852	5.11	5.83	2.31	3.59	0.20	0.07	0.25	0.04
230-248	FIADKTGAGERGARGIVAL	18	1902.0556	4.6925	2.52	3.67	0.81	1.11	0.10	0.08	0.10	0.17
231-248	IADKTGAGERGARGIVAL	17	1754.9872	4.3937	2.49	3.60	0.90	1.22	0.03	0.06	0.10	0.03
249-260	LGPNNKAERIVV	10	1309.7587	4.3174	1.22	1.42	1.05	1.26	0.07	0.07	0.07	0.01
261-270	IYLRDTPASM	8	1166.5874	4.8494	1.51	2.43	0.73	1.00	0.03	0.04	0.07	0.02
271-281	AERNQQIAGIG	10	1156.6069	4.0781	0.22	0.89	0.11	0.25	0.02	0.02	0.01	0.02
271-283	AERNQQIAGIGAA	12	1298.6811	4.1753	0.32	1.09	0.11	0.17	0.05	0.01	0.03	0.08

Supplementary Table S3. HDX-MS dataset for PC1/BLIP

					Deuterium uptake / Da				Standard deviation / Da			
	Sequence	MaxUptake	MHP	RT / min	Free		Bound		Free		Bound	
					1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
23-36	MHHHHHHMKELNDL	3.05	1815.8278	4.87	1.47	1.74	1.36	1.33	0.14	0.10	0.08	0.24
37-46	EKKYNAHIGV	3.62	1158.6266	3.74	0.84	1.02	0.75	0.74	0.02	0.00	0.06	0.13
47-66	YALDTKSGKEV-KFNSDKRF	4.37	2233.1612	4.21	4.37	5.74	4.38	6.20	0.14	0.14	0.13	0.15
69-80	ASTSKAINSAIL	1.84	1175.663	5.07	1.35	2.40	0.69	2.14	0.07	0.10	0.09	0.09
80-90	LEQVPYNKL	3.32	1103.6095	5.07	2.27	3.28	2.08	3.12	0.04	0.05	0.02	0.05
92-104	NKKVHINKDDIVA	2.91	1493.8435	3.55	2.76	3.18	2.90	3.73	0.09	0.08	0.15	0.02
110-116	EKYVGKD	3.67	838.4305	2.72	2.13	2.54	1.88	2.43	0.01	0.01	0.05	0.01
110-119	EKYVGKDITL	3.82	1165.6463	4.78	2.38	3.21	1.95	3.06	0.02	0.02	0.15	0.04
111-119	KYVGKDITL	1.61	1036.6037	4.64	2.08	2.79	2.13	2.53	0.02	0.03	0.07	0.03
125-141	ASMTYSDNTANNKIIKE	2.91	1899.9117	4.19	1.87	2.15	1.20	1.50	0.02	0.01	0.06	0.07
128-141	TYSDNTANNKIIKE	0.42	1610.802	3.82	1.36	1.46	0.92	1.04	0.02	0.01	0.04	0.06
129-141	YSDNTANNKIIKE	3.64	1509.7544	3.73	0.96	1.07	0.60	0.57	0.03	0.01	0.04	0.07
170-185	NYYSPKSKKDTSTPAA	1.35	1757.8705	3.20	3.18	3.80	2.43	2.70	0.03	0.04	0.02	0.13
170-186	NYYSPKSKKDTSTPAAF	3.11	1904.9389	4.23	3.02	3.59	2.45	2.77	0.09	0.12	0.07	0.08
187-193	GKTLNKL	1.18	773.488	3.57	0.43	0.48	0.23	0.24	0.03	0.03	0.04	0.05
194-209	IANGKLSKENKKFLLD	6.01	1818.0484	4.375	4.14	4.37	4.60	5.07	0.06	0.03	0.04	0.12
194-210	IANGKLSKENKKFLLDL	0.15	1931.1324	5.363	4.08	4.31	4.40	4.93	0.09	0.04	0.09	0.07
211-220	MLNNKSGDTL	2.51	1092.5354	4.4316	4.25	4.20	3.11	3.53	0.02	0.02	0.13	0.02
221-238	IKDGVPKDYKVADKSGQA	1.12	1919.0233	3.3864	3.92	5.17	3.06	3.95	0.02	0.13	0.14	0.08
221-240	IKDGVPKDYKVADKSGQAIT	3.61	2133.155	3.7137	5.79	7.46	4.02	5.71	0.02	0.12	0.03	0.25
239-249	ITYASRNDVAF	0.44	1256.627	5.2463	1.28	2.13	0.59	0.77	0.02	0.02	0.01	0.02
241-248	YASRNDVA	1.71	895.4268	3.1884	0.82	1.13	0.55	0.62	0.03	0.02	0.02	0.07
241-249	YASRNDVAF	3.94	1042.4952	4.9827	0.88	1.17	0.71	0.71	0.03	0.02	0.02	0.06
242-249	ASRNDVAF	3.65	879.4319	4.7614	0.68	0.91	0.85	0.81	0.03	0.02	0.02	0.06
249-261	FVYPKGQSEPIVL	4.35	1476.8097	5.8909	3.10	3.08	2.64	2.75	0.03	0.04	0.07	0.04
250-261	VYPKGQSEPIVL	4.78	1329.7413	5.4584	3.23	3.18	2.65	2.76	0.03	0.02	0.06	0.01
262-279	VIFTNKDNKSDKPNDKLI	1.61	2089.1288	4.4452	3.63	4.02	2.64	2.94	0.03	0.03	0.07	0.14
263-278	IFTNKDNKSDKPNDKL	3.05	1876.9763	3.8067	3.78	4.16	3.17	3.67	0.02	0.03	0.15	0.09
264-278	FTNKDNKSDKPNDKL	3.72	1763.8923	3.8074	3.15	3.44	2.56	2.73	0.02	0.04	0.20	0.13
269-286	NKSDKPNDKLISETAKSV	9.95	1974.0502	3.8068	3.95	4.38	3.56	3.90	0.03	0.05	0.09	0.04
279-287	ISETAKSVM	2.51	965.4972	4.1515	0.67	0.75	0.39	0.39	0.02	0.01	0.01	0.05
279-290	ISETAKSVMKEF	5.56	1369.7032	5.2264	1.56	2.37	0.58	1.62	0.03	0.01	0.09	0.05

Supplementary Table S4. HDX-MS dataset for TEM1 /BLIP-Y50A

	Sequence	MaxUptake	MHP / Da	RT / min	Deuterium uptake / Da				Standard deviation / Da			
					Free		Bound		Free		Bound	
					1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
25-46	MHPETLVKVKDAEDQLGARVGY	20	2456.260	5.68	2.66	4.14	4.28	5.18	0.06	0.05	0.13	0.12
31-46	VKVKDAEDQLGARVGY	15	1747.934	5.18	2.43	3.13	3.05	3.65	0.07	0.07	0.07	0.08
50-57	DLNSGKIL	7	859.488	5.71	2.30	2.60	2.49	2.79	0.02	0.09	0.04	0.03
50-58	DLNSGKILE	8	988.531	5.45	2.36	2.67	2.56	2.86	0.01	0.06	0.03	0.03
50-59	DLNSGKILES	9	1075.563	5.37	2.96	3.31	3.18	3.47	0.02	0.08	0.03	0.01
50-66	DLNSGKILESFRPEERF	15	2037.040	6.43	4.23	4.82	4.54	5.05	0.03	0.11	0.07	0.04
50-69	DLNSGKILESFRPEERFPMM	17	2396.174	6.90	3.77	4.33	4.16	4.63	0.04	0.10	0.07	0.02
50-71	DLNSGKILESFRPEERFPMMST	19	2584.253	6.71	4.48	5.04	4.71	5.36	0.02	0.12	0.08	0.03
52-69	NSGKILESFRPEERFPMM	15	2168.063	6.46	2.94	3.50	3.32	3.85	0.05	0.08	0.04	0.02
59-69	SFRPEERFPMM	8	1426.661	6.35	1.42	1.68	1.56	1.87	0.02	0.02	0.02	0.01
60-69	FRPEERFPMM	7	1339.629	6.31	0.77	1.01	0.95	1.23	0.01	0.03	0.06	0.02
60-71	FRPEERFPMMST	9	1527.708	6.04	1.33	1.67	1.49	1.76	0.03	0.04	0.02	0.02
60-72	FRPEERFPMMSTF	10	1674.777	6.87	1.32	1.67	1.51	1.73	0.02	0.07	0.05	0.01
72-78	FKVLLCG	6	779.448	6.56	0.14	0.20	0.15	0.15	0.00	0.01	0.01	0.01
79-91	AVLSRVDAGQEQL	12	1385.738	5.45	2.07	2.88	2.08	2.78	0.08	0.07	0.03	0.03
82-91	SRVDAGQEQL	9	1102.549	4.75	2.13	2.90	2.19	2.98	0.03	0.08	0.01	0.04
82-100	SRVDAGQEQLGRRIHYSQN	18	2214.112	4.50	4.86	5.83	4.92	5.78	0.04	0.08	0.09	0.01
92-102	GRRIHYSQNDL	10	1358.692	4.45	1.83	2.12	1.62	1.72	0.02	0.05	0.05	0.02
92-104	GRRIHYSQNDLVE	12	1586.803	4.60	2.55	2.80	1.82	1.96	0.01	0.09	0.06	0.03
101-118	DLVEYSPVTEKHLTDGMT	16	2034.969	5.88	3.79	4.62	2.10	2.53	0.04	0.19	0.03	0.02
103-117	VEYSPVTEKHLTDGM	13	1705.810	5.41	3.11	3.87	2.19	2.40	0.02	0.14	0.03	0.01
103-118	VEYSPVTEKHLTDGMT	14	1806.858	5.27	3.20	4.06	2.11	2.52	0.02	0.07	0.03	0.02
103-124	VEYSPVTEKHLTDGMTVRELCS	20	2494.195	5.93	1.78	2.59	1.00	1.55	0.09	0.08	0.03	0.14
105-117	YSPVTEKHLTDGM	11	1477.699	5.18	1.84	2.53	1.80	2.04	0.02	0.09	0.04	0.03
105-118	YSPVTEKHLTDGMT	12	1578.747	5.05	1.80	2.70	1.81	2.23	0.01	0.06	0.03	0.02
119-125	VRELCSA	6	777.392	4.70	0.28	0.30	0.63	0.62	0.01	0.01	0.07	0.02
119-126	VRELCSAA	7	848.430	4.73	0.34	0.36	0.31	0.29	0.00	0.01	0.01	0.00
125-137	AAITMSDNTAANL	12	1292.615	5.86	0.92	1.53	0.48	0.56	0.10	0.03	0.02	0.01
125-138	AAITMSDNTAANLL	13	1405.699	6.70	1.05	1.64	0.66	0.75	0.07	0.01	0.01	0.01
125-139	AAITMSDNTAANLLL	14	1518.783	7.31	0.61	1.12	0.25	0.42	0.08	0.01	0.00	0.02
126-137	AITMSDNTAANL	11	1221.578	5.83	1.22	1.78	0.73	0.82	0.04	0.03	0.02	0.01
126-138	AITMSDNTAANLL	12	1334.662	6.70	1.28	1.84	0.98	1.04	0.01	0.06	0.04	0.01
126-139	AITMSDNTAANLLL	13	1447.746	7.31	0.66	1.16	0.36	0.42	0.08	0.01	0.02	0.02
127-137	ITMSDNTAANL	10	1150.541	5.71	1.26	1.80	0.98	1.01	0.03	0.04	0.02	0.01
127-138	ITMSDNTAANLL	11	1263.625	6.63	1.19	1.70	0.98	1.05	0.01	0.06	0.02	0.01
127-139	ITMSDNTAANLLL	12	1376.709	7.27	0.79	1.28	0.56	0.61	0.06	0.00	0.03	0.01
129-139	MSDNTAANLLL	10	1162.577	7.07	0.30	0.46	0.30	0.32	0.03	0.03	0.00	0.01
138-148	LLTTIGGPKEL	9	1141.683	5.99	1.99	2.41	2.08	2.45	0.01	0.10	0.03	0.00
138-150	LLTTIGGPKELTA	11	1313.768	5.83	1.96	2.63	2.16	2.66	0.03	0.07	0.03	0.02
138-151	LLTTIGGPKELTAF	12	1460.836	6.79	1.71	2.24	1.88	2.28	0.00	0.04	0.02	0.02
138-152	LLTTIGGPKELTAFL	13	1573.920	7.55	1.72	2.42	1.96	2.42	0.01	0.05	0.03	0.02
138-162	LLTTIGGPKELTAFLHNMGDHVTRL	23	2734.471	7.19	2.06	2.93	2.26	2.90	0.01	0.09	0.02	0.05
139-151	LTTIGGPKELTAF	11	1347.752	6.48	1.77	2.40	2.00	2.41	0.02	0.06	0.07	0.02
139-152	LTTIGGPKELTAFL	12	1460.836	7.32	1.82	2.48	2.02	2.48	0.00	0.06	0.03	0.02
139-162	LTTIGGPKELTAFLHNMGDHVTRL	22	2621.387	7.05	2.09	2.91	2.32	2.95	0.04	0.08	0.07	0.02
140-150	TTIGGPKELTA	9	1087.599	5.02	1.91	2.22	2.01	2.18	0.01	0.06	0.03	0.02
140-151	TTIGGPKELTAF	10	1234.668	6.26	1.71	2.02	1.98	2.07	0.02	0.06	0.07	0.03
140-152	TTIGGPKELTAFL	11	1347.752	7.15	1.73	2.11	1.88	2.02	0.01	0.04	0.04	0.02
140-162	TTIGGPKELTAFLHNMGDHVTRL	21	2508.303	6.93	1.92	2.66	1.92	2.65	0.05	0.00	0.03	0.07
149-162	TAFLHNMGDHVTRL	13	1611.806	5.61	1.01	1.35	1.12	1.32	0.01	0.05	0.03	0.02
151-162	FLHNMGDHVTRL	11	1439.721	5.19	0.92	1.18	0.93	1.20	0.02	0.03	0.01	0.02
152-161	LHNMGDHVTR	9	1179.569	4.62	0.69	0.82	0.79	0.88	0.02	0.02	0.03	0.00
152-162	LHNMGDHVTRL	10	1292.653	4.62	0.69	0.91	0.80	0.96	0.03	0.03	0.04	0.02
153-162	HNMGDHVTRL	9	1179.569	4.40	0.72	0.99	0.83	1.08	0.00	0.03	0.02	0.01
163-169	DRWEPEL	5	944.447	6.40	0.76	0.86	0.68	0.82	0.01	0.03	0.01	0.01
163-170	DRWEPELN	6	1058.490	5.76	1.00	1.33	0.89	1.26	0.00	0.01	0.02	0.02
170-182	NEAIPNDERDTTM	11	1505.654	4.85	2.01	2.67	1.78	2.56	0.01	0.07	0.03	0.02
170-186	NEAIPNDERDTTMPAAM	14	1875.821	5.48	2.39	2.98	2.27	2.88	0.01	0.10	0.05	0.03
189-198	TLRKLLTGEL	9	1143.710	6.60	1.71	2.06	1.99	1.98	0.03	0.03	0.09	0.05
190-198	LRKLLTGEL	8	1042.662	5.67	2.05	2.20	2.20	2.30	0.01	0.04	0.03	0.01
191-198	RKLLTGEL	7	929.578	5.35	1.98	2.08	2.06	2.14	0.01	0.05	0.02	0.01
199-207	LTLASRQQL	8	1029.605	5.38	1.52	1.80	1.57	1.64	0.04	0.03	0.04	0.01
199-210	LTLASRQQLIDW	11	1443.795	6.98	1.26	1.47	1.31	1.35	0.02	0.04	0.04	0.02
202-209	ASRQQLID	7	930.500	4.77	0.28	0.24	0.48	0.45	0.00	0.05	0.12	0.03
202-211	ASRQQLIDWM	9	1247.620	7.07	0.17	0.15	0.33	0.30	0.00	0.05	0.05	0.02
210-221	WMEADKVAGPLL	10	1329.687	6.88	4.40	4.48	3.90	4.41	0.01	0.11	0.03	0.02
211-221	MEADKVAGPLL	9	1143.608	6.21	4.26	4.35	3.86	4.32	0.02	0.08	0.08	0.04
212-221	EADKVAGPLL	8	1012.567	5.98	4.41	4.23	3.78	4.36	0.03	0.04	0.07	0.03
213-221	ADKVAGPLL	7	883.525	5.90	3.86	3.65	3.48	3.91	0.04	0.11	0.01	0.04
214-221	DKVAGPLL	6	812.488	5.94	3.47	3.35	3.04	3.42	0.02	0.05	0.03	0.03
214-229	DKVAGPLLRSALPAGW	13	1650.933	6.76	6.87	7.60	6.42	7.81	0.03	0.08	0.03	0.05
222-229	RSALPAGW	6	857.463	6.24	2.33	3.22	2.22	3.11	0.01	0.06	0.01	0.01
222-230	RSALPAGWF	7	1004.531	7.18	2.30	3.28	2.21	3.15	0.01	0.05	0.01	0.06
230-248	FIADKSGAGERGSRGIIAA	18	1876.004	4.76	3.78	4.15	2.66	2.81	0.01	0.11	0.04	0.04
231-246	IADKSGAGERGSRGII	15	1586.861	4.50	3.81	4.18	2.80	3.07	0.06	0.11	0.01	0.01
231-260	IADKSGAGERGSRGIIAALGPDGKPSRIVV	27	2947.644	5.46	3.87	4.97	2.82	3.70	0.03	0.12	0.03	0.01
232-260	ADKSGAGERGSRGIIAALGPDGKPSRIVV	26	2834.560	5.53	3.58	4.75	2.99	4.06	0.03	0.21	0.07	0.08
249-260	LGPDGKPSRIVV	9	1237.726	5.12	1.05	1.67	1.29	1.86	0.02	0.06	0.06	0.02
261-267	IYTTGSQ	6	769.373	4.09	1.81	1.76	1.61	1.76	0.01	0.05	0.03	0.02
261-270	IYTTGSQATM	9	1072.498	5.11	3.94	3.83	3.76	3.85	0.01	0.10	0.03	0.01
261-279	IYTTGSQATMDERNRQIAE	18	2184.035	4.67	4.63	5.74	4.39	5.11	0.05	0.11	0.10	0.07
268-279	ATMDERNRQIAE	11	1433.680	4.11	1.86	2.83	1.72	2.25	0.03	0.15	0.02	0.06
271-281	DERNRQIAEIG	10	1300.660	4.82	1.04	1.91	0.84	1.34	0.03	0.05	0.01	0.01
271-283	DERNRQIAEIGAS	12	1458.730	4.75	1.07	2.00	0.88	1.37	0.02	0.06	0.03	0.01
271-284	DERNRQIAEIGASL	13	1571.814	6.00	1.04	2.04	0.76	1.29	0.07	0.07	0.12	0.04

Supplementary Table S5. HDX-MS dataset for SHV1/BLIP-E73M

	Sequence	MaxUptake	MHP	RT / min	Deuterium uptake / Da				Standard deviation / Da			
					Free		Bound		Free		Bound	
					1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
26-32	SPQPLEQ	4	798.399	3.59	1.32	1.37	1.40	1.46	0.02	0.03	0.00	0.03
33-46	IKLSESQLSGRVGM	13	1504.815	4.64	1.75	2.58	2.01	2.91	0.06	0.04	0.01	0.03
49-59	MDLASGRTLTA	10	1135.578	4.62	3.01	3.69	3.05	3.82	0.07	0.07	0.01	0.03
50-59	DLASGRTLTA	9	1004.537	4.17	2.92	3.66	2.81	3.69	0.07	0.07	0.00	0.03
50-66	DLASGRTLTAWRADERF	16	1964.994	5.27	3.35	4.11	3.57	4.35	0.16	0.07	0.05	0.02
52-58	ASGRTLT	6	705.389	4.17	2.00	2.47	1.99	2.35	0.09	0.07	0.05	0.09
52-71	ASGRTLTAWRADERFPMMST	18	2284.096	5.34	2.58	3.57	2.69	3.38	0.06	0.04	0.03	0.07
60-66	WRADERF	6	979.474	4.60	0.53	0.80	0.57	0.87	0.02	0.01	0.02	0.03
60-69	WRADERFPMM	8	1338.608	5.71	0.55	0.78	0.62	0.87	0.03	0.01	0.00	0.00
60-71	WRADERFPMMST	10	1526.688	5.42	1.09	1.51	1.01	1.38	0.04	0.03	0.01	0.02
60-72	WRADERFPMMSTF	11	1673.756	6.26	1.00	1.42	0.78	1.21	0.01	0.00	0.13	0.03
82-91	ARVDAGDEQL	9	1073.522	4.02	1.38	1.64	1.38	1.64	0.01	0.03	0.02	0.04
92-104	ERKIHYRQQDLVD	12	1699.887	3.50	1.88	2.42	1.70	2.14	0.03	0.04	0.01	0.07
105-116	YSPVSEKHLADG	10	1302.633	3.58	2.32	2.94	1.61	2.40	0.06	0.08	0.00	0.02
105-118	YSPVSEKHLADGMT	12	1534.721	4.17	2.58	3.32	2.18	3.09	0.09	0.07	0.07	0.10
105-122	YSPVSEKHLADGMTVGEL	16	1932.937	5.22	2.52	3.43	2.10	3.04	0.09	0.06	0.01	0.01
125-138	AAITMSDNSAANLL	13	1391.684	5.83	1.09	1.53	0.97	1.34	0.08	0.02	0.02	0.03
130-138	SDNSAANLL	8	904.437	5.26	0.47	0.71	0.31	0.45	0.02	0.04	0.03	0.04
130-139	SDNSAANLLL	9	1017.521	6.02	0.43	0.70	0.32	0.52	0.01	0.03	0.02	0.04
138-148	LLATVGGPAGL	9	968.578	5.66	1.29	1.67	1.23	1.61	0.02	0.05	0.04	0.00
139-151	LATVGGPAGLTAF	11	1174.647	6.17	1.15	1.77	1.13	1.80	0.01	0.02	0.01	0.02
140-151	ATVGGPAGLTAF	10	1061.563	5.95	1.33	1.58	1.22	1.47	0.06	0.04	0.02	0.02
152-162	LRQIGDNVTRL	10	1284.738	4.43	1.39	1.83	1.35	1.87	0.02	0.02	0.02	0.00
152-169	LRQIGDNVTRLDRWETEL	17	2214.163	5.54	2.21	2.59	2.01	2.43	0.03	0.06	0.02	0.06
153-162	RQIGDNVTRL	9	1171.654	4.14	1.32	1.78	1.33	1.71	0.04	0.03	0.01	0.01
163-169	DRWETEL	6	948.442	5.38	1.15	1.14	1.13	1.15	0.01	0.03	0.01	0.00
170-186	NEALPGDARDTTTPASM	14	1746.796	4.31	1.66	2.31	1.64	2.33	0.00	0.00	0.01	0.00
187-199	AATLRKLLTSQRL	12	1470.912	4.94	2.36	2.57	2.30	2.58	0.04	0.05	0.05	0.06
187-201	AATLRKLLTSQRLSA	14	1628.981	4.75	2.88	3.27	2.79	3.20	0.03	0.07	0.01	0.02
188-199	ATLRKLLTSQRL	11	1399.874	4.94	2.11	2.31	2.11	2.37	0.06	0.07	0.00	0.03
188-201	ATLRKLLTSQRLSA	13	1557.944	4.75	2.36	2.71	2.40	2.77	0.06	0.08	0.05	0.07
189-199	TLRKLLTSQRL	10	1328.837	4.94	2.05	2.26	1.93	2.20	0.03	0.04	0.04	0.09
189-201	TLRKLLTSQRLSA	12	1486.906	4.75	2.47	2.81	2.34	2.71	0.04	0.05	0.03	0.03
190-199	LRKLLTSQRL	9	1227.790	4.01	2.11	2.35	2.11	2.37	0.04	0.08	0.04	0.02
190-201	LRKLLTSQRLSA	11	1385.859	3.87	2.49	2.79	2.47	2.88	0.06	0.04	0.01	0.04
191-201	RKLLTSQRLSA	10	1272.775	3.54	2.46	2.80	2.39	2.76	0.05	0.04	0.01	0.01
200-210	SARSQRQLLQW	10	1372.744	5.41	0.35	0.61	0.28	0.60	0.01	0.01	0.00	0.02
202-210	RSQRQLLQW	8	1214.675	5.42	0.01	0.09	0.11	0.18	0.00	0.01	0.03	0.02
210-225	WMVDDRVAGPLIRSVL	14	1826.995	6.40	4.61	5.29	3.33	5.03	0.12	0.13	0.01	0.03
211-225	MVDDRVAGPLIRSVL	13	1640.915	5.80	4.71	5.36	3.19	5.00	0.10	0.12	0.01	0.05
212-225	VDDRVAGPLIRSVL	12	1509.875	5.66	4.64	5.23	3.11	4.97	0.12	0.10	0.00	0.03
212-229	VDDRVAGPLIRSVLPAGW	15	1921.065	6.46	5.61	6.40	4.00	6.24	0.14	0.15	0.07	0.07
213-225	DDRVAGPLIRSVL	11	1410.806	5.80	3.62	4.16	2.60	3.98	0.11	0.11	0.03	0.03
214-225	DRVAGPLIRSVL	10	1295.779	5.43	3.87	4.38	2.62	4.21	0.09	0.09	0.02	0.04
214-229	DRVAGPLIRSVLPAGW	13	1706.970	6.33	4.95	5.81	3.71	5.53	0.13	0.09	0.02	0.01
214-230	DRVAGPLIRSVLPAGWF	14	1854.039	6.90	4.77	5.72	3.51	5.75	0.14	0.12	0.02	0.04
215-225	RVAGPLIRSVL	9	1180.753	5.16	3.44	4.04	2.24	3.78	0.08	0.07	0.00	0.00
215-229	RVAGPLIRSVLPAGW	12	1591.943	6.11	4.07	4.93	2.94	4.66	0.10	0.12	0.02	0.03
215-230	RVAGPLIRSVLPAGWF	13	1739.012	6.72	4.08	4.94	2.99	4.69	0.11	0.12	0.01	0.01
230-248	FIADKTGAGERGARGIVAL	18	1902.056	4.65	4.09	4.47	2.87	4.24	0.07	0.12	0.01	0.09
231-248	IADKTGAGERGARGIVAL	17	1754.987	4.36	4.01	4.34	2.91	4.14	0.07	0.10	0.03	0.04
232-248	ADKTGAGERGARGIVAL	16	1641.903	4.65	3.52	3.84	2.37	3.56	0.04	0.13	0.01	0.11
248-260	LLGPNNKAERIVV	11	1422.843	4.61	0.72	0.97	0.86	1.11	0.05	0.03	0.01	0.16
249-260	LGPNNKAERIVV	10	1309.759	4.29	1.11	1.15	1.18	1.34	0.01	0.02	0.01	0.01
249-262	LGPNNKAERIVVIY	12	1585.906	5.08	0.91	0.97	0.95	1.09	0.02	0.01	0.01	0.01
250-260	GPNNKAERIVV	9	1196.675	4.29	0.83	0.92	0.89	1.06	0.03	0.03	0.01	0.00
251-260	PNNKAERIVV	8	1139.653	4.29	0.94	0.96	0.99	1.09	0.01	0.02	0.00	0.01
261-270	IYLRDTPASM	8	1166.587	4.84	1.97	2.55	1.72	2.27	0.03	0.06	0.01	0.02
271-281	AERNQQIAGIG	10	1156.607	4.07	0.90	1.39	0.85	1.11	0.01	0.01	0.02	0.06
271-283	AERNQQIAGIGAA	12	1298.681	4.16	1.11	1.53	1.06	1.42	0.05	0.04	0.00	0.05
273-279	RNQQIAG	6	786.422	4.65	0.97	1.37	0.98	1.44	0.04	0.01	0.04	0.01

Supplementary Table S6. HDX-MS dataset for PC1/BLIP-K74G

	Sequence	MaxUptake	MHP	RT / min	Deuterium uptake / Da				Standard deviation / Da			
					Free		Bound		Free		Bound	
					1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
23-36	MHHHHHHMKELNDL	13	1815.828	3.34	0.55	0.93	1.00	1.26	0.08	0.14	0.08	0.06
37-46	EKKYNAHIGV	9	1158.627	4.08	0.76	0.76	0.72	0.78	0.03	0.04	0.06	0.06
47-76	YALDTKSGKEVKFNSDKRFAYASTSKAINS	29	3326.702	4.95	4.22	5.70	3.84	4.90	0.17	0.06	0.06	0.18
50-79	DTKSGKEVKFNSDKRFAYASTSKAINSAIL	29	3276.722	6.40	3.03	4.83	2.84	3.23	0.34	0.01	0.13	0.05
80-88	LEQVPYNKL	7	1103.610	6.40	1.65	2.56	1.97	2.93	0.07	0.07	0.04	0.07
81-101	EQVPYNKLNKKVHINKDDIVA	19	2465.351	4.66	3.53	4.52	3.40	4.73	0.33	0.26	0.07	0.12
102-113	YSPILEKYVGKD	10	1411.747	6.59	2.53	3.27	2.13	3.41	0.16	0.11	0.03	0.08
102-116	YSPILEKYVGKDITL	13	1738.963	8.56	2.80	4.16	1.79	3.79	0.08	0.06	0.08	0.08
102-118	YSPILEKYVGKDITLKA	15	1938.095	7.44	2.66	3.96	1.80	3.72	0.17	0.17	0.19	0.17
104-116	PILEKYVGKDITL	11	1488.867	8.57	2.50	3.60	1.61	3.74	0.12	0.08	0.32	0.08
107-116	EKYVGKDITL	9	1165.646	6.14	2.16	3.19	1.49	3.05	0.04	0.07	0.08	0.09
107-118	EKYVGKDITLKA	11	1364.778	4.79	1.88	2.87	1.41	2.86	0.07	0.05	0.03	0.07
114-121	ITLKALIE	7	900.576	8.32	-0.05	0.01	0.10	0.14	0.02	0.01	0.04	0.05
117-124	KALIEASM	7	862.470	6.07	-0.03	0.00	0.14	0.14	0.08	0.09	0.02	0.02
117-125	KALIEASMT	8	963.518	5.60	0.11	0.10	0.14	0.14	0.06	0.05	0.03	0.03
119-125	LIEASMT	6	764.386	5.46	0.11	0.14	0.09	0.10	0.03	0.03	0.03	0.02
122-138	ASMTYSDNTANNKIIKE	16	1899.912	4.88	1.26	1.58	0.81	1.17	0.05	0.06	0.10	0.11
139-162	IGGIKKVKQRLKELGDKVTNPVRY	22	2739.636	5.82	3.30	3.94	3.28	4.39	0.29	0.20	0.05	0.13
162-191	YEIELNYYSPKSKKDTSTPAAFGKTLNKLI	27	3419.810	6.51	2.67	4.09	2.32	3.37	0.25	0.22	0.31	0.17
183-207	FGKTLNKLIANGKLSKENKKFLLDL	24	2832.671	7.81	1.99	2.22	2.30	2.71	0.07	0.08	0.14	0.13
184-207	GKTLNKLIANGKLSKENKKFLLDL	23	2685.603	7.15	1.81	2.58	2.08	2.23	0.03	0.04	0.04	0.08
191-207	IANGKLSKENKKFLLDL	16	1931.132	7.26	1.37	1.55	1.50	1.66	0.20	0.08	0.05	0.07
208-217	MLNNKSGDTL	9	1092.535	5.13	3.12	3.23	2.69	3.23	0.18	0.05	0.04	0.14
208-225	MLNNKSGDTLIKDGVPKD	16	1945.006	5.07	4.83	6.12	3.74	4.73	0.14	0.01	0.01	0.11
208-237	MLNNKSGDTLIKDGVPKDYKVADKSGQAIT	28	3206.673	5.24	7.55	11.66	5.80	7.99	0.09	0.09	0.13	0.16
218-238	IKDGVPKDYKVADKSGQAITY	19	2296.218	4.95	3.41	5.71	3.09	4.87	0.22	0.17	0.04	0.08
218-245	IKDGVPKDYKVADKSGQAITYASRNDVA	26	3009.564	4.80	4.71	7.51	3.86	5.51	0.10	0.10	0.21	0.23
218-246	IKDGVPKDYKVADKSGQAITYASRNDVAF	27	3156.632	5.91	5.48	7.02	4.00	5.67	0.19	0.05	0.15	0.15
238-246	YASRNDVAF	8	1042.495	6.10	0.66	0.82	0.38	0.44	0.08	0.09	0.08	0.07
238-258	YASRNDVAFVYPKGQSEPIVL	18	2353.219	7.87	3.21	3.21	3.11	3.12	0.14	0.06	0.14	0.19
239-246	ASRNDVAF	7	879.432	5.72	0.53	0.61	0.42	0.46	0.09	0.07	0.08	0.05
239-258	ASRNDVAFVYPKGQSEPIVL	17	2190.155	7.76	3.25	3.14	3.09	3.04	0.12	0.05	0.07	0.16
246-257	FVYPKGQSEPIV	9	1363.726	6.32	2.73	2.80	2.91	2.69	0.13	0.02	0.09	0.19
259-275	VIFTNKDNKSDKPNDKL	15	1976.045	4.03	2.27	3.01	2.33	2.43	0.36	0.06	0.05	0.17
260-278	IFTNKDNKSDKPNDKLISE	17	2206.135	4.57	3.38	4.30	2.85	3.80	0.26	0.12	0.15	0.21
260-287	IFTNKDNKSDKPNDKLISETAKSVMKEF	26	3227.662	8.06	3.64	4.92	2.33	3.47	0.27	0.21	0.13	0.11
261-275	FTNKDNKSDKPNDKL	13	1763.892	4.03	1.99	2.69	2.00	2.20	0.27	0.04	0.12	0.12
261-284	FTNKDNKSDKPNDKLISETAKSVM	22	2710.372	5.96	2.66	3.03	0.93	1.20	0.34	0.31	0.10	0.15
261-287	FTNKDNKSDKPNDKLISETAKSVMKEF	25	3114.578	8.06	3.07	4.24	1.44	1.99	0.31	0.16	0.11	0.21
277-287	SETAKSVMKEF	10	1256.619	7.06	1.06	1.91	1.01	1.67	0.05	0.09	0.05	0.06
279-287	TAKSVMKEF	8	1040.545	5.76	1.19	1.83	1.09	1.86	0.10	0.01	0.05	0.05

Supplementary Table S7. HDX-MS dataset for BLIP/TEM1

	Sequence	Modification	MaxUptake	MHP	RT / min	Deuterium uptake / Da				Standard deviation / Da			
						Free		Bound		Free		Bound	
						1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
1-9	AGVMTGAKF		8	881.455	4.54	3.18	4.24	2.90	3.76	0.02	0.01	0.03	0.01
1-14	AGVMTGAKFTQIQF		13	1498.7723	5.75	4.18	5.80	2.97	4.29	0.06	0.03	0.13	0.05
10-22	TQIQFGMTRQQVL		12	1549.8155	5.29	1.86	2.48	1.98	2.63	0.01	0.09	0.16	0.04
15-22	GMTRQQVL		7	932.4982	3.98	1.02	1.58	1.07	1.48	0.01	0.03	0.02	0.02
15-23	GMTRQQVLD		8	1047.5252	3.63	0.91	1.43	1.08	1.45	0.04	0.03	0.09	0.07
66-85	KVDSKSQEKLLAPSAPTLTL		17	2126.2067	4.96	4.15	5.61	3.71	4.34	0.05	0.08	0.06	0.19
74-85	KLLAPSAPTLTL		9	1224.7562	5.46	4.06	5.35	3.69	4.56	0.16	0.05	0.08	0.03
86-94	AKFNQVTVG		8	963.5258	3.92	1.91	2.85	1.80	2.83	0.02	0.04	0.06	0.06
86-95	AKFNQVTVGM		9	1094.5663	4.58	1.72	2.77	1.64	2.67	0.09	0.02	0.12	0.07
86-98	AKFNQVTVGMTRA		12	1422.7522	4.09	3.20	4.26	3.88	4.90	0.03	0.01	0.04	0.02
86-101	AKFNQVTVGMTRAQVL		15	1762.9632	4.79	3.74	5.06	3.99	5.50	0.00	0.03	0.09	0.01
89-101	NQVTVGMTRAQVL		12	1416.7628	4.74	4.63	5.45	3.84	4.77	0.07	0.04	0.10	0.10
95-101	MTRAQVL		6	818.4553	3.97	2.30	2.25	2.43	2.33	0.04	0.01	0.04	0.01
137-149	MTRAQVL		12	1445.6808	4.46	3.30	3.54	1.99	2.56	0.07	0.12	0.04	0.03
143-149	YRGSAHL		6	803.4159	3.18	0.71	1.09	0.41	0.64	0.03	0.01	0.02	0.01
150-156	WFTDGVL		6	837.4141	6.80	1.06	1.11	1.05	1.12	0.01	0.01	0.00	0.01
150-165	WFTDGVLQGKRQWDLV		15	1948.0076	6.18	2.53	3.53	1.55	2.18	0.04	0.02	0.07	0.03
157-165	QGKRQWDLV		8	1129.6113	4.79	1.97	2.92	1.17	1.56	0.03	0.01	0.02	0.00

Supplementary Table S8. HDX-MS dataset for BLIP/SHV1

	Sequence	Modification	MaxUptake	MHP	RT / min	Deuterium uptake / Da				Standard deviation / Da			
						Free		Bound		Free		Bound	
						1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
0-9	FAGVMTGAKF		9	1028.5234	5.43	1.10	1.51	1.07	1.55	0.04	0.00	0.02	0.00
1-9	AGVMTGAKF		8	881.455	4.80	1.38	2.04	1.27	1.93	0.01	0.01	0.00	0.03
1-14	AGVMTGAKFTQIQF		13	1498.7723	5.98	1.84	2.52	1.60	2.49	0.01	0.01	0.09	0.04
10-22	TQIQFGMTRQQVL		12	1549.8155	5.51	0.83	1.06	0.77	1.03	0.02	0.01	0.05	0.02
15-22	GMTRQQVL		7	932.4982	4.24	0.35	0.61	0.36	0.56	0.00	0.02	0.06	0.02
41-57	HCRGHAAGDYYAYATFG	Carbamidomethyl C (2)	16	1916.8133	3.92	2.63	2.55	2.85	2.77	0.09	0.04	0.01	0.07
51-61	YAYATFGFTSA		10	1198.5415	6.49	0.69	0.79	0.47	0.89	0.02	0.01	0.18	0.07
66-85	KVDSKSQEKLLAPSAPTLTL		17	2126.2067	5.15	1.69	2.65	1.54	2.65	0.18	0.09	0.07	0.01
74-85	KLLAPSAPTLTL		9	1224.7562	5.66	1.46	2.13	1.45	2.18	0.08	0.05	0.03	0.00
86-94	AKFNQVTVG		8	963.5258	4.16	0.74	1.25	0.57	1.04	0.03	0.02	0.01	0.05
86-98	AKFNQVTVGMTRA		12	1422.7522	4.30	1.64	2.32	1.47	2.29	0.02	0.04	0.03	0.06
86-101	AKFNQVTVGMTRAQVL		15	1762.9632	4.99	1.87	2.59	1.71	2.59	0.08	0.01	0.07	0.00
89-101	NQVTVGMTRAQVL		12	1416.7628	4.98	2.06	2.70	1.85	2.71	0.05	0.00	0.02	0.02
95-101	MTRAQVL		6	818.4553	4.22	0.98	1.02	0.99	1.11	0.02	0.01	0.00	0.01
137-149	YSSTGFYRGSAHL		12	1445.6808	4.66	1.67	1.77	1.42	1.64	0.13	0.01	0.07	0.08
143-149	YRGSAHL		6	803.4159	3.26	0.24	0.42	0.23	0.34	0.00	0.00	0.00	0.00
150-165	WFTDGVLQGKRQWDLV		15	1948.0076	6.39	1.06	1.54	0.85	1.28	0.04	0.04	0.02	0.02
157-165	QGKRQWDLV		8	1129.6113	5.04	0.72	1.13	0.65	1.01	0.02	0.02	0.00	0.00

Supplementary Table S9. HDX-MS dataset for BLIP/PC1

	Sequence	Modification	MaxUptake	MHP	RT / min	Deuterium uptake / Da				Standard deviation / Da			
						Free		Bound		Free		Bound	
						1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
0-9	FAGVMTGAKF		9	1028.5234	5.53	1.01	1.49	1.19	1.65	0.03	0.01	0.12	0.09
1-9	AGVMTGAKF		8	881.455	4.90	1.20	1.95	1.00	1.70	0.03	0.04	0.08	0.00
1-14	AGVMTGAKFTQIQF		13	1498.7723	6.06	1.42	2.50	1.25	2.32	0.10	0.08	0.12	0.11
10-22	TQIQFGMTRQQVL		12	1549.8155	5.61	0.66	1.03	0.89	1.05	0.02	0.03	0.04	0.03
15-22	GMTRQQVL		7	932.4982	4.34	0.30	0.59	0.35	0.65	0.01	0.02	0.02	0.01
41-57	HCRGHAAGDYYAYATFG	Carbamidomethyl C (2)	16	1916.8133	4.00	2.46	2.53	2.64	2.96	0.02	0.07	0.17	0.07
51-61	YAYATFGFTSA		10	1198.5415	6.57	0.45	0.75	0.59	0.95	0.01	0.05	0.06	0.15
66-85	KVDSKSQEKLLAPSAPTLTL		17	2126.2067	5.23	1.63	2.70	1.47	2.49	0.11	0.10	0.09	0.15
74-85	KLLAPSAPTLTL		9	1224.7562	5.75	1.24	2.04	1.11	1.96	0.04	0.01	0.04	0.00
86-98	AKFNQVTVGMTRA		12	1422.7522	4.40	1.30	2.08	1.17	2.10	0.07	0.02	0.14	0.03
86-101	AKFNQVTVGMTRAQVL		15	1762.9632	5.09	1.69	2.63	1.54	2.62	0.05	0.07	0.17	0.07
89-101	NQVTVGMTRAQVL		12	1416.7628	5.07	1.62	2.51	1.57	2.43	0.02	0.00	0.12	0.07
95-101	MTRAQVL		6	818.4553	4.32	0.88	1.01	1.10	1.30	0.01	0.00	0.07	0.01
137-149	YSSTGFYRGSAHL		12	1445.6808	4.78	1.16	1.53	1.16	1.47	0.02	0.03	0.08	0.13
143-149	YRGSAHL		6	803.4159	3.42	0.16	0.37	0.15	0.22	0.00	0.02	0.01	0.03
150-165	WFTDGVLQGKRQWDLV		15	1948.0076	6.46	0.859	1.494	0.5	0.914	0.03	0.032	0.02	0.036
157-165	QGKRQWDLV		8	1129.6113	5.1413	0.642	1.191	0.45	0.811	0.012	0.061	0.02	0.042

Supplementary Table S10. HDX-MS dataset for BLIP-Y50A/TEM1

	Sequence	MaxUptake	MHP	RT / min	Deuterium uptake / Da				Standard deviation / Da			
					Free		Bound		Free		Bound	
					1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
1-7	AGVMTGA	6	606.292	4.78	3.66	3.53	3.56	3.41	0.00	0.03	0.02	0.01
1-9	AGVMTGAKF	8	881.455	4.54	3.77	4.44	3.30	3.86	0.05	0.02	0.03	0.03
1-11	AGVMTGAKFTQ	10	1110.561	5.17	6.22	6.36	6.10	6.18	0.06	0.05	0.07	0.03
1-14	AGVMTGAKFTQIQF	13	1498.772	6.77	4.98	6.14	3.50	4.49	0.14	0.06	0.06	0.02
1-19	AGVMTGAKFTQIQFGMTRQ	18	2072.042	6.19	6.43	8.07	5.61	6.89	0.22	0.12	0.14	0.10
3-9	VMTGAKF	6	753.396	5.24	2.32	2.72	1.99	2.31	0.01	0.02	0.04	0.01
3-14	VMTGAKFTQIQF	11	1370.714	6.66	4.00	4.99	2.63	3.26	0.06	0.07	0.10	0.07
3-22	VMTGAKFTQIQFGMTRQQVL	19	2284.194	6.46	4.86	6.87	3.71	5.64	0.13	0.14	0.08	0.10
7-14	AKFTQIQF	7	982.536	6.44	1.68	2.92	0.95	1.72	0.00	0.05	0.04	0.02
7-19	AKFTQIQFGMTRQ	12	1555.805	5.80	4.28	5.17	4.34	4.24	0.12	0.11	0.24	0.14
7-22	AKFTQIQFGMTRQQVL	15	1896.016	6.29	4.60	5.66	4.41	4.67	0.35	0.07	0.10	0.10
9-19	FTQIQFGMTRQ	10	1356.673	6.11	3.09	4.03	3.14	3.46	0.09	0.02	0.16	0.06
9-22	FTQIQFGMTRQQVL	13	1696.884	6.66	3.02	4.19	1.94	3.04	0.21	0.07	0.13	0.07
10-19	TQIQFGMTRQ	9	1209.605	5.61	3.29	3.64	3.12	2.97	0.16	0.03	0.08	0.06
10-22	TQIQFGMTRQQVL	12	1549.816	6.34	3.40	4.03	3.18	3.08	0.25	0.03	0.12	0.07
10-23	TQIQFGMTRQQVLD	13	1664.843	6.08	3.69	4.34	3.65	3.58	0.27	0.02	0.15	0.05
12-29	IQFGMTRQQVLDIAGAEN	17	1991.002	7.67	3.40	4.64	1.70	2.33	0.08	0.05	0.07	0.04
15-22	GMTRQQVL	7	932.498	5.10	2.02	2.57	2.02	2.21	0.07	0.05	0.07	0.07
15-23	GMTRQQVLD	8	1047.525	4.73	2.32	2.78	2.16	2.18	0.06	0.05	0.07	0.04
23-29	DIAGAEN	6	689.310	3.95	2.87	2.78	2.87	2.76	0.00	0.04	0.00	0.06
36-45	FGDSIHCRGH	9	1128.500	6.81	3.74	4.13	3.34	3.89	0.16	0.09	0.05	0.06
36-50	FGDSIHCRGHAAGDA	14	1513.660	6.66	6.16	6.48	6.32	6.43	0.04	0.05	0.09	0.17
53-61	YATFGFTSA	8	964.441	6.92	2.26	2.69	2.09	2.38	0.07	0.00	0.03	0.00
53-63	YATFGFTSAAA	10	1106.515	6.81	3.60	3.94	3.55	3.81	0.10	0.05	0.08	0.04
53-65	YATFGFTSAAADA	12	1292.579	6.81	3.61	4.05	3.46	3.67	0.19	0.06	0.03	0.05
57-63	GFTSAAA	6	624.299	4.81	2.85	2.94	2.83	2.96	0.07	0.03	0.04	0.02
66-85	KVDSKSQEKLLAPSAPTLTL	17	2126.207	5.94	5.07	6.67	3.89	4.75	0.07	0.02	0.03	0.01
74-85	KLLAPSAPTLTL	9	1224.756	6.47	4.41	5.68	3.80	4.75	0.02	0.02	0.04	0.00
76-85	LAPSAPTLTL	7	983.577	6.63	2.70	3.47	2.39	3.02	0.00	0.05	0.03	0.02
83-100	LTLAKFNQVTVGMTRAQV	17	1977.095	6.10	5.54	7.18	5.04	6.32	0.19	0.10	0.05	0.01
85-101	LAKFNQVTVGMTRAQVL	16	1876.047	6.00	4.91	6.46	4.40	5.94	0.08	0.09	0.07	0.07
86-94	AKFNQVTVG	8	963.526	4.99	1.88	2.97	1.62	2.40	0.06	0.02	0.05	0.06
86-95	AKFNQVTVGM	9	1094.566	5.65	2.25	3.36	1.76	2.63	0.07	0.04	0.02	0.05
86-98	AKFNQVTVGMTRA	12	1422.752	5.10	4.11	5.40	3.77	4.90	0.10	0.02	0.02	0.04
86-101	AKFNQVTVGMTRAQVL	15	1762.963	5.79	4.64	6.00	3.97	5.10	0.16	0.07	0.05	0.06
89-98	NQVTVGMTRA	9	1076.552	4.78	3.87	4.73	3.60	4.83	0.03	0.04	0.03	0.13
89-101	NQVTVGMTRAQVL	12	1416.763	5.80	4.14	5.08	4.08	5.02	0.15	0.05	0.05	0.06
95-101	MTRAQVL	6	818.455	5.09	2.36	2.49	2.32	2.28	0.01	0.00	0.04	0.03
116-127	YPAYPSTAGVTL	9	1239.626	6.50	2.99	3.85	2.64	3.21	0.09	0.04	0.06	0.03
133-142	DVDGYSSTGF	9	1047.427	6.21	3.89	3.89	3.01	3.53	0.00	0.05	0.02	0.03
137-149	YSSTGFYRGSAHL	12	1445.681	5.47	2.88	3.34	2.19	2.31	0.20	0.09	0.05	0.06
143-149	YRGSAHL	6	803.416	4.10	0.63	1.00	0.28	0.42	0.02	0.01	0.01	0.01
150-156	WFTDGVL	6	837.414	7.92	0.48	0.56	0.41	0.45	0.01	0.00	0.03	0.02
150-158	WFTDGVLQG	8	1022.494	7.25	0.94	1.32	0.72	0.83	0.02	0.02	0.01	0.05
150-162	WFTDGVLQGKRQW	12	1620.828	6.66	1.72	2.42	0.97	1.20	0.19	0.02	0.09	0.10
150-165	WFTDGVLQGKRQWDLV	6	837.414	6.80	2.64	3.84	1.46	2.03	0.07	0.05	0.16	0.11
155-165	VLQGKRQWDLV	10	1341.764	6.01	3.09	3.81	2.02	2.31	0.07	0.08	0.13	0.04
157-165	QGKRQWDLV	8	1129.611	4.79	2.16	3.01	1.12	1.47	0.04	0.03	0.11	0.01

Supplementary Table S11. HDX-MS dataset for BLIP-E73M/SHV1

						Deuterium uptake / Da				Standard deviation / Da			
	Sequence	Modification	MaxUptake	MHP	RT / min	Free		Bound		Free		Bound	
						1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
0-9	FAGVMTGAKF		9	1028.5234	6.01	2.84	3.26	2.57	2.94	0.03	0.01	0.13	0.01
1-9	AGVMTGAKF		8	881.455	5.25	3.12	3.53	2.90	3.47	0.05	0.01	0.13	0.01
1-14	AGVMTGAKFTQIQF		13	1498.7723	6.65	4.65	5.58	4.22	5.39	0.08	0.01	0.09	0.01
3-9	VMTGAKF		6	753.3964	4.83	2.04	2.30	2.03	2.24	0.08	0.04	0.06	0.09
7-14	AKFTQIQF		7	982.5356	6.23	2.81	3.04	2.96	3.13	0.09	0.10	0.11	0.06
10-19	TQIQFGMTRQ		9	1209.6045	5.20	3.41	3.50	3.30	3.50	0.10	0.01	0.14	0.16
10-22	TQIQFGMTRQQVL		12	1549.8155	6.08	4.24	4.46	4.10	4.41	0.11	0.04	0.09	0.01
15-22	GMTRQQVL		7	932.4982	4.68	2.39	2.57	2.36	2.66	0.04	0.00	0.06	0.02
53-61	YATFGFTSA		8	964.4411	6.86	2.31	2.58	2.25	2.58	0.10	0.07	0.05	0.04
66-85	KVDSKSQEKLLAPSAPTLTL		17	2128.2046	5.80	4.43	5.77	3.64	4.52	0.19	0.17	0.11	0.18
74-85	KLLAPSAPTLTL		9	1224.7562	6.26	3.96	4.99	3.65	4.65	0.06	0.02	0.09	0.03
86-94	AKFNQVTVG		8	963.5258	4.52	1.93	2.69	1.56	2.25	0.10	0.02	0.07	0.03
86-98	AKFNQVTVGMTRA		12	1422.7522	4.58	3.41	4.38	3.21	4.40	0.05	0.02	0.09	0.03
86-101	AKFNQVTVGMTRAQVL		15	1762.9632	5.40	3.89	5.11	3.83	4.86	0.06	0.03	0.15	0.05
89-101	NQVTVGMTRAQVL		12	1416.7628	5.42	4.16	5.06	4.23	5.38	0.05	0.02	0.07	0.01
133-142	DVDGYSSTGF		9	1047.4265	5.99	3.11	3.10	2.81	3.31	0.05	0.05	0.09	0.08
137-149	YSSTGFYRGSAHL		12	1445.6808	5.0133	2.322	2.799	2.11	2.411	0.121	0.072	0.11	0.081
143-149	YRGSAHL		6	803.4159	4.1315	0.503	0.8	0.38	0.402	0.045	0.018	0.04	0.004
150-165	WFTDGVLQGKRQWDLV		15	1948.0076	7.1214	2.813	3.814	2.15	2.731	0.075	0.06	0.16	0.016
157-165	QGKRQWDLV		8	1129.6113	5.5108	1.836	2.57	1.32	1.739	0.064	0.068	0.08	0.026

Supplementary Table S12. HDX-MS dataset for BLIP-K74G/PC1

						Deuterium uptake / Da				Standard deviation / Da			
	Sequence	Modification	MaxUptake	MHP	RT / min	Free		Bound		Free		Bound	
						1 min	10 min	1 min	10 min	1 min	10 min	1 min	10 min
1-9	AGVMTGAKF		6	752.3097	5.57	1.98	2.03	2.23	2.17	0.02	0.02	0.02	0.01
1-19	AGVMTGAKFTQIQFGMTRQ		8	881.455	5.27	2.82	3.22	2.91	3.35	0.04	0.00	0.10	0.04
10-22	TQIQFGMTRQQVL		18	2072.0416	5.91	6.01	6.80	5.08	6.37	0.13	0.06	0.13	0.16
15-22	GMTRQQVL		12	1549.8155	6.12	3.99	4.39	3.68	4.45	0.01	0.12	0.10	0.05
66-85	KVDSKSQEGLLAPSAPTLTL		7	932.4982	4.69	2.10	2.43	2.10	2.58	0.01	0.04	0.06	0.02
86-98	AKFNQVTVGMTRA		17	2055.1332	6.08	6.06	6.40	5.31	6.28	0.06	0.05	0.16	0.08
86-101	AKFNQVTVGMTRAQVL		12	1422.7522	4.59	4.06	4.70	3.55	4.69	0.06	0.05	0.14	0.09
89-101	NQVTVGMTRAQVL		15	1762.9632	5.42	5.03	5.80	4.42	5.70	0.01	0.06	0.09	0.05
95-101	MTRAQVL		12	1416.7628	5.45	4.91	5.31	4.83	5.73	0.01	0.01	0.17	0.03
133-142	DVDGYSSTGF		6	818.4553	4.70	2.11	2.29	2.12	2.22	0.02	0.01	0.00	0.00
137-149	YSSTGFYRGSAHL		9	1047.4265	6.03	2.97	2.90	3.17	3.09	0.07	0.01	0.02	0.03
143-149	YRGSAHL		12	1445.6808	5.02	2.77	3.10	2.74	2.95	0.06	0.02	0.08	0.09
150-156	WFTDGVL		6	803.4159	4.14	0.75	0.99	0.62	0.77	0.01	0.00	0.01	0.01
150-165	WFTDGVLQGKRQWDLV		6	837.4141	8.18	1.27	1.28	1.08	1.52	0.01	0.04	0.06	0.12
157-165	QGKRQWDLV		15	1948.0076	7.17	4.22	4.67	3.31	4.35	0.06	0.10	0.14	0.04